

Majordome

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Majordome

Getting Started

If you do not intend to build/develop the package, you can find us on [Majordome on PyPI](#) and install it with pip:

```
pip install majordome
```

To register majordome as a Jupyter kernel, run:

```
majordome --install-kernel
```

You can customize the kernel identity if needed:

```
majordome --install-kernel `
  --kernel-name majordome-dev `
  --display-name "Majordome (dev)"
```

If you don't fit in the first group, you can build the package from source. For that, please refer to the development section below.

Tip

If you just prefer to compile locally without delving into the development environment, you can install both Rust and uv and then run `uv build --wheel` to generate the wheel for your platform. This will generate a wheel that you can install with `uv pip install dist/*.whl` or `pip install dist/*.whl` in the current environment. Note that this wheel will be specific to your platform and may not be compatible with other platforms, so it is recommended to use the release wheels for distribution.

Part I

Majordome API

1 Engineering

aka majordome.engineering

1.1 Reactor models

1.1.1 Warnings

Module warnings can be controlled through the following attributes:

i `toggle_reactor_warnings`

`toggle_reactor_warnings(**kwargs : Any): -> None:`

Reverse truth value of warning flags.

Parameters

`toggle_non_key_value : bool | None = None`

If true, reverse truth value of warning about non key-value composition.

`toggle_missing_species_name : bool | None = None`

If true, reverse truth value of warning about missing species name.

`toggle_unknown_species : bool | None = None`

If true, reverse truth value of warning about unknown species.

1.1.2 Utilities

Let's start by creating a standard Cantera solution:

```
solution = ct.Solution("airish.yaml")  
solution.TPY = 273.15, ct.one_atm, "N2: 0.78 O2: 0.21, AR: 0.01"
```

Conversion of composition strings with name filtering is available:

composition_to_dict

```
composition_to_dict(  
    Y : str,  
    species_names : list[str] = []  
) -> dict[str, float]:
```

Convert a Cantera composition string to dictionary.

Parameters

Y : str

Composition specification string, e.g. "O2: 1, N2: 3".

species_names : list[str] = []

List of valid species names for validation. If provided, only species in this list will be included in the output dictionary. If not provided, all species will be included.

```
composition_to_dict("O2: 1, teste: 1", solution.species_names)
```

```
{'O2': 1.0}
```

It is also possible to set unit composition with species names; if you do not provide a validation list, all are are kept:

```
composition_to_dict("O2, N2, hello")
```

```
{'O2': 1.0, 'N2': 1.0, 'hello': 1.0}
```

When working with arrays, take care not to end up in the following situation:

```
composition_to_array("", teste: 1", solution.species_names)
```

```
array([0., 0., 0.])
```

There is also a helper for generating inputs for use with `tabulate.tabulate`:

solution_report

```
solution_report(  
    sol : cantera.composite.Solution | cantera.composite.Quantity,  
    specific_props : bool = True,  
    composition_spec : str = 'mass',  
    selected_species : list[str] = [],  
    **kwargs : Any
```



```
) -> list[tuple[str, str, typing.Any]]:
```

Generate a solution report for tabulation.

Parameters

`sol : cantera.composite.Solution | cantera.composite.Quantity`

Cantera solution object for report generation.

`specific_props : bool = True`

If true, add specific heat capacity and enthalpy.

`composition_spec : str = 'mass'`

Composition units specification, mass or mole.

`selected_species : list[str] = []`

Selected species to display; return all if a composition specification was provided.

`show_mass : bool = False`

If true, add mass of quantity to report if `sol` is a `ct.composite.Quantity`.

```
data = solution_report(solution, specific_props=True,  
                        composition_spec="mass", selected_species=[])  
print(tabulate(data))
```

```
-----
```

Temperature	K	273.15
Pressure	Pa	101325
Density	kg/m ³	1.28735
Specific enthalpy	J/(kg.K)	-25114.9
Specific heat capacity	J/(kg.K)	1004.95
mass: AR	-	0.01
mass: N2	-	0.78
mass: O2	-	0.21

```
-----
```

Because sometimes Cantera lacks hard-copy utilities for certain classes, we provide simple wrappers that create new instances and set the state to the same of the source object. Nothing checked, nothing tested. A first of this kind is `copy_solution`, illustrated below:

i copy_solution

copy_solution(sol : Solution): -> Solution:

Makes a hard copy of a Solution object.

Parameters

sol : Solution

Solution to be copied.

```
newairs = copy_solution(solution)
newairs.TPX = 373.15, None, newairs.X
print(tabulate(solution_report(newairs)))
```

```
-----
```

Temperature	K	373.15
Pressure	Pa	101325
Density	kg/m ³	0.942356
Specific enthalpy	J/(kg.K)	75902.1
Specific heat capacity	J/(kg.K)	1015.82
mass: AR	-	0.01
mass: N2	-	0.78
mass: O2	-	0.21

```
-----
```

```
print(tabulate(solution_report(solution)))
```

```
-----
```

Temperature	K	273.15
Pressure	Pa	101325
Density	kg/m ³	1.28735
Specific enthalpy	J/(kg.K)	-25114.9
Specific heat capacity	J/(kg.K)	1004.95
mass: AR	-	0.01
mass: N2	-	0.78
mass: O2	-	0.21

```
-----
```

In addition to this, there is `copy_quantity`, which proves quite useful in establishing an *algebra* of mixtures.

i copy_quantity

copy_quantity(qty : Quantity): -> Quantity:

Makes a hard copy of a `ct.composite.Quantity` object.

Parameters

qty : Quantity

Quantity to be copied.

```
air = copy_solution(solution)

air1 = ct.Quantity(air, mass=1.0)
air1.TPX = 373.15, None, newairs.X

air2 = copy_quantity(air1)
air2.TPX = 273.15, None, air2.X

mixair = air1 + air2
air1.T, air2.T, mixair.T

(373.15, 273.15, 323.337485365699)
```

1.1.3 NormalFlowRate

Common daily work activity for the process engineer is to perform mass balances, but wait, ..., gas flow rates are generally provided under normal conditions, and compositions may vary, so you need to compute normal densities first... whatever. This class provides a calculator wrapping a Cantera solution object so that your life gets easier.

NormalFlowRate

```
NormalFlowRate(
    mech : str | pathlib.Path,
    *,
    X : str | dict[str, float] | None = None,
    Y : str | dict[str, float] | None = None,
    T_ref : float = 273.15,
    P_ref : float = 101325.0,
    name : str | None = None
) -> None:
```

Compute normal flow rate for a given composition.

This class makes use of the user defined state to create a function object that converts industrial scale flow rates in normal cubic meters per hour to kilograms per second. Nothing more, nothing less, it aims at helping the process engineer in daily life for this quite repetitive need when performing mass balances.

Parameters

mech : str | pathlib.Path

Path to Cantera mechanism used to compute mixture properties.

```
X : str | dict[str, float] | None = None
```

Composition specification in mole fractions. Notice that both X and Y are mutually exclusive keyword arguments.

```
Y : str | dict[str, float] | None = None
```

Composition specification in mass fractions. Notice that both X and Y are mutually exclusive keyword arguments.

```
T_ref : float = 273.15
```

Reference temperature of the system. If your industry does not use the same standard as the default values, and only in that case, please consider updating this keyword.

```
P_ref : float = 101325.0
```

Reference pressure of the system. If your industry does not use the same standard as the default values, and only in that case, please consider updating this keyword.

```
name : str | None = None
```

Name of phase in mechanism, if more than one are specified within the same Cantera YAML database file.

Its simplest use case is as follows:

```
nfr = NormalFlowRate("airish.yaml")
print(f"Convert 1000 Nm3/h to {nfr(1000.0):.5f} kg/s")
```

Convert 1000 Nm³/h to 0.35903 kg/s

If the database file default composition does not suit you, no problems:

```
nfr = NormalFlowRate("airish.yaml", X="N2: 1")
print(f"Convert 1000 Nm3/h to {nfr(1000.0):.5f} kg/s")
```

Convert 1000 Nm³/h to 0.34718 kg/s

You can also print a nice report to inspect the details of internal state. For more, please check its API documentation.

```
print(nfr.report())
```

-----	-----	-----
Temperature	K	273.15
Pressure	Pa	101325
Density	kg/m ³	1.24985
Specific enthalpy	J/(kg.K)	-25864.9
Specific heat capacity	J/(kg.K)	1035.52
mass: N2	-	1

1.1.4 Plug-flow reactors

1.1.5 PlugFlowAxialSources

PlugFlowAxialSources

```
PlugFlowAxialSources(  
    n_reactors : int,  
    n_species : int  
    ) -> None:
```

Provides a data structure for use with PlugFlowChainCantera.
Helper data class for use with the solution method loop of the plug-flow reactor implementation.
It provides the required memory for storage of source terms distributed along the reactor.

Parameters

`n_reactors` : int

Number of reactors in chain.

`n_species` : int

Number of species in mechanism.

`Q` : NDArray[np.float64] = None

Array of external heat source [W].

`m` : NDArray[np.float64] = None

Array of axial mass source terms [kg/s].

`h` : NDArray[np.float64] = None

Array of enthalpy of axial mass source terms [J/kg].

`Y` : NDArray[np.float64, np.float64] = None

Array of mass fractions of axial mass source terms [-].

1.1.5.1 PlugFlowChainCantera

PlugFlowChainCantera

```
PlugFlowChainCantera(  
    mechanism : str,  
    phase : str,  
    z : numpy.ndarray[tuple[typing.Any, ...], numpy.dtype[numpy.float64]],  
    V : numpy.ndarray[tuple[typing.Any, ...], numpy.dtype[numpy.float64]],  
    P : float = 101325.0,
```

```
K : float = 1.0,
smoot_flux : bool = False,
cantera_steady : bool = True
) -> None:
```

Plug-flow reactor as a chain of 0-D reactors with Cantera.

Parameters

`mechanism` : `str`

Name or path to Cantera mechanism to be used.

`phase` : `str`

Name of phase to simulate (not inferred, even if a single is present!).

`z` : `numpy.ndarray[tuple[typing.Any, ...], numpy.dtype[numpy.float64]]`

Spatial coordinates of reactor cells [m].

`V` : `numpy.ndarray[tuple[typing.Any, ...], numpy.dtype[numpy.float64]]`

Volumes of reactor cells [m³].

`P` : `float` = 101325.0

Reactor operating pressure [Pa].

`K` : `float` = 1.0

Valve response constant (do not use unless simulation fails).

`smoot_flux` : `bool` = False

Apply a smoot transition function when internal stepping is performed; this is intended to avoid unphysical steady state approximations.

`cantera_steady` : `bool` = True

If true, use Cantera's `advance_to_steady_state` to solve problem; otherwise advance over meaningful time-scale of the problem.

i get_reactor_data

```
get_reactor_data(pfr : PlugFlowChainCantera): -> PlugFlowAxialSources:
```

Wrapper to allocate properly dimensioned solver data.

Parameters

pfr : PlugFlowChainCantera

Reactor for which data is to be allocated.

1.2 Energy models

This module is concerned with providing classes for computation of energy sources; these are mostly related to estimation of fuel consumption or integration to another simulation modules. Hereafter we illustrate the in cascading complexity the available energy sources. Some of these are simple wrapper around features provided by the combustion utilities.

1.2.1 CombustionAtmosphereCHON

A common need in industry is to evaluate the real heating value from a gas composition reported by the supplier. This is the main goal of CombustionAtmosphereCHON, which has a very simple API provided below:

```
gas = CombustionAtmosphereCHON("gri30.yaml")
lhv = gas.solution_heating_value("CH4: 1", "O2: 1")
print(f"Solution heating value: {lhv:.2f} MJ/kg")
```

Solution heating value: 50.03 MJ/kg

1.2.2 CombustionPowerSupply

Built upon CombustionAtmosphereCHON, CombustionPowerSupply performs the basic calculations for retrieving required flow rates from a given power specification; this proves very handy for the CFD engineer.

```
supply = CombustionPowerSupply(500.0, 1.0, "CH4: 1", "O2: 1", "gri30.yaml")
print(supply.report())
```

Property	Unit	Value
Required power	kW	500
Lower heating value	MJ/kg	50.0254
Fuel mass flow rate	kg/h	35.9817
Oxidizer mass flow rate	kg/h	143.532

Total mass flow rate	kg/h	179.514	
Fuel volume flow rate	Nm ³ /h	50.2707	
Oxidizer volume flow rate	Nm ³ /h	100.541	
Total volume flow rate	Nm ³ /h	150.812	
Water production	kg/h	80.8092	
Carbon dioxide production	kg/h	98.7047	
Total emissions	kg/h	179.514	

1.2.3 Energy sources

1.2.3.1 HeatedGasEnergySource

```
source = HeatedGasEnergySource("airish.yaml", 500.0, mass_flow_rate=1.0)
print(source.report())
```

Property	Unit	Value	
-----	-----	-----	
Source kind		HeatedGasEnergySource	
Provided power	kW	500.0	
Source		airish.yaml	
Phase		air	
Reference area	m ²	1.0	
Mass flow rate	kg/s	1.0	
Volume flow rate	m ³ /s	2.2051590808893957	
Momentum flux	kg.m/s ²	2.2051590808893957	
Reference temperature	K	298.15	
Reference pressure	Pa	101325.0	
Temperature	K	778.5219246537463	
Pressure	Pa	101324.99999999999	
Density	kg/m ³	0.453482022529039	
Specific enthalpy	J/(kg.K)	500038.49937969144	
Specific heat capacity	J/(kg.K)	1091.748083028256	
mass: AR	-	0.013790127718329308	
mass: N2	-	0.7542602692440457	
mass: O2	-	0.23194960303762516	

```
source = HeatedGasEnergySource("airish.yaml", 500.0, mass_flow_rate=1.0,
                                cross_area=0.1, Y="N2: 0.79, O2: 0.21")
print(source.report())
```

Property	Unit	Value	
-----	-----	-----	
Source kind		HeatedGasEnergySource	
Provided power	kW	500.0	
Source		airish.yaml	
Phase		air	
Reference area	m ²	0.1	
Mass flow rate	kg/s	1.0	
Volume flow rate	m ³ /s	2.2090693490835482	
Momentum flux	kg.m/s ²	22.090693490835484	

Reference temperature	K	298.15	
Reference pressure	Pa	101325.0	
Temperature	K	774.4142430033143	
Pressure	Pa	101324.99999999999	
Density	kg/m ³	0.45267931512193527	
Specific enthalpy	J/(kg.K)	500040.3236104384	
Specific heat capacity	J/(kg.K)	1100.3346580471573	
mass: N2	-	0.79	
mass: O2	-	0.21	

1.2.3.2 CombustionEnergySource

```

fuel_state = StateType("CH4: 1", 300, 101325)
oxid_state = StateType("N2: 0.79, O2: 0.21", 300, 101325)

ops = CombustionPowerOp(500.0, 1.0, fuel_state, oxid_state)
source = CombustionEnergySource("ch4/bfer.yaml", operation=ops, cross_area=0.1)

# Recover values for other tests that follow
# TODO wrap these in properties for clean API:
mdot_fuel = source._qty_fuel.mass
mdot_oxid = source._qty_oxid.mass

nfr_fuel = NormalFlowRate.new_from_solution(source._qty_fuel)
nfr_oxid = NormalFlowRate.new_from_solution(source._qty_oxid)

qdot_fuel = 3600 * mdot_fuel / nfr_fuel.density
qdot_oxid = 3600 * mdot_oxid / nfr_oxid.density

print(source.report())

```

Property	Unit	Value	
Source kind		CombustionEnergySource	
Provided power	kW	500.0	
Source		ch4/bfer.yaml	
Phase		CH4_BFER	
Reference area	m ²	0.1	
Mass flow rate	kg/s	0.18117768922524702	
Volume flow rate	m ³ /s	1.2210278725486656	
Momentum flux	kg.m/s ²	2.2122300842798666	
***** FLUE:			
Temperature	K	2257.7969133307734	
Pressure	Pa	101325.00000819027	
Density	kg/m ³	0.14838128866548536	
Specific enthalpy	J/(kg.K)	-254492.89776989678	
Specific heat capacity	J/(kg.K)	1516.597922218735	
mass: CH4	-	2.7662274427222582e-17	
mass: CO	-	0.010559409120413024	
mass: CO2	-	0.13474113875913193	
mass: H2O	-	0.12389490081229192	

mass: N2	-	0.7247731344386522	
mass: O2	-	0.006031416869510841	
***** FUEL:			
Temperature	K	300.0	
Pressure	Pa	101325.00000000001	
Density	kg/m ³	0.6516985521312658	
Specific enthalpy	J/(kg.K)	-4645856.881890704	
Specific heat capacity	J/(kg.K)	2229.0429122783953	
mass: CH4	-	1.0	
***** OXIDIZER:			
Temperature	K	300.0	
Pressure	Pa	101325.00000000001	
Density	kg/m ³	1.171970349439655	
Specific enthalpy	J/(kg.K)	1907.6015935135354	
Specific heat capacity	J/(kg.K)	1010.0686132988887	
mass: N2	-	0.7670907820415769	
mass: O2	-	0.2329092179584231	

```
ops = CombustionFlowOp("mass", mdot_fuel, mdot_oxid, fuel_state, oxid_state)
source = CombustionEnergySource("ch4/bfer.yaml", operation=ops, cross_area=0.1)
print(source.report())
```

Property	Unit	Value	
----- ----- -----			
Source kind		CombustionEnergySource	
Provided power	kW	500.0	
Source		ch4/bfer.yaml	
Phase		CH4_BFER	
Reference area	m ²	0.1	
Mass flow rate	kg/s	0.18117768922524702	
Volume flow rate	m ³ /s	1.2210278725486656	
Momentum flux	kg.m/s ²	2.2122300842798666	
***** FLUE:			
Temperature	K	2257.7969133307734	
Pressure	Pa	101325.00000819027	
Density	kg/m ³	0.14838128866548536	
Specific enthalpy	J/(kg.K)	-254492.89776989678	
Specific heat capacity	J/(kg.K)	1516.597922218735	
mass: CH4	-	2.7662274427222582e-17	
mass: CO	-	0.010559409120413024	
mass: CO2	-	0.13474113875913193	
mass: H2O	-	0.12389490081229192	
mass: N2	-	0.7247731344386522	
mass: O2	-	0.006031416869510841	
***** FUEL:			
Temperature	K	300.0	
Pressure	Pa	101325.00000000001	
Density	kg/m ³	0.6516985521312658	
Specific enthalpy	J/(kg.K)	-4645856.881890704	
Specific heat capacity	J/(kg.K)	2229.0429122783953	
mass: CH4	-	1.0	
***** OXIDIZER:			

Temperature	K	300.0	
Pressure	Pa	101325.00000000001	
Density	kg/m ³	1.171970349439655	
Specific enthalpy	J/(kg.K)	1907.6015935135354	
Specific heat capacity	J/(kg.K)	1010.0686132988887	
mass: N2	-	0.7670907820415769	
mass: O2	-	0.2329092179584231	

```
ops = CombustionFlowOp("volume", qdot_fuel, qdot_oxid, fuel_state, oxid_state)
source = CombustionEnergySource("ch4/bfer.yaml", operation=ops, cross_area=0.1)
print(source.report())
```

Property	Unit	Value	
-----	-----	-----	
Source kind		CombustionEnergySource	
Provided power	kW	500.0	
Source		ch4/bfer.yaml	
Phase		CH4_BFER	
Reference area	m ²	0.1	
Mass flow rate	kg/s	0.18117768922524705	
Volume flow rate	m ³ /s	1.2210278725486658	
Momentum flux	kg.m/s ²	2.212230084279867	
***** FLUE:			
Temperature	K	2257.7969133307734	
Pressure	Pa	101325.00000819027	
Density	kg/m ³	0.14838128866548536	
Specific enthalpy	J/(kg.K)	-254492.89776989678	
Specific heat capacity	J/(kg.K)	1516.597922218735	
mass: CH4	-	2.7662274427222582e-17	
mass: CO	-	0.010559409120413024	
mass: CO2	-	0.13474113875913193	
mass: H2O	-	0.12389490081229192	
mass: N2	-	0.7247731344386522	
mass: O2	-	0.006031416869510841	
***** FUEL:			
Temperature	K	300.0	
Pressure	Pa	101325.00000000001	
Density	kg/m ³	0.6516985521312658	
Specific enthalpy	J/(kg.K)	-4645856.881890704	
Specific heat capacity	J/(kg.K)	2229.0429122783953	
mass: CH4	-	1.0	
***** OXIDIZER:			
Temperature	K	300.0	
Pressure	Pa	101325.00000000001	
Density	kg/m ³	1.171970349439655	
Specific enthalpy	J/(kg.K)	1907.6015935135354	
Specific heat capacity	J/(kg.K)	1010.0686132988887	
mass: N2	-	0.7670907820415769	
mass: O2	-	0.2329092179584231	

1.3 Numerical utilities

1.3.1 Relaxation and convergence check utilities

Simplest of relaxation methods; assume you have a new updated solution A_{new}^* for a problem whose past state was A_{old} , then the manager will ensure the following relaxation will be applied to compute the next solution state A_{new} to be used in whatever you are computing:

$$A_{new} = \alpha A_{old} + (1 - \alpha) A_{new}^*$$
$$A_{old} = A_{new}$$

Notice that in this formulation, α (or alpha in the API) represents the fraction of old solution to be used in the *smearing* process. Below we illustrate the effect of a step function H valued at 10 from the beginning over consecutive updates (here we do not test for convergence, as that is problem specific and for this simple case the required number of steps could be evaluated by hand, take some time to try!).

```
alpha = 0.76
niter = 50

single = np.ones(1)
relaxer = RelaxUpdate(single, alpha)

opts = dict(n_vars=1, max_iter=niter, patience=3, rtol=0.001)
converged = StabilizeNvarsConvergenceCheck(**opts)

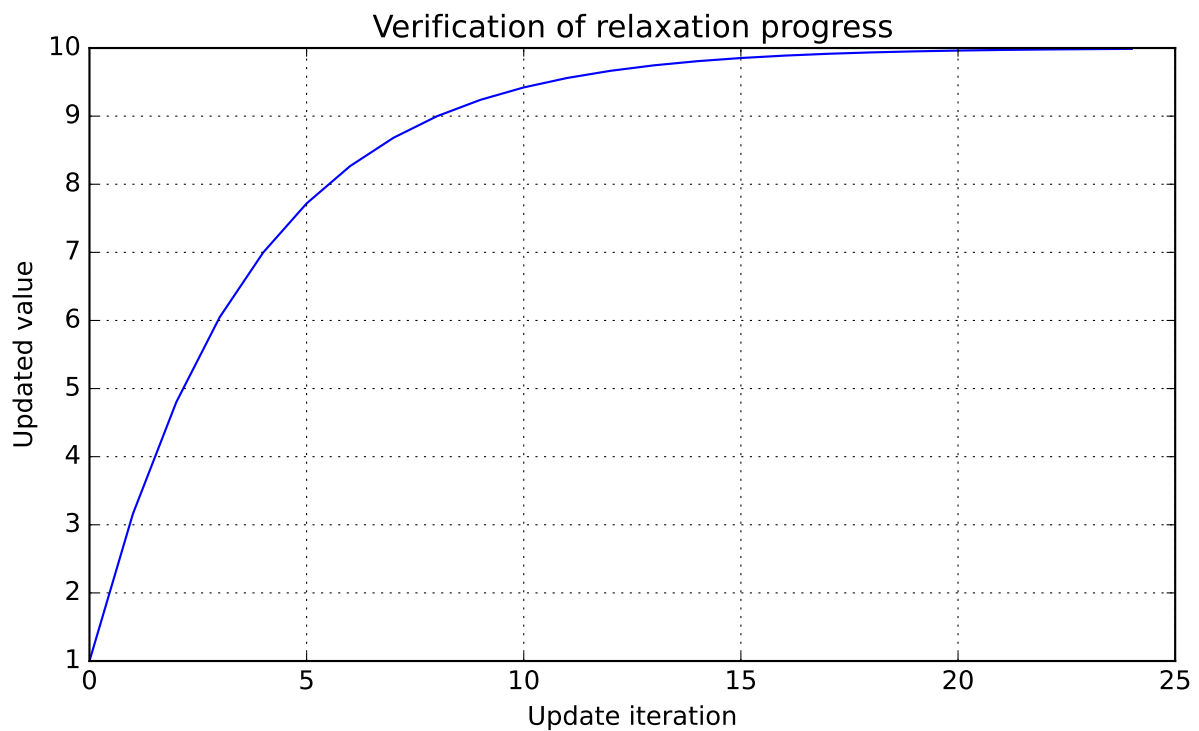
history = np.zeros(niter+1)
history[0] = single[0]

H = np.asarray([10])

for n in range(niter):
    single[:] = relaxer(H)
    history[n+1] = single[0]

    if converged(single[0]):
        history = history[:n+2]
        break

_ = plot_history(history)
```



1.3.1.1 RelaxUpdate

i RelaxUpdate

```
RelaxUpdate(
    v_ini : numpy.ndarray[tuple[typing.Any, ...], numpy.dtype[numpy.float64]],
    alpha : float = 0.5
) -> None:
```

Relax solution for updating new iteration.

Parameters

`v_ini` : `numpy.ndarray[tuple[typing.Any, ...], numpy.dtype[numpy.float64]]`

Initial guess of solution.

`alpha` : `float = 0.5`

Fraction of old solution to use at updates.

i update

```
update(  
    self : Any,  
    alpha : float  
    ) -> None:
```

Update relaxation coefficients.

Parameters

alpha : float

Fraction of old solution to use at updates.

i call

```
__call__(  
    self : Any,  
    v_new : numpy.ndarray[tuple[typing.Any, ...], numpy.dtype[numpy.float64]]  
    ) -> numpy.ndarray[tuple[typing.Any, ...], numpy.dtype[numpy.float64]]:
```

Evaluate new relaxed solution estimate.

Parameters

v_new : numpy.ndarray[tuple[typing.Any, ...], numpy.dtype[numpy.float64]]

New solution estimate to be relaxed.

1.3.1.2 StabilizeNvarsConvergenceCheck

i StabilizeNvarsConvergenceCheck

```
StabilizeNvarsConvergenceCheck(  
    *,  
    n_vars : int,  
    min_iter : int = 1,  
    max_iter : int = 1000000,  
    patience : int = 10,  
    rtol : float = 1e-10,  
    atol : float = 1e-20,  
    equal_nan : bool = False,  
    log_iter : bool = False
```

```
) -> None:
```

Check stabilization towards a constant value along iterations.

Parameters

```
n_vars : int
```

Number of variables to be checked in problem.

```
min_iter : int = 1
```

Minimum number of iterations before considering converged.

```
max_iter : int = 1000000
```

Maximum number of iterations before considering failure.

```
patience : int = 10
```

Number of consecutive convergences before declaring convergence.

```
rtol : float = 1e-10
```

See `numpy.isclose` for details.

```
atol : float = 1e-20
```

See `numpy.isclose` for details.

```
equal_nan : bool = False
```

See `numpy.isclose` for details.

```
log_iter : bool = False
```

If true, log convergence message when achieved.

call

```
__call__(  
    self : Any,
```



```
state : numpy.ndarray[tuple[typing.Any, ...], numpy.dtype[numpy.float64]]
) -> bool:
```

Check if all variables have stabilized at current iteration.

Parameters

state : numpy.ndarray[tuple[typing.Any, ...], numpy.dtype[numpy.float64]]

Current solution state to be checked for convergence.

n_iterations

n_iterations(self : Any): -> int:

Provides access to number of iterations performed.

1.3.1.3 ComposedStabilizedConvergence

ComposedStabilizedConvergence

```
ComposedStabilizedConvergence(
    n_arrs : int,
    **kwargs : Any
):
```

Wrapper for checking stabilization of several arrays.

See `StabilizeNvarsConvergenceCheck` for keyword arguments; these are shared by all tested arrays. It is always possible to compose your own convergence checker using individual instances for more control over setup.

Parameters

n_arrs : int

Number of arrays to be checked for convergence.

kwargs : = None

See `StabilizeNvarsConvergenceCheck` for details.

i call

```
__call__(  
    self : Any,  
    *arrs : tuple[numpy.ndarray[tuple[typing.Any, ...], numpy.dtype[numpy.float64]], ...]  
    ) -> bool:
```

Check if all arrays have stabilized at current iteration.

Parameters

*arrs : tuple[numpy.ndarray[tuple[typing.Any, ...], numpy.dtype[numpy.float64]], ...]

Arrays to be checked for convergence.

i n_iterations

```
n_iterations(self : Any): -> int:
```

Provides access to number of iterations performed.

1.4 Symbolic functions

This module contains utilities for working with symbolic functions, especially in the context of engineering problems treated by Majordome models. It is mostly a wrapper around [CasADi](#), organizing symbolic operations and providing additional functionality for function manipulation in a physical context.

1.4.1 PiecewiseFunction

A major issue when working with symbolic expressions and real-world data is that they do not support flow-control constructs like `if` statements. This is a problem when we want to represent functions that are defined piecewise, especially when algorithmic differentiation is involved.

As an example, take the parameterization of [thermodynamic properties](#) of chemical species. For quantitative purposes, to ensure high accuracy, researchers have found that it is best to use different polynomial fits for different temperature ranges. This means that the thermodynamic properties of a species are defined as piecewise functions of temperature, which is a problem when we want to use them in a symbolic context.

To tackle this issue in *certain* contexts, `PiecewiseSymbolicFunction` provides an interpolation between different ranges of the domain of a function. Internally it makes use of Heaviside functions to ensure that the function is continuous. This is not a perfect solution, as it does not guarantee differentiability at the breakpoints, but it is a good compromise between accuracy and simplicity. It is also worth noting that this class is not meant to be used in a general context, but rather in specific contexts where the function is known to be well-behaved.

PiecewiseSymbolicFunction

```
PiecewiseSymbolicFunction(  
    breakpoints : list[float],  
    functions : list[typing.Any]  
) -> None:
```

Compose a symbolic piecewise function with CasADi.

Parameters

`breakpoints : list[float]`

List of breakpoints where the function changes.

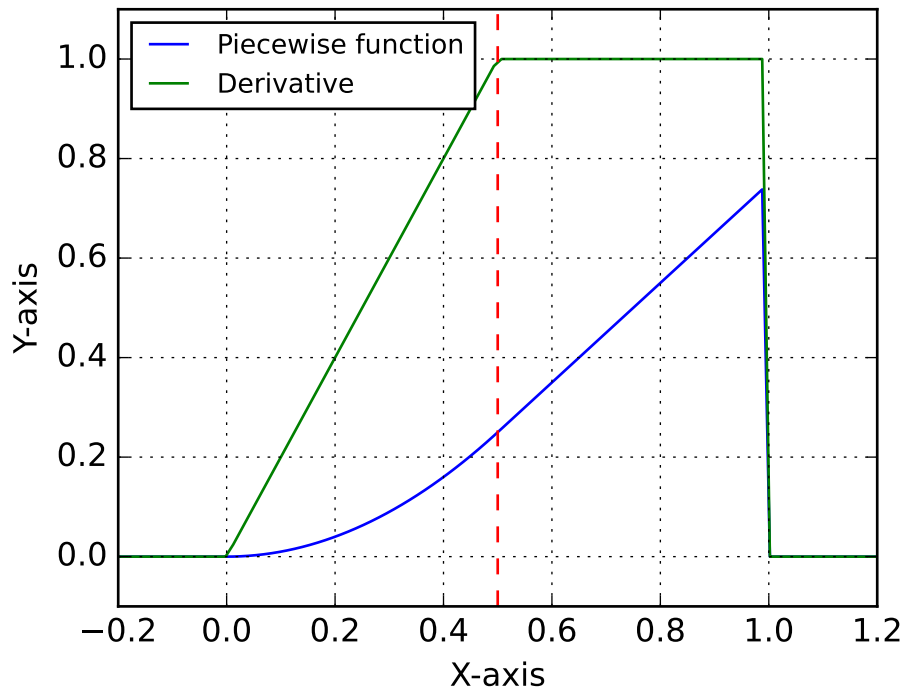
`functions : list[typing.Any]`

List of functions to apply between breakpoints. The number of functions must be one less than the number of breakpoints.

Let's illustrate its use with a simple example. We define two functions f and g , and we want to create a piecewise function that is equal to f for $x < 0.5$ and equal to g for $x \geq 0.5$. Functions have been created such that they evaluate to the same value at the breakpoint, as the practical case that led to the class development, *i.e.* NASA-parameterization of thermochemical properties. Below we show how to create the piecewise function, evaluate it numerically and symbolically, and compute its derivative symbolically.

```
# Define two simple functions:  
f = lambda x: x**2  
g = lambda x: f(0.5) + (x - 0.5)  
  
# Create the piecewise function:  
F = PiecewiseSymbolicFunction([0, 0.5, 1], [f, g])  
  
# Evaluate numerically:  
x = np.linspace(-0.2, 1.2, 100)  
y = F(x)  
  
# Evaluate symbolically:  
T = SX.sym("T")  
Y = F(T)  
  
# Create a function for evaluating the derivative:  
ydot = Function("fdot", [T], [jacobian(Y, T)])
```

Below we plot the piecewise function and its derivative. The function is continuous, but the derivative is not, as expected, although the transition is handled smoothly by the Heaviside functions. The red dashed line indicates the breakpoint at $x = 0.5$, where the function transitions from f to g .



1.4.2 Species thermodynamics

i symbolic_thermo_factory

```
symbolic_thermo_factory(
    species : Species,
    T : SX
) -> AbstractSymbolicThermo:
```

Create an AbstractSymbolicThermo object.

Parameters

species : Species

Cantera species object, with NASA7 thermodynamic data.

T : SX

Temperature variable (symbolic).

Suppose you need to create a symbolic representation of the NASA7 thermodynamic parameterization of molecular nitrogen, species of index 47 in the GRI-Mech 3.0 mechanism shipped with Cantera. Let's start by loading and retrieving the thermodynamic data for this species.

```
gas = ct.Solution("gri30.yaml")
```

```
species = gas.species()[47]
thermo = species.thermo
```

Below we instantiate `SymbolicThermo` with the input data of the species. It is important to emphasize here that when building more complex systems, generally one should share the symbolic variables, e.g. temperature, across the different components of the system. For this reason, the constructor of symbolic thermodynamic classes expect the temperature symbol to be provided, instead of trying to create it internally; this is a design choice that will improve code robustness when handling mixtures and avoid many of the pitfalls of symbolic programming.

```
T = SX.sym("T")
nasa7 = symbolic_thermo_factory(species, T)
```

i Practical usage

You could instead directly instantiate `Nasa7Thermo` with the input data, but using the factory method `SymbolicThermo.from_species` is a better choice, as it will allow you to easily switch to a different thermodynamic model. Furthermore, it is the safe choice for loading entire databases.

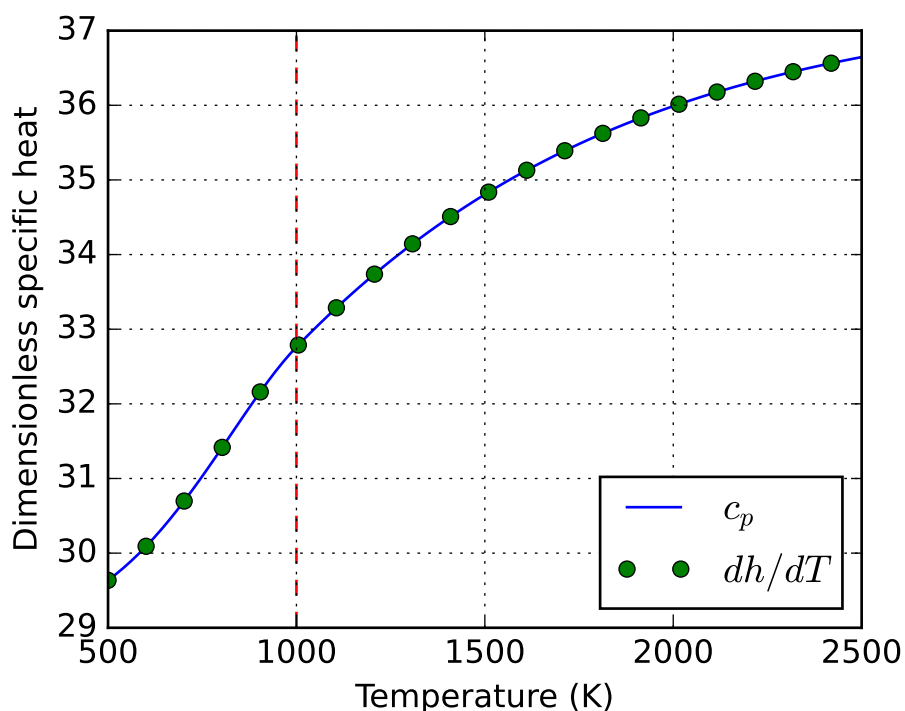
```
nasa7 = Nasa7Thermo(T, species.thermo.input_data)

# This is just syntactic sugar for the above:
nasa7 = Nasa7Thermo.from_species(species, T)
```

Here we illustrate the use of algorithmic differentiation to compute the derivative of the specific enthalpy with respect to temperature.

```
hdot = Function("hdot", [T], [jacobian(nasa7.h(T), T)])
```

The above derivative is then confronted to the specific heat below:



Finally, we compare the deviations between symbolic and numerical (Cantera) evaluation of the thermodynamic functions. This is not for error analysis, but a proof of correctness of the symbolic implementation.

```
# XXX: Cantera evaluates quantities per kmol! Divide by 1000 to get
# per mol, as the symbolic implementation does.
c_ct = np.array([0.001 * species.thermo.cp(T) for T in T_num])
h_ct = np.array([0.001 * species.thermo.h(T) for T in T_num])
s_ct = np.array([0.001 * species.thermo.s(T) for T in T_num])

error_c = np.mean(np.abs(nasa7._cp(T_num) - c_ct))
error_h = np.mean(np.abs(nasa7._h(T_num) - h_ct))
error_s = np.mean(np.abs(nasa7._s(T_num) - s_ct))

error_c, error_h, error_s

(np.float64(4.334310688136611e-15),
 np.float64(6.311893230304122e-12),
 np.float64(3.2400748750660566e-14))
```

1.4.2.1 Nasa7Thermo

Nasa7Thermo

```
Nasa7Thermo(
    T : SX,
    input_data : dict[str, typing.Any]
) -> None:
```

NASA7 thermodynamic parameterization.

This class does not implement Horner polynomial evaluation, as the main use case is to create symbolic expressions that are then evaluated by CasADi, which can handle polynomial evaluation efficiently. This is intentional and allows for easy verification.

It aims at providing a similar interface as SpeciesThermo from Cantera, from which it retrieves the data. That means, molar properties are provided through cp, h, and s properties.

Parameters

T : SX

Temperature variable (symbolic).

input_data : dict[str, typing.Any]

NASA7 thermodynamic data, as provided by Cantera's SpeciesThermo property input_data.

specific_heat

```
specific_heat(  
  T : Any,  
  a : list[float],  
  symbolic : bool = False  
) -> casadi.casadi.Function | casadi.casadi.SX:
```

Compose NASA7 specific heat parameterization.

Parameters

T : Any

Temperature variable (symbolic or numeric).

a : list[float]

List of 7 NASA7 coefficients.

symbolic : bool = False

Whether to return a symbolic expression.

enthalpy

```
enthalpy(  
  T : Any,  
  a : list[float],  
  symbolic : bool = False  
) -> casadi.casadi.Function | casadi.casadi.SX:
```

Compose NASA7 specific enthalpy parameterization.

Parameters

T : Any

Temperature variable (symbolic or numeric).

a : list[float]

List of 7 NASA7 coefficients.

```
symbolic : bool = False
```

Whether to return a symbolic expression.

i entropy

```
entropy(  
    T : Any,  
    a : list[float],  
    symbolic : bool = False  
    ) -> casadi.casadi.Function | casadi.casadi.SX:
```

Compose NASA7 specific entropy parameterization.

Parameters

T : Any

Temperature variable (symbolic or numeric).

a : list[float]

List of 7 NASA7 coefficients.

symbolic : bool = False

Whether to return a symbolic expression.

i compose

```
compose(  
    cls : Any,  
    name : str,  
    T : casadi.casadi.SX | numpy.ndarray[tuple[typing.Any, ...], numpy.dtype[numpy.float64],  
    data : list[list[float]],  
    symbolic : bool = False  
    ) -> list[casadi.casadi.Function | casadi.casadi.SX | numpy.ndarray[tuple[typing.Any,
```

Compose a list of NASA7 functions for given data.

Parameters

name : str

Name of the function (“specific_heat”, “enthalpy”, or “entropy”, *i.e* the static methods of this class).

`T : casadi.casadi.SX | numpy.ndarray[tuple[typing.Any, ...], numpy.dtype[numpy.float64]]`

Temperature variable (symbolic).

`data : list[list[float]]`

List of NASA7 coefficient sets for each temperature range.

`symbolic : bool = False`

Whether to return symbolic expressions or CasADi functions. Only relevant if T is symbolic.

from_species

```
from_species(  
    cls : Any,  
    species : Species,  
    T : SX  
    ) -> typing.Self:
```

Create a Nasa7Thermo object from a Cantera species.

Parameters

`species : Species`

Cantera species object, with NASA7 thermodynamic data.

`T : SX`

Temperature variable (symbolic).

1.4.3 Species transport

symbolic_transport_factory

```
symbolic_transport_factory(  
    species : Species,  
    T : SX  
    ) -> AbstractSymbolicTransport:
```

Create an `AbstractSymbolicTransport` object.

Parameters

`species` : `Species`

Cantera species object, with transport data.

`T` : `SX`

Temperature variable (symbolic).

1.4.4 Kinetics parameterization

1.4.5 Solution properties

1.5 Transport properties

1.5.1 EffectiveThermalConductivity

In some fields of research, such as porous media or composite materials, the evaluation of effective thermal properties are key for simulation of macroscopic application cases. Class `EffectiveThermalConductivity` implements (static) thermal conductivity models for this sort of applications, including:

- Maxwell-Garnett approximation (`EffectiveThermalConductivity.maxwell_garnett`) as exposed in (Hanein, Glasser, and Bannerman 2017); further discussion of its origin and applicability is provided by (Kiradjev et al. 2019).
- For porous media (packed bed in the context) with enhanced radiative effects the model by (Singh and Kaviany 1994) as discussed by (Kee et al. 2017) is implemented (`EffectiveThermalConductivity.singh1994`).

Its name is quite long, let's start by getting an alias before evaluating the desired models:

```
etc = EffectiveThermalConductivity()
```

Using [Maxwell](#) approximation, we could estimate the the effective thermal conductivity of a packed bed of particles in a *matrix* of air; assume particles loosely embeded in air and the following properties; the computed effective thermal conductivity is shown to approach the air limit:

```
phi = 0.30    # Loosely packed solids [30%v]
k_g = 0.025   # Air thermal conductivity [W/(m.K)]
k_s = 1.000   # Solids thermal conductivity [W/(m.K)]

etc.maxwell_garnett(phi, k_g, k_s)
```

0.05396039603960396

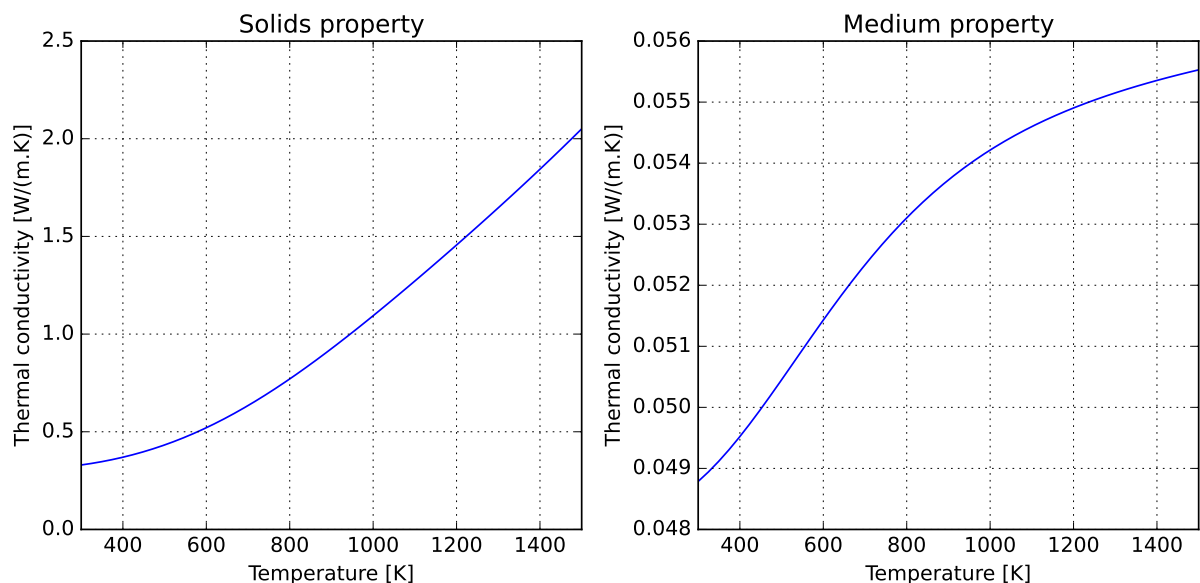
For the limit of high temperatures, it is usually important to account for particle-particle radiation heat transfer; this introduces a T^3 dependence on temperature, as one should expect by linearizing Stefan-Boltzmann law.

Please notice that these models compute different things; while Maxwell approximation computes the medium properties (to approximate matrix-inclusion as a single domain), Singh's model accounts only for solids properties. One might wish to combine them (warning: unverified validity!) to evaluate overall medium thermal conductivity. See the references in the class documentation for further discussion, specially the extension proposed by Kiradjiev (2019), which leads to a result similar to the asymptotic behavior displayed below.

```
d_p = 0.005
eps = 0.9

T = np.linspace(300, 1500, 50)
k_eff_s = etc.singh1994(T, phi, d_p, k_s, eps)
k_eff_m = etc.maxwell_garnett(phi, k_g, k_eff_s)

plot_etc((T, k_eff_s, k_eff_m)).resize(10, 5)
```

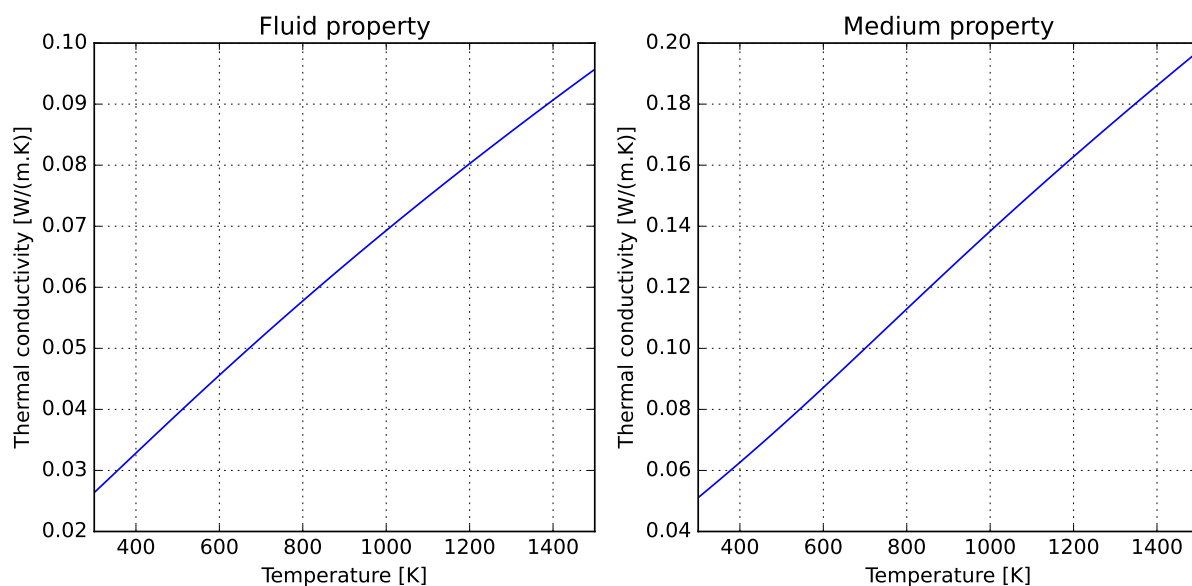


Due to kinetic theory implications, gas thermal conductivity tend to have a positive slope in temperature; the following illustrates how this can be accounted for and the roughly linear behaviour introduced when computing medium properties within air.

```
gas = ct.Solution("airish.yaml")
sol = ct.SolutionArray(gas, (T.shape[0],))
sol.TP = T, None

k_gas = sol.thermal_conductivity
k_eff_m = etc.maxwell_garnett(phi, k_gas, k_eff_s)

plot_etc((T, k_gas, k_eff_m)).resize(10, 5)
```



1.5.2 SolutionDimless

A dimensionless numbers calculator is provided for gas flows; it currently has a certain number of groups which are all evaluated by definition (which might change according to your field, please check the docs). The mechanics of using the class can be resumed to:

1. loading a mechanism file (Cantera YAML)
2. setting the state of the solution
3. evaluating the required properties
4. (optional) displaying a report

The meaning of the tuple of arguments provided to `set_state` is specified by `tuple_name="TPX"`, which defaults to temperature, pressure, and molar proportions. Any triplet allowed by [Cantera](#) can be specified here.

```
Tw = 1000.0 # Wall temperature [K]
U = 10.0    # Characteristic velocity [m/s]
D = 0.05    # Pipe diameter [m]
L = 1.0     # Pipe length [m]
```

```
calculator = SolutionDimless("airish.yaml")
calculator.set_state(300.0, 101325.0, "N2: 1", tuple_name="TPX")
```

```
Re = calculator.reynolds(U, D)
Pr = calculator.prandtl()
Sc = calculator.schmidt()
print(calculator.report())
```

```
-----
Reynolds  31460.9      U=10.0, L=0.05
Prandtl   0.709327
Schmidt   0.761752    mix_diff_coeffs
-----
```

If you prefer to have direct access to the internal solution, you can set properties as usual in Cantera, but you need to keep in mind to call `update()` to refresh the internal state of the calculator. Every time the properties are updated, the internal buffer of computed dimensionless numbers is refreshed, as you might notice in the following table.

```
calculator.solution.TPX = 300.0, 101325.0, "N2: 1"
calculator.update()

Pe_m = calculator.peclet_mass(U, L)
Pe_h = calculator.peclet_heat(U, L)
Gr    = calculator.grashof(Tw, D)
Ra    = calculator.rayleigh(Tw, D)
print(calculator.report())
```

-----	-----	-----
Péclet (mass)	479308	U=10.0, L=1.0, mix_diff_coeffs
Péclet (heat)	446321	U=10.0, L=1.0
Grashof	1.13242e+07	Tw=1000.0, H=0.05, g=9.80665
Rayleigh	8.03259e+06	Tw=1000.0, H=0.05, g=9.80665
-----	-----	-----

1.5.3 SutherlandFitting

To the author's knowledge, there is no standard tool to convert Cantera transport data to Sutherland parameters for use with OpenFOAM, what led to the motivation to develop `SutherlandFitting`. This simple class wraps a Cantera solution object, which it makes use for fitting Sutherland parameters to export as a table (to be used elsewhere), and also allows for retrieving a converted database in OpenFOAM compatible format. The following example should be self-explanatory:

```
T = np.linspace(500, 2500, 100)

sutherland = SutherlandFitting("airish.yaml")
sutherland.fit(T, species_names=["O2", "N2"])

coef = sutherland.coefs_table
coef
```

	species	As [uPa.s]	Ts [K]	RMSE [uPa.s]
0	O2	1.873876	229.321211	0.527484
1	N2	1.619350	226.141806	0.470152

If needed, it is also possible to have direct access to the viscosity data used in parameter fitting:

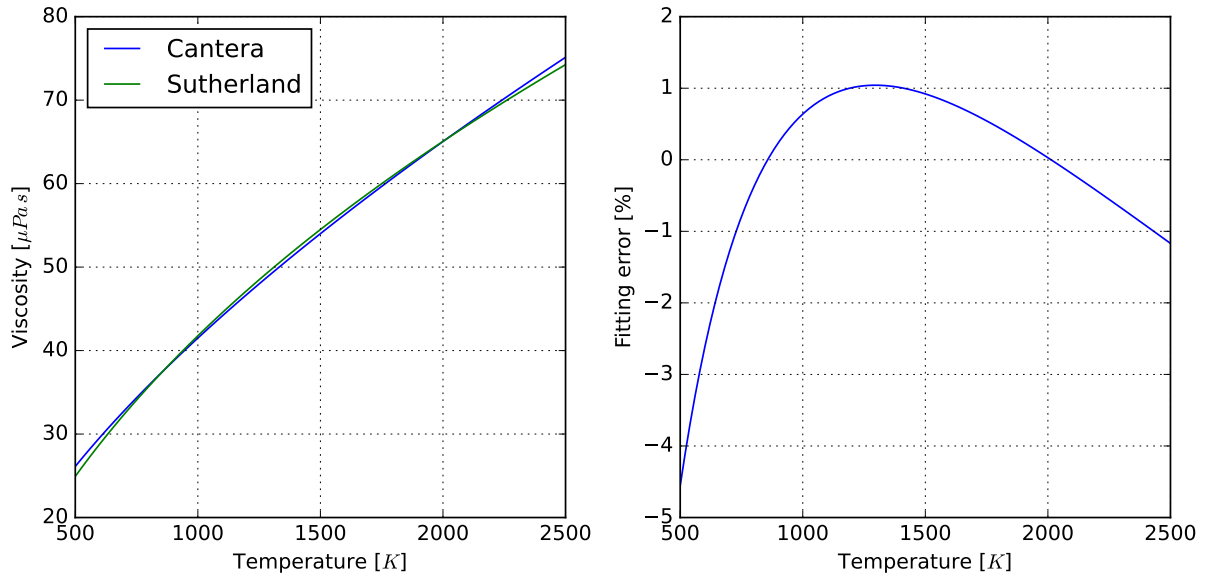
```
sutherland.viscosity.head()
```

	T	O2	N2
0	500.000000	30.045456	26.123142
1	520.202020	30.886884	26.845049
2	540.404040	31.713820	27.554780

	T	O2	N2
3	560.606061	32.527146	28.253075
4	580.808081	33.327662	28.940602

Because RMSE compresses all the error in a single value, you can check graphically where the deviations happen in the interval:

```
plot = sutherland.plot_species("N2")
```



Finally, it *also* responds to its initial goal of exporting values in OpenFOAM format:

```
print(sutherland.to_openfoam())
```

```
/* --- RMSE 0.5274842993902649 ---*/
"O2"
{
    transport
    {
        As  1.8738763185e-06;
        Ts  2.2932121082e+02;
    }
}
/* --- RMSE 0.4701517988286822 ---*/
"N2"
{
    transport
    {
        As  1.6193495479e-06;
        Ts  2.2614180620e+02;
    }
}
```

1.5.4 WSGGRadlibBordbar2020

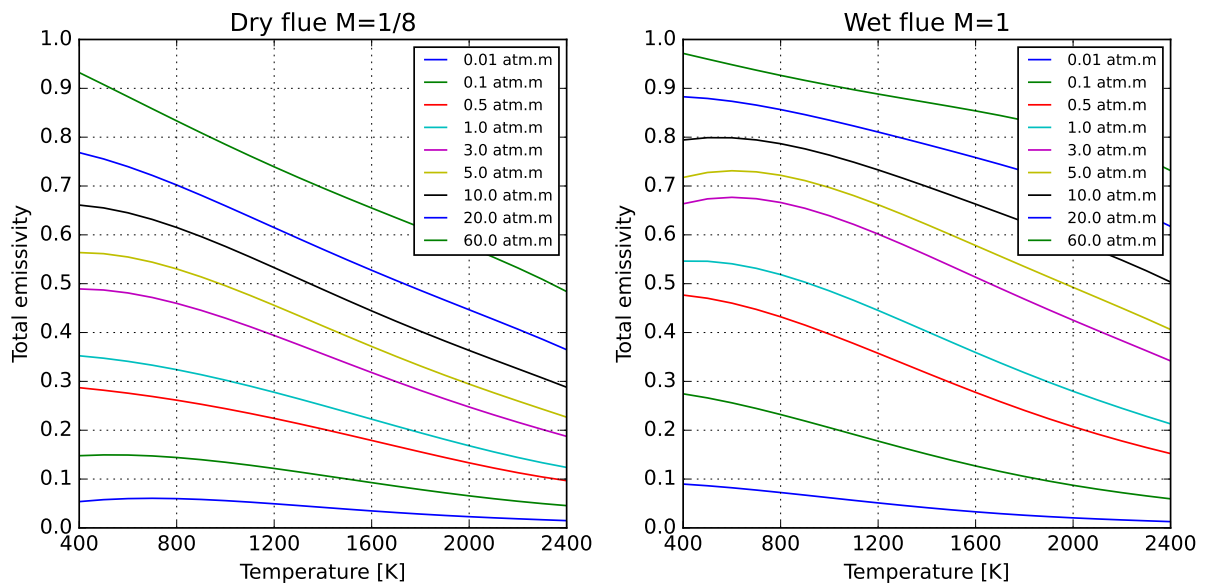
Radiative properties of gases are often required when participating media characteristics imply so. Here we implement the WSGG model of Sadeghi et al. (2021) in class WSGGRadlibBordbar2020. The basic usage of the model to estimate total emissivity is done as follows:

```
model = WSGGRadlibBordbar2020()
model(L=1, T=1000, P=101325, x_h2o=0.18, x_co2=0.08, fvsoot=0.0)

np.float64(0.18368577452587137)
```

1.5.4.1 Validation of Bordbar WSGG implementation

Evaluation of the model against original source of Bordbar (2014) is satisfactory, as follows:



Absorption coefficients for pure substances only reproduce approximately values reported by Bordbar (2020).

```
model(L=1, T=300, P=101325, x_h2o=0, x_co2=1)
model.absorption_coefs[1:]

array([3.388079e-02, 4.544269e-01, 4.680226e+00, 1.038439e+02])

model(L=1, T=300, P=101325, x_h2o=1, x_co2=0)
model.absorption_coefs[1:]

array([ 0.07703541,  0.8242941 ,  6.854761  , 65.93653   ])
```

1.5.4.2 Application to radiative combustion

Below we illustrate the application of `WSGGRadlibBordbar2020` to predict the emissivity of $x\text{H}_2\text{O} - (1-x)\text{CO}_2$ mixtures over a broad temperature range applicable to the analysis of combustion processes. One observes an important participation of flue gases, especially at lower temperatures.

```
wsgg = WSGGRadlibBordbar2020()
```

```
@np.vectorize
```

```
def emissivity(T, X, L=1, P=101325):
```

```
    return wsgg(L=L, T=T, P=P, x_h2o=X, x_co2=1-X, fvsoot=0.0)
```

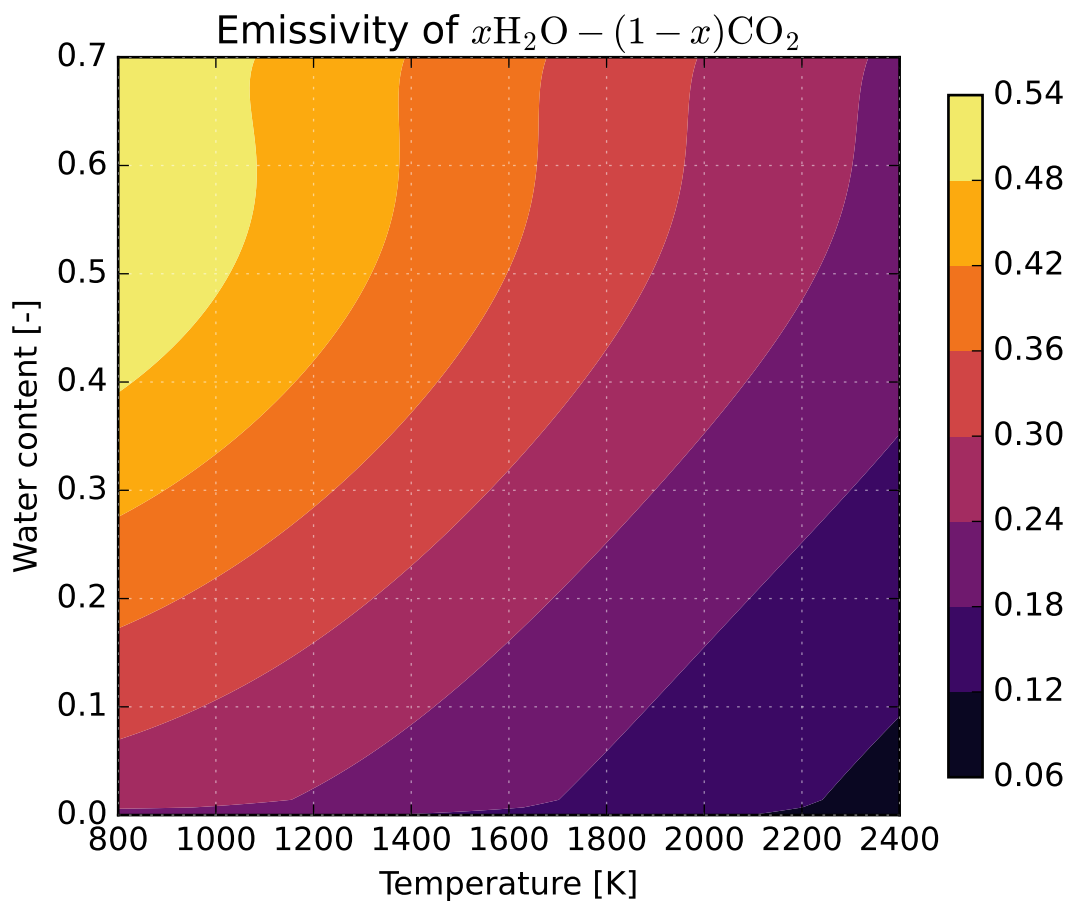
```
T = np.linspace(800, 2400, 100)
```

```
X = np.linspace(0, 0.7, 100)
```

```
T, X = np.meshgrid(T, X)
```

```
eps = emissivity(T, X)
```

```
plot_emissivity(T, X, eps)
```



1.6 Computer vision

Upcoming...

2 Simulation

aka majordome.simulation

2.1 Elmer Multiphysics

2.1.1 TimeStepAccumulator

TimeStepAccumulator

```
TimeStepAccumulator(*time_steps : ConstantTimeStepInterval): -> None:
```

Helper for creating cumulative time steps in Elmer SIF files.

Parameters

`*time_steps : ConstantTimeStepInterval`

Variable number of time step intervals to accumulate.

add_block

```
add_block(  
    self : Any,  
    *,  
    duration : float | None = None,  
    end_time : float | None = None,  
    **kws : Any  
) -> None:
```

Add a new block of time steps with the given parameters.

Parameters

`duration : float | None = None`

Duration of the new time step block. If not provided, then `end_time` must be provided.

`end_time : float | None = None`

End time of the new time step block. If not provided, then duration must be provided. If provided, the duration of the new block will be calculated as `end_time - self.duration`.

`kws : = None`

Additional keyword arguments to pass to the `ConstantTimeStepInterval` constructor.

`append`

```
append(  
    self : Any,  
    time_step : ConstantTimeStepInterval  
    ) -> None:
```

Append a new time step to the accumulator.

Parameters

`time_step : ConstantTimeStepInterval`

The time step interval to append to the accumulator.

`to_dataframe`

```
to_dataframe(self : Any): -> DataFrame:
```

Convert the accumulated time steps into a table.

In general the usage of the `TimeStepAccumulator` is as follows:

```
! >> Total duration: 0.001 << !  
! Step Duration(3)      =      1e-05      9e-05      0.0009  
Timestep Sizes(3)      =  1.000000e-06  1.000000e-05  1.000000e-04  
Timestep Intervals(3)  =      10      9      9  
Output Intervals(3)    =      10      9      9
```

2.1.1.1 `ConstantTimeStepInterval`

In some special cases, when using `TimeStepAccumulator` you might want to add time steps one by one, which can be done using the `append` method in combination with the `ConstantTimeStepInterval` described below.

`ConstantTimeStepInterval`

```
ConstantTimeStepInterval(  
    duration : float,
```

```

time_step : float | None = None,
save_interval : float | None = None,
*,
min_steps : int = 2
) -> None:

```

Create a constant time step interval for a simulation.

Parameters

`duration : float`

Total duration of the time step interval.

`time_step : float | None = None`

Size of each time step. If not provided, it will be calculated based on the duration and `min_steps` parameters.

`save_interval : float | None = None`

Interval at which to save results. If not provided, it will be set equal to `time_step`.

`min_steps : int = 2`

Minimum number of time steps to use if `time_step` is not provided. Must be at least 1. Default is 2.

`into_formatted_data`

`into_formatted_data(self : Any): -> list[str]:`

Format the time step data for output in an Elmer SIF file.

2.1.2 ElmerConvergenceData

`ElmerConvergenceData`

`ElmerConvergenceData(file_path : str | pathlib.Path): -> None:`

Class for handling convergence data from Elmer simulations.

Parameters

```
file_path : str | pathlib.Path
```

Path to the file containing convergence data, typically output by Elmer's SaveLine solver.

i get_solver_data

```
get_solver_data(  
    self : Any,  
    solver_id : int  
    ) -> DataFrame:
```

Get convergence data for a specific solver.

Parameters

```
solver_id : int
```

The ID of the solver to filter by.

i plot_solver_convergence

```
plot_solver_convergence(  
    self : Any,  
    *,  
    solver_id : int,  
    final_iter : bool = False,  
    time_axis : numpy.ndarray[tuple[typing.Any, ...], numpy.dtype[~_ScalarT]] | None = None  
    plot : majordome.utilities.MajordomePlot | None = None  
    ) -> MajordomePlot:
```

Plot convergence data for a specific solver.

Parameters

```
solver_id : int
```

The ID of the solver to plot.

```
final_iter : bool = False
```

If True, only plot the final iteration for each timestep.

```
time_axis : numpy.ndarray[tuple[typing.Any, ...], numpy.dtype[~_ScalarT]] | None = None
```

Optional array of time values to use as the x-axis. If not provided, the timestep numbers will be used.

```
plot : majordome.utilities.MajordomePlot | None = None
```

Placeholder for plot object provided by decorator.

tracked_solvers

```
tracked_solvers(self : Any): -> list[int]:
```

List of unique solvers tracked in the data.

2.1.3 ElmerTabularData

ElmerTabularData

```
ElmerTabularData(  
    fname : str | pathlib.Path,  
    *,  
    fmeta : str | pathlib.Path | None = None  
):
```

Class to represent the data of a SaveLine results file.

Parameters

```
fname : str | pathlib.Path
```

The path to the data file output of Elmer's SaveLine solver.

```
fmeta : str | pathlib.Path | None = None
```

The path to the associated metadata file. If not provided, the class will attempt to find a metadata file in the same directory as the data file, with the same name but with a .names extension. If still not found, it will look for the metadata file one level above the data file directory.

metadata

```
metadata(self : Any): -> ElmerTabularMetadata:
```

Provides access to the metadata of the save line data.

data

```
data(self : Any): -> DataFrame:
```

Provides access to the data of the save line.

2.1.4 ElmerTabularMetadata

ElmerTabularMetadata

```
ElmerTabularMetadata(filepath : str | pathlib.Path)::
```

Class to hold metadata from Elmer SaveLine .names file.

Parameters

```
filepath : str | pathlib.Path
```

Path to the .names metadata file.

metadata

```
metadata(self : Any): -> dict:
```

Return the metadata as a dictionary.

columns

```
columns(self : Any): -> dict:
```

Return the column definitions as a dictionary.

2.2 Ansys Fluent

2.2.1 FluentFvParticlesParser

This class provides functionality for parsing particle track files. Fluent export DPM results in several formats; majordome supports *fieldview* particle tracks through `FluentFvParticlesParser`. Loading a file is as simple as:

You can retrieve the names of columns and number of individual tracks with the following properties:

```
(['x', 'y', 'z', 'residence_time', 'time_step'], 2)
```

For convenience, tracks can be recovered as data frames through object indexing:

	x	y	z	residence_time	time_step
0	0.128501	0.537214	-0.622358	0.000000	0.000000e+00
1	0.128493	0.537929	-0.619682	0.000107	1.697420e-07
2	0.128493	0.537964	-0.619554	0.000112	2.738057e-07
3	0.128488	0.538534	-0.617419	0.000176	4.008972e-07
4	0.128477	0.539842	-0.612481	0.000299	7.041471e-07

2.2.2 FluentInterpolationParser

Sometimes it is also useful to parse interpolation files. It enables one to compute new quantities for an interpolation file may be useful, especially in research settings. Class `FluentInterpolationParser` is able to read the contents of an interpolation file exported in plain text format. Below you find an example of parsing.

```

x0 in range (      11:      16)
x1 in range (      17:      22)
pressure in range (    23:      28)
x-velocity in range (   29:      34)
y-velocity in range (   35:      40)
temperature in range (   41:      46)
k in range (      47:      52)
omega in range (     53:      58)

```

Names of variables are provided by property `variable_names`:

```

['x0',
 'x1',
 'pressure',
 'x-velocity',
 'y-velocity',
 'temperature',
 'k',
 'omega']

```

Valid names can be used to retrieve data for a given variable:

```
array([0.1, 0.1, 0.2, 0.3, 0.4])
```

The whole data table can be retrieved with data property, where each variable corresponds to a column:

```

array([[0.1, 1. , 9. , 9. , 9. , 9. , 9. , 9. ],
       [0.1, 2. , 1. , 1. , 1. , 1. , 1. , 1. ],
       [0.2, 3. , 2. , 2. , 2. , 2. , 2. , 2. ],
       [0.3, 4. , 3. , 3. , 3. , 3. , 3. , 3. ],
       [0.4, 5. , 4. , 4. , 4. , 4. , 4. , 4. ]])

```

2.3 OpenFOAM

2.4 Meshing

2.4.1 GmshOCCModel

GmshOCCModel

```
GmshOCCModel(  
    *,  
    render : bool = False,  
    name : str = 'domain',  
    **kws : Any  
    ) -> None:
```

Wrapper to manage OCC models with an OOP approach.

Parameters

render : bool = False

Whether to launch the Gmsh GUI after building the model.

name : str = 'domain'

Name of the Gmsh model.

ks : dict = None

Provide configuration for gmsh internals.

3 Utilities

aka majordome.utilities

This module provides a set of utilities that are whether too general purpose for integrating another module or still waiting to find their definitive home. These include constants, type aliases, and functionalities of different kinds. Regarding this last kind of components, this tutorial aims at illustrating their use for practical documentation.

3.1 Plotting

3.1.1 MajordomePlot

One might ask why a wrapper around Matplotlib's `Figure` and `Axes` is necessary, and I would strongly agree that it isn't. However, keeping all the plots in a report consistent is a huge effort, and being able to standardize the plots without having to write new themes is a nice feature. That's the main goal of `MajordomePlot`: simplifying the plotting of publication-quality figures across Majordome (and beyond!). Below we illustrate how to use this functionality to ensure all the plots in a project look the same.

MajordomePlot

```
MajordomePlot(  
    shape : tuple[int, int] = (1, 1),  
    style : str = 'classic',  
    *,  
    size : tuple[float, float] | None = None,  
    xlabel : str | list[str] | None = None,  
    ylabel : str | list[str] | None = None,  
    opts : dict[str, typing.Any] = {'facecolor': 'white'}  
) -> None:
```

Handles the creation and management of plots.

Provides a handle for creating and managing plots using Matplotlib. It is aimed at standardizing the plot creation process across the Majordome framework.

Parameters

shape : **tuple**[**int**, **int**] = (1, 1)

Shape of the plot as (n_rows, n_cols).

style : **str** = 'classic'

Matplotlib style to use for the plot.

```
size : tuple[float, float] | None = None
```

Size of the plot in inches as (width, height). If None, the default size will be used.

```
xlabel : str | list[str] | None = None
```

Label(s) for the x-axis. If a list is provided, each subplot will get its own label.

```
ylabel : str | list[str] | None = None
```

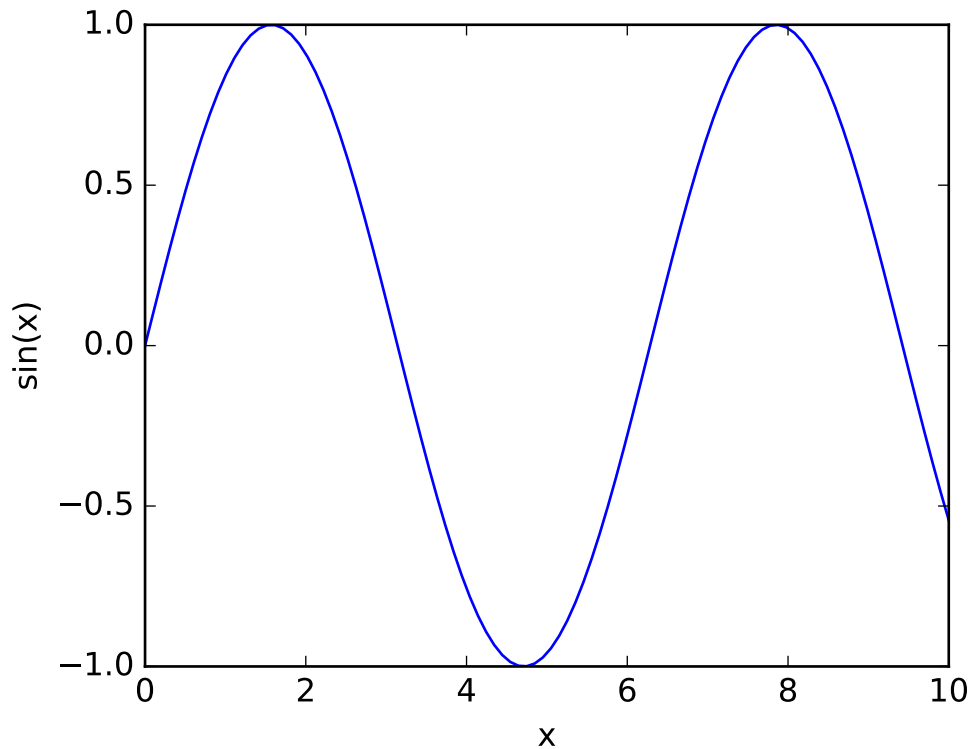
Label(s) for the y-axis. If a list is provided, each subplot will get its own label.

```
opts : dict[str, typing.Any] = {'facecolor': 'white'}
```

Additional options to pass to `plt.subplots()`.

Let's see how one would expect to use it from the above constructor signature. This is in fact almost the same as calling `plt.subplots()`, and that is what the constructor does under the hood. This is a *negative* example actually, the author has *never* used the constructor directly, but instead uses the new classmethod, which is a much more convenient way to create a plot. More on that later.

```
x = np.linspace(0, 10, 100)
mp = MajordomePlot(shape=(1, 1), size=(5, 4))
mp.axes[0].plot(x, np.sin(x))
mp.axes[0].set_xlabel("x")
mp.axes[0].set_ylabel("sin(x)")
pass
```



The story behind this class dates a few years back, when I was learning about Python decorators. As an exercise, I coded what later became the new classmethod, which is a decorator that creates a new `MajordomePlot` and passes it to the decorated function. This is a very convenient way to create plots, as it allows you to focus on the plotting code and not worry about the boilerplate of creating the plot and setting the labels.

i new

```
new(
    cls : Any,
    _func : typing.Optional[typing.Callable[~Params, typing.Any]] = None,
    *,
    shape : tuple[int, int] = (1, 1),
    style : str = 'classic',
    sharex : bool = True,
    facecolor : str = 'white',
    grid : bool = True,
    size : tuple[float, float] | None = None,
    xlabel : str | list[str] | None = None,
    ylabel : str | list[str] | None = None
) -> typing.Callable[[typing.Callable[~Params, typing.Any]], typing.Any]:
```

Wraps a function for ensuring a standardized plot.

Parameters

`_func : typing.Optional[typing.Callable[~Params, typing.Any]] = None`

Function to wrap, if any.

```
shape : tuple[int, int] = (1, 1)
```

Shape of the plot as (n_rows, n_cols).

```
style : str = 'classic'
```

Matplotlib style to use for the plot. By default “classic”.

```
sharex : bool = True
```

Whether to share x-axis among subplots.

```
facecolor : str = 'white'
```

Face color of the figure.

```
grid : bool = True
```

Whether to display grid lines.

```
size : tuple[float, float] | None = None
```

Size of the plot in inches as (width, height). If None, the default size will be used. By default None.

```
xlabel : str | list[str] | None = None
```

Label(s) for the x-axis. If a list is provided, each subplot will get its own label. By default None.

```
ylabel : str | list[str] | None = None
```

Label(s) for the y-axis. If a list is provided, each subplot will get its own label. By default None.

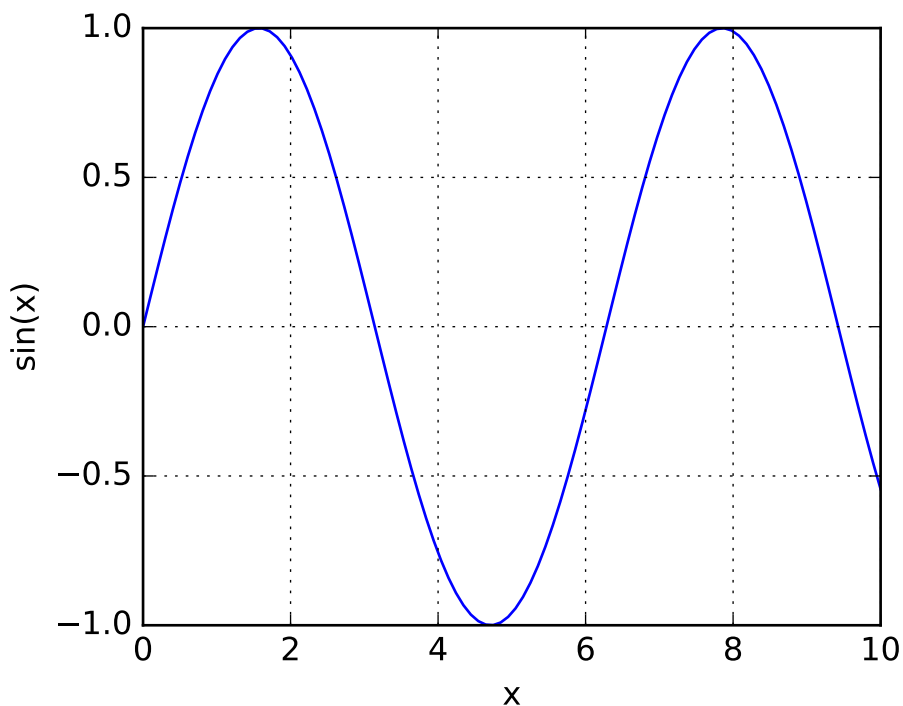
In the following example, we make use of this decorator to parametrize the labels of the plot. Internally, we must expect a *keyword-only* argument `plot`, which is the `MajordomePlot` instance created by the decorator, from which we can access the `Figure` and `Axes` objects as usual. The rest of the code is just standard Matplotlib code, with the exception that the axes is raveled, so we can access it with `ax[0]` instead of `ax[0, 0]`. Returning `plot` is optional (it is done by the decorator already), but it avoids linting errors if you want to annotate the return type as `MajordomePlot`.

Here we create a function that will be used below in the examples. In summary, to get it working, you need to:

- Use decorator `@MajordomePlot.new` with all its configurable attributes; for details, please refer to its API documentation.
- Have a function with signature `func(*args, plot=None, **kwargs) -> None`, where it is recommended (for linter) to provide explicitly keyword `plot=None`.
- Unpack `fig, ax = plot.subplots()` or just `_`, `ax = plot.subplots()`, as needed, inside the figure; these contain standard matplotlib figure and axes.

```
@MajordomePlot.new(size=(5, 4), xlabel="x", ylabel="sin(x)")
def plot_sin(x, *, plot=None, **kwargs) -> MajordomePlot:
    _, ax = plot.subplots()
    ax[0].plot(x, np.sin(x))
    return plot
```

```
plot = plot_sin(np.linspace(0, 10, 100))
```



The following elements document the members and properties of the class.

`resize`

```
resize(
    self : Any,
    w : float,
    h : float
) -> None:
```

Resize a plot with width and height in inches.

Parameters

w : float

Width of the plot in inches.

h : float

Height of the plot in inches.

i savefig

```
savefig(  
    self : Any,  
    filename : str | pathlib.Path,  
    **kwargs : Any  
) -> None:
```

Wrapper for saving a figure to file.

Parameters

filename : str | pathlib.Path

Path to save the figure to.

kwargs : None = None

Additional keyword arguments to pass to Figure.savefig().

i subplots

```
subplots(self : Any): -> tuple[matplotlib.figure.Figure, list[matplotlib.axes._axes.Axes]]
```

Provides access to underlying figure and axes.

i show

```
show(self : Any): -> None:
```

Display the plot.

i figure

```
figure(self : Any): -> Figure:
```

Provides access to underlying figure.

i axes

```
axes(self : Any): -> list[matplotlib.axes._axes.Axes]:
```

Provides access to underlying axes.

3.1.2 PowerFormatter

i PowerFormatter

```
PowerFormatter(**kwargs : Any): -> None:
```

Formatter for power of ten in numerical axes.

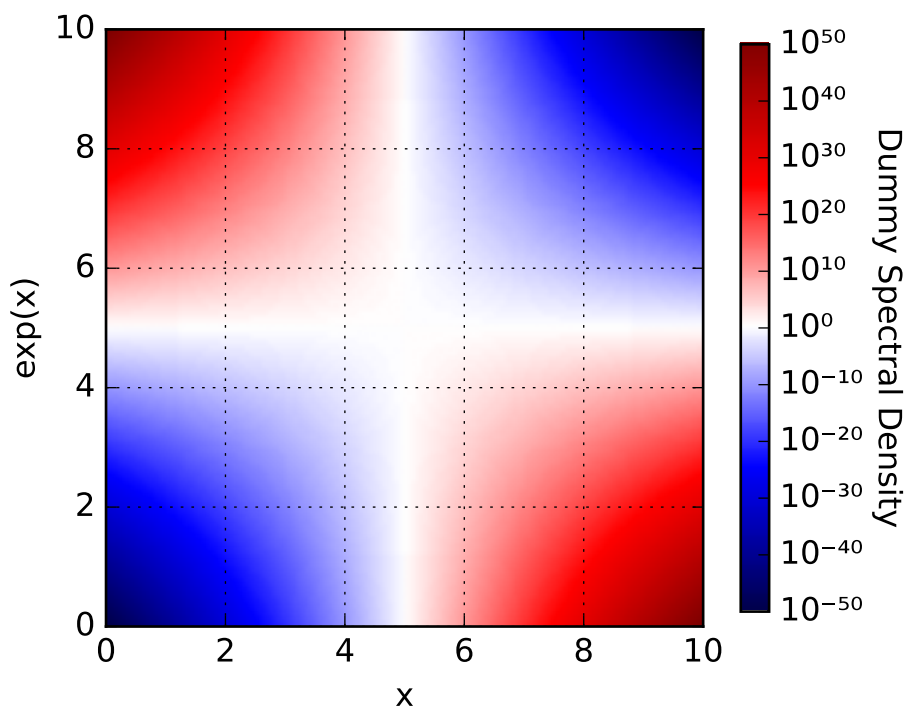
Parameters

```
values : str = None
```

String of characters to be replaced by their superscript counterparts. By default "0123456789-".

```
supers : str = None
```

String of superscript characters corresponding to values. By default " ¹²³ ".



3.1.3 General utilities

centered_colormap

```
centered_colormap(  
  name : str,  
  vmin : float,  
  vmax : float,  
  vcenter : float = 0.0  
) -> LookupTable:
```

Ensure the center of a colormap is at zero.

Parameters

name : str

Name of the colormap to use.

vmin : float

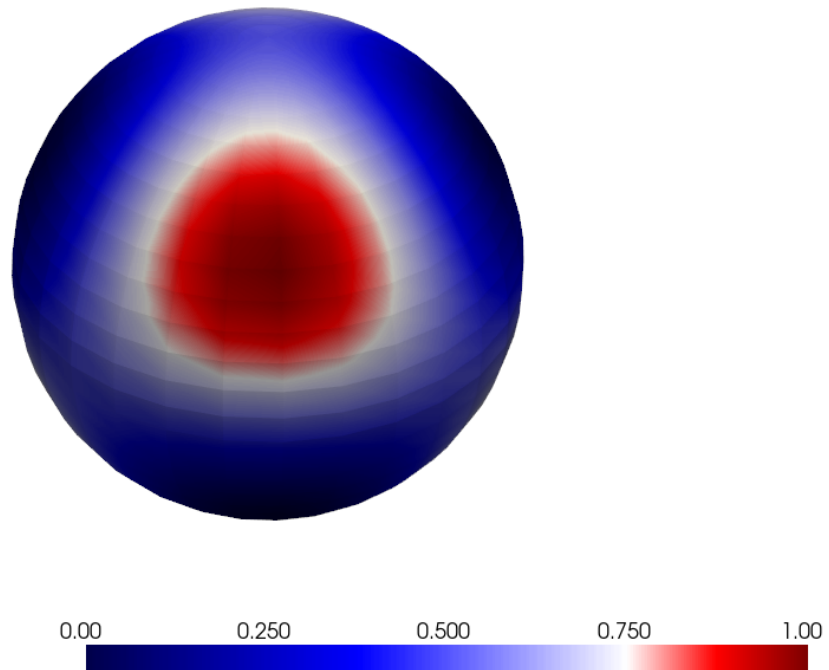
Minimum value of the data range.

vmax : float

Maximum value of the data range.

vcenter : float = 0.0

Center value of the colormap, by default 0.



3.2 LaTeX

Upcoming...

3.3 Argument parsing

3.3.1 FuncArguments

```
def f(*args, **kwargs):
    """ Arbitrary interface function for illustration. """
    f.parser.update(*args, **kwargs)

    try:
        print(f"a = {f.parser.get('a')}, b = {f.parser.get('b')}")
        f.parser.close()
    except Exception as err:
        print(err)
        pass
```

- Arguments are both positional-mandatory

```
f.parser = FuncArguments()
f.parser.add("a", index=0)
f.parser.add("b", index=1)
```

```
f(3)
f(3, 4)
```

Cannot retrieve mandatory positional argument at position 1
a = 3, b = 4

- Arguments are both keyword-only

```
f.parser = FuncArguments()
f.parser.add("a", default="foo")
f.parser.add("b", default="bar")
```

```
f(3)
f(3, 4)
f(3, b=4)
f(a=3, b=4)
```

```
a = foo, b = bar
Too many positional arguments, got 1 but only 0 were used.
a = foo, b = bar
Too many positional arguments, got 2 but only 0 were used.
a = foo, b = 4
Too many positional arguments, got 1 but only 0 were used.
a = 3, b = 4
```

- One positional and another (maybe) positional

```
f.parser = FuncArguments()
f.parser.add("a", index=0)
f.parser.add("b", index=1, default=6)
```

```
f(3)
f(3, 4)
f(3, 4, b=4)
```

```
a = 3, b = 6
a = 3, b = 4
Cannot have both positional and keyword version of b (1) simultaneously
```

- Badly configured

```
f.parser = FuncArguments()
```

```
try:
    f.parser.add("a")
    f.parser.add("b")
except Exception as err:
    print(err)
```

Argument must be either positional or keyword, cannot be neither: a

3.3.2 Creating class constructors

You should probably not do this, but it is quite fun!

```
def _init_some_class(cls):
    """ Decorator to enhance SomeClass with argument parsing. """
    orig_init = cls.__init__

    # Maybe stash for later use?
    # cls.__orig_init__ = orig_init

    parser = FuncArguments(greedy_args=False, pop_kw=True)
    parser.add("a", 0)
    parser.add("x", default=None)

    @wraps(orig_init)
    def new_init(self, *args, **kwargs):
        parser.update(*args, **kwargs)
        a = parser.get("a")
        x = parser.get("x")

        orig_init(self, *parser.args, **parser.kwargs)
        parser.close()

        # -- logic goes here
        print(f"some a = {a}")
        print(f"some x = {x}")

        return None

    cls.__init__ = update_wrapper(new_init, orig_init)
    return cls


class BaseClass:
    def __init__(self, *args, **kwargs) -> None:
        print(f"args    = {args}")
        print(f"kwargs  = {kwargs}")

@_init_some_class
class SomeClass(BaseClass):
    def __init__(self, *args, **kwargs) -> None:
        super().__init__(*args, **kwargs)

some = SomeClass(1, 2, 3, x=1, y=2)

args    = (1, 2, 3)
kwargs  = {'y': 2}
some a = 1
some x = 1
```

3.4 PDF Tools

3.4.1 PdfToTextConverter

Below we illustrate the usage of PdfToTextConverter. Please notice that data curation of extracted texts is still required if readability is a requirement. If quality of automated extractions is often poor for a specific language, you might want to search the web how to *train tesseract*, that topic is not covered here.

Note: this note assumes [tesseract](#) and [poppler](#), and [ImageMagick](#) are available in system path. Under Windows you might struggle to get them all working together, please check Majordome's Kompanion for automatic installation.

Install dependencies on Ubuntu 22.04:

```
sudo apt install tesseract-ocr imagemagick poppler-utils
```

In case of Rocky Linux 9:

```
sudo dnf install tesseract tesseract-langpack-eng ImageMagick poppler-utils
```

Assuming the dependencies are found in the path, it is simply a matter of creating a converter:

```
converter = PdfToTextConverter()
```

For generated PDF (not scanned documents), it is much faster to avoid using OCR; below we show the metadata from a paper:

```
data = converter("data/pdftools/paper.pdf", use_ocr=False)
data.meta
```

```
{'/Author': 'W. Dal'Maz Silva',
  '/CreationDate': 'D:20170403224009+05'30'',
  '/Creator': 'Elsevier',
  '/CrossMarkDomains[1]': 'elsevier.com',
  '/CrossMarkDomains[2]': 'sciencedirect.com',
  '/CrossmarkDomainExclusive': 'true',
  '/CrossmarkMajorVersionDate': '2010-04-23',
  '/ElsevierWebPDFSpecifications': '6.5',
  '/Keywords': 'Hardness measurement; Martensite; Low-alloy steel; Precipitation',
  '/ModDate': 'D:20170403224009+05'30'',
  '/Subject': 'Materials Science & Engineering A, 693 (2017) 225-232. doi:10.1016/j.msea.2017',
  '/Title': 'Carbonitriding of low alloy steels_ Mechanical and metallurgical responses',
  '/doi': '10.1016/j.msea.2017.03.077',
  '/robots': 'noindex'}
```

For scanned documents, by default if OCR is not enabled it will be used as a fallback method for text extraction:

```
data = converter("data/pdftools/scanned.pdf", last_page=1)
data.content[:500]
```

```
'549\n\n5.. Uber die von der molekularkinetischen Theorie\nder Wirme geforderte Bewegung von\n\kinetischen Theorie der Warme in Flissigkeiten suspendierte\nKorper von mikroskopisch sich\n\nkularbewegung der Wirme Bewegungen von solcher GrifSe\nausfiihren miissen, daB diese Beweg
```

3.5 Markdown

3.5.1 Normalize LaTeX delimiters

i apply

```
apply(  
    cls : Any,  
    text : str  
    ) -> str:
```

Apply normalization to the given text.

Parameters

text : str

The input Markdown text to normalize.

i apply_token_list

```
apply_token_list(  
    cls : Any,  
    tokens : list[markdown_it.token.Token],  
    inplace : bool = True  
    ) -> tuple[list[markdown_it.token.Token], int]:
```

Apply LaTeX delimiter normalization to all applicable tokens.

Parameters

tokens : list[markdown_it.token.Token]

The token stream to process.

inplace : bool = True

Whether to modify the tokens in place or return a new list (default is True).

The following illustrates the use of the functionality:

```
sample = r"""  
Inline good: \(\ x^2 + y^2 \)  
  
Inline missing slash: \((a+b)
```

Inline missing slash close: $(c+d\backslash)$

Inline parenthetized: $(\backslash(c+d\backslash))$

Inline breaking case: $\backslash(V_{IJ}^{\{(n)\}} \backslash)$

Block inline: $\backslash[\int_0^1 f(x)\backslash, dx]$

Fragilized block:

$\backslash[\int_0^1 f(x)\backslash, dx]$

I noticed that sometimes links may be fragile, like in:

`![diagram showing something](assets/img/favicon-32x32.png)`

What about this doubly escaped inline math: $\backslash\backslash(\backslash\Delta J\backslash\backslash)$?

```

$$\backslash\backslash[$$

$$W_{\backslash\backslash\text{point}}^{\{(n)\}} = \backslash\backslash\left( W_{AA}^{\{(n)\}} + W_{BB}^{\{(n)\}} - \right.$$

$$\left. 2W_{AB}^{\{(n)\}} \backslash\backslash\right)$$

$$\backslash\backslash]$$

$$"""$$

```

```
normalizer = LatexDelimiterNormalizer()
output = normalizer.apply(sample)
display(Markdown(output))
```

Inline good: $x^2 + y^2$

Inline missing slash: $a + b$

Inline missing slash close: $c + d$

Inline parenthetized: $(c + d)$

Inline breaking case: $V_{IJ}^{(n)}$

Block inline:

$$\int_0^1 f(x) dx$$

Fragilized block:

$$\int_0^1 f(x) dx$$

I noticed that sometimes links may be fragile, like in:



Figure 3.1: diagram showing something

What about this doubly escaped inline math: ΔJ ?

$$W_{\text{point}}^{(n)} = (W_{AA}^{(n)} + W_{BB}^{(n)} - 2W_{AB}^{(n)})$$

3.5.2 Strip Markdown links

This is a sample markdown with some links and images.

<removed-link>

Here is a [link to Google](https://www.google.com) and another [link to OpenAI](https://www.openai.com).

3.6 Misc

3.6.1 ColorPrint

```
ColorPrint.red("This is a red message.")
ColorPrint.green("This is a green message.")
ColorPrint.blue("This is a blue message.")
ColorPrint.yellow("This is a yellow message.")
ColorPrint.cyan("This is a cyan message.")
```

```
This is a red message.
This is a green message.
This is a blue message.
This is a yellow message.
This is a cyan message.
```

4 Documents

aka majordome.documents

SignatureEntry

```
SignatureEntry(  
    fn : type,  
    *,  
    greedy : bool = True  
) -> None:
```

Builds the representation of a function signature.

Parameters

`fn : type`

The function to build the signature for.

`greedy : bool = True`

Whether to raise errors for missing docstrings or parameters.

documentation

```
documentation(  
    self : Any,  
    **kwargs : Any  
) -> None:
```

Generates the markdown documentation for the function.

Parameters

`title : str = None`

The title to use for the section header and callout. By default, the function name is used.

4.1 Tutorial

```
class Dummy:
    """ A dummy class for testing purposes. """
    def method(self, a, b):
        """ This is a method.

        Parameters
        -----
        a : int
            This is the first parameter.
        b : int
            This is the second parameter.
        """
        pass

    @staticmethod
    def static_method(a, b):
        """ This is a static method.

        Parameters
        -----
        a : int
            This is the first parameter.
        b : int
            This is the second parameter.
        """
        pass

    @classmethod
    def class_method(cls, a, b):
        """ This is a class method.

        Parameters
        -----
        a : int
            This is the first parameter.
        b : int
            This is the second parameter.
        """
        pass

    @property
    def prop(self) -> int:
        """ This is a property. """
        return 42
```

Documentation of all members at once is discouraged, but it is possible to do so by calling all methods of 'DocumentedClass':

```
doc = DocumentedClass(Dummy)
doc.constructor()
```

```
doc.members()
doc.properties()
doc.classmethods()
doc.staticmethods()
```

For generating more organized documentation, it is possible to selectively include or exclude static methods, class methods and properties. This allows to generate separate sections for each type of member, which can be more readable and easier to navigate.

The constructor:

Dummy

Dummy() :

A dummy class for testing purposes.

Only normal methods:

method

```
method(
    self : Any,
    a : Any,
    b : Any
):
```

This is a method.

Parameters

a : Any

This is the first parameter.

b : Any

This is the second parameter.

Only static methods:

static_method

```
static_method(
    a : Any,
    b : Any
):
```

This is a static method.

Parameters

a : Any

This is the first parameter.

b : Any

This is the second parameter.

Only class methods:

class_method

```
class_method(  
    cls : Any,  
    a : Any,  
    b : Any  
):
```

This is a class method.

Parameters

a : Any

This is the first parameter.

b : Any

This is the second parameter.

Only properties:

prop

```
prop(self : Any): -> int:
```

This is a property.

4.2 Examples

A minimal function with no parameters and no docstring:

```
def awesome_useless():
    pass
```

```
i awesome_useless
awesome_useless():
(missing summary)
```

A function with variadic parameters and no documented parameters:

```
def awesome_useless(*args, **kwargs):
    """ Some useless function. """
    pass
```

```
i awesome_useless
awesome_useless(
    *args : Any,
    **kwargs : Any
):
```

Some useless function.

Parameters

*args : Any

(missing description)

A function with variadic parameters with type annotation and explicitly documented keyword parameter:

```
def awesome_useless(*args: tuple[int, ...], **kwargs):
    """ Some useless function.

    Parameters
    -----
    args : tuple
        Positional arguments. This is intended to be a catch-all for
        any number of positional arguments. Let's write a long text.
    some_kwarg : int = 1
        Keyword argument. This is intended to be a catch-all for any
        number of keyword arguments. Let's write a long text.
    """
    pass
```

awesome_useless

```
awesome_useless(  
    *args : tuple[int, ...],  
    **kwargs : Any  
):
```

Some useless function.

Parameters

`*args : tuple[int, ...]`

Positional arguments. This is intended to be a catch-all for any number of positional arguments. Let's write a long text.

`some_kwarg : int = '1'`

Keyword argument. This is intended to be a catch-all for any number of keyword arguments. Let's write a long text.

A function with a positional-only parameter, a keyword-only parameter and variadic parameters:

```
def awesome_useless(a: int, /, b, **kwargs):  
    """ Some useless function. """  
    pass
```

awesome_useless

```
awesome_useless(  
    a : int,  
    /,  
    b : Any,  
    **kwargs : Any  
):
```

Some useless function.

Parameters

`a : int`

(missing description)

`b : Any`

(missing description)

Function with all kinds of parameters:

```
def awesome_useless(x, /, y: int = 0, *, z: int = 0, **kwargs) -> None:
    """ Some useless function.

    Parameters
    -----
    x : int
        Nothing to say about this.
    y : int
        Nothing to say about this.
    z : int
        Nothing to say about this.
    xx : int
        Nothing to say about this.
    """
    pass
```

awesome_useless

```
awesome_useless(
    x : Any,
    /,
    y : int = 0,
    *,
    z : int = 0,
    **kwargs : Any
) -> None:
```

Some useless function.

Parameters

x : Any

Nothing to say about this.

y : int = 0

Nothing to say about this.

z : int = 0

Nothing to say about this.

xx : int = None

Nothing to say about this.

5 Extensions

aka majordome.magic

On top of your notebook, simply load the extension:

```
%load_ext majordome.magic
```

For JupyterLab 4, a dedicated frontend extension is now available in this repository at [jupyterlab/majordome-mdmagic](https://github.com/jupyterlab/majordome-mdmagic).

```
cd jupyterlab/majordome-mdmagic
npm install
npm run build
jupyter labextension link .
```

After this, start JupyterLab as usual and the extension will switch `%%md` code cells to Markdown highlighting (or LaTeX for pure `$$...$$` display-math bodies).

5.1 Skipping execution

SkipperMagic

SkipperMagic():

An extension for skipping the execution of code cells.

The module provides a simple extension for skipping the execution of code cells in a notebook. It does that and only that. The goal is to control notebook execution from the environment, which is useful when running notebooks with long-running cells, as numerical simulations.

```
%skipper_verbosity 1
```

Setting skipper verbosity to 1

Here we will use `sleep` to emulate a long-running cell, and we will control its execution with an environment variable. Generally speaking, the goal is to export environment variables in the terminal before running the notebook, but for demonstration purposes, we will set it in the notebook itself.

```
from time import sleep
import os
```

The following (case insensitive) options are accepted by the `%%skipper` magic command. Please, notice that the *truthy* values mean *skip this cell* (as the extension name implies, but even myself can be confused by the double negative here).

```
{'true': True, '1': True, 'yes': True, 'false': False, '0': False, 'no': False}
```

Below we set the environment variable `SOLVE_LONNNNG` to `true`, which means that the cell will be skipped. You can change it to `false` to execute the cell, or even set it to any other value, as long as it is in the lookup table.

```
os.environ["SOLVE_LONNNNG"] = "true"
```

```
did_you_skip_it = True
```

If everything is file, the following cell will be skipped:

```
%%skipper SOLVE_LONNNNG
sleep(100)
did_you_skip_it = False
```

Skipping cell: SOLVE_LONNNNG=True

And here we check if it was actually skipped:

```
print("Did you skip it?", did_you_skip_it)
```

```
Did you skip it? True
```

You can also avoid the *Skipping cell* message by setting the verbosity to 0:

```
%%skipper_verbosity 0

%%skipper SOLVE_LONNNNG
sleep(100)
did_you_skip_it = False
```

5.2 Markdown and LaTeX interpolation

MdMagic

`MdMagic()`:

An extension for interpolating code in Markdown.

Cell magic extension to interpolate Markdown and LaTeX. It must be used from code cells and the content is rendered as Markdown. As such, it is expected to be used only when required, as without some frontend tweaks you will have undesirable syntax highlighting in the cell, as it is not interpreted as text. Nonetheless, this proves very useful for complex reporting with dynamic values.

- Variables are interpolated from the IPython user namespace using Python format placeholders, e.g. `{variable}`.
- If a variable is not found or evaluation fails, the original placeholder is left unchanged in the output, e.g. `{variable}`. For details, check `eval_placeholder` method.

- Curly braces inside LaTeX math blocks ($\dots$ or $\dots$) are automatically escaped before interpolation so expressions such as $x_{\{i\}}$ render correctly.
- You can override the interpretation of curly braces inside LaTeX math blocks by providing a format spec, e.g. `{variable:.2f}`.

Declare some values in the user namespace:

```
variable = "hello"
some_value = 12.2
```

Below we have a Markdown cell with some variables and LaTeX expressions. The variables will be replaced by their values, and the LaTeX will be rendered correctly.

```
%%md
```

I have this string `with` a variable: `{variable}`.

There `is` also some latex: $\int_a^b f(x) \, dx$.

- Some formatter value `{some_value:.2f}`

- Some block math:

```
$$
```

$$\int_0^\infty e^{-x^2} dx = \frac{\sqrt{\pi}}{2}$$

```
$$
```

I have this string with a variable: `hello`.

There is also some latex: $\int_a^b f(x) dx$.

- Some formatter value 12.20
- Some block math:

$$\int_0^\infty e^{-x^2} dx = \frac{\sqrt{\pi}}{2}$$

! About Quarto rendering

Please notice that when working in VSCode Jupyter extension, the following currently fails, *i.e.* it does not recognize the `%%md` magic command as it is not on the top of the cell; if you change the order, the Quarto annotation will be interpreted as markdown. For properly rendering Quarto documents, consider adding a global configuration for the `echo` option set to `false` and explicitly set it to `true` in the cells where you want to see the code. This is a known issue that will (maybe) be fixed in the future, but for now, it is a good workaround (that actually enforces a good practice of hiding code cells in the final document, which is what you usually want when writing reports).

```
#| echo: false
%%md
```

The following is also possible in math blocks:

```
%%md
```

```
$$  
F(x) = {some_value:.2f} \cdot x^2  
$$
```

But there **is** no way to use the plain `some_value` without a **format** specifier, as it would be

```
$$  
F(x) = {some_value} \cdot x^2  
$$
```

$$F(x) = 12.20 \cdot x^2$$

But there is no way to use the plain `some_value` without a format specifier, as it would be indistinguishable from a regular LaTeX expression.

$$F(x) = some_value \cdot x^2$$

SymPy expressions are also supported, and they will be rendered as LaTeX. In general rendering is done by displaying some expression as provided below:

```
import sympy as sp  
  
x = sp.symbols("x")  
f_x = some_value * x**2  
f_x  
  
12.2x2
```

With help of the extension we can also render it in markdown:

Now we can also talk about the function we just defined: $12.2x^2$! If you need to inject the expression inside a math block, you can do so by calling `sp.latex` and using the string formatter: $f = 12.2x^2$. It should also work with block math:

$$f = 12.2x^2$$

Part II

Reactor Engineering

6 Acetylene pyrolysis

 Warning

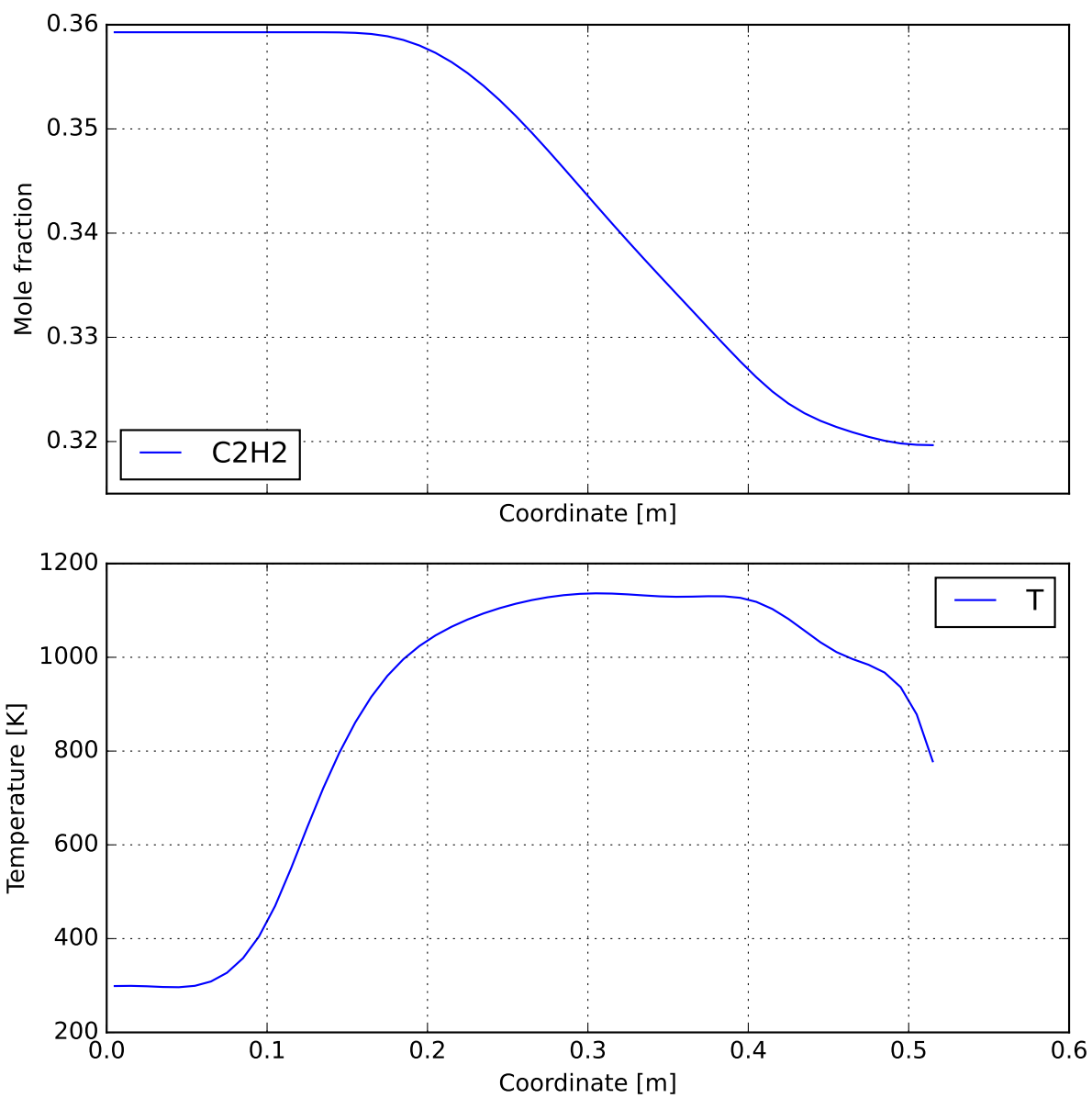
Work in progress!

6.1 Toolbox

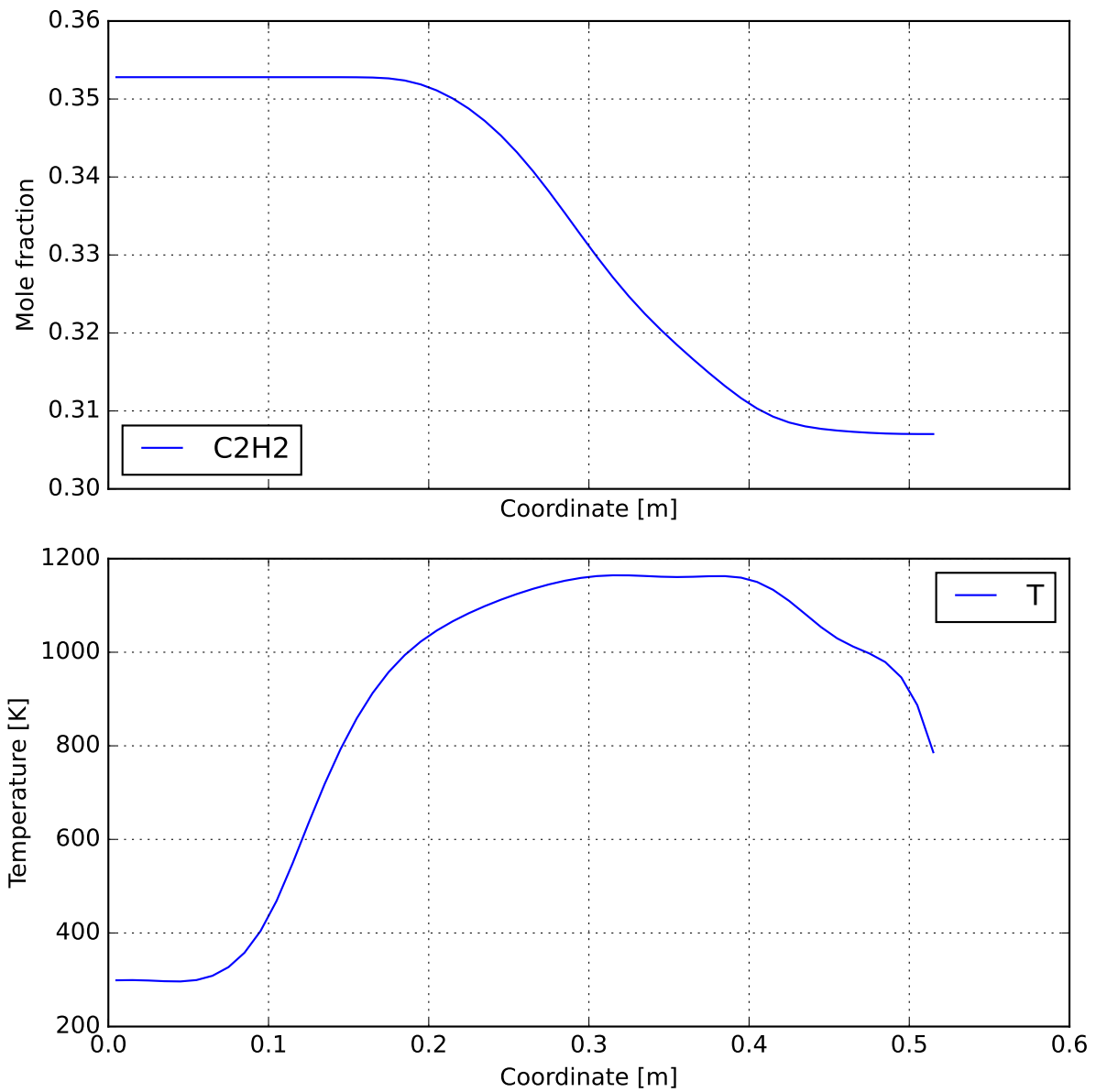
6.2 Reference case

CPU times: total: 781 ms

Wall time: 790 ms



CPU times: total: 1.89 s
Wall time: 1.9 s



6.3 Scan table

Working on case no. 1
Working on case no. 2
Working on case no. 3
Working on case no. 4
Working on case no. 5
Working on case no. 6
Working on case no. 7
Working on case no. 8
Working on case no. 9
Working on case no. 10
Working on case no. 11
Working on case no. 12
Working on case no. 13

Working on case no. 14
CPU times: total: 26 s
Wall time: 25.9 s

Working on case no. 1
Working on case no. 2
Working on case no. 3
Working on case no. 4
Working on case no. 5
Working on case no. 6
Working on case no. 7
Working on case no. 8
Working on case no. 9
Working on case no. 10
Working on case no. 11
Working on case no. 12
Working on case no. 13
Working on case no. 14
CPU times: total: 10.7 s
Wall time: 10.7 s

	N	P	Q	T	X_meas	X_dalmazsi_2017	err_dalmazsi_2017	err_graf_2007
0	1	50	222	773	0.352	0.352786	0.223212	2.032933
1	2	50	222	873	0.364	0.352540	-3.148330	-1.550413
2	3	50	222	973	0.364	0.350258	-3.775208	-2.602299
3	4	50	222	1073	0.346	0.339812	-1.788298	-1.188026
4	5	50	222	1123	0.312	0.326556	4.665512	6.179187
5	6	50	222	1173	0.307	0.307034	0.010994	4.120820
6	7	50	222	1273	0.288	0.290866	0.995060	6.743916
7	8	30	222	1173	0.323	0.329143	1.901911	5.859843
8	9	30	222	1223	0.314	0.318436	1.412680	7.828360
9	10	100	222	1173	0.249	0.242491	-2.613868	4.125649
10	11	100	222	1223	0.226	0.230372	1.934472	5.944962
11	12	100	222	1273	0.201	0.221251	10.075293	12.714431
12	13	100	378	1023	0.343	0.342777	-0.065090	0.254930
13	14	100	378	1123	0.298	0.305793	2.615003	6.255879

	err_dalmazsi_2017	err_graf_2007
0	3.53261	5.758262

7 Counter-current PFR

Warning

Work in progress!

7.1 Introduction

The goal of this tutorial is to discuss a sequence of increasingly complex plug-flow reactor (PFR) models. These are all built upon `PlugFlowChainCantera` which approximates a PFR as a cascade of perfect-stirred reactors (PSR), as illustrated [here](#). Care must be taken when using such an approach because this zero-dimensional modeling does not formally represent a PFR, but approaches one - so grid convergence studies might be required to validate a model, what is beyond our scope here (and has been done during the conception of this report), which limits itself to studying the coupling of reactors. Nonetheless, this representation proves very interesting when assembling complex systems as it remains modular, allowing for extensibility of models.

In what follows we will be guided through the following models:

- A single PFR with possibility of distributed mass sources along its length; this case is often encountered in industry when modeling reactors coupled through a permeable membrane, or in a rotary kiln, where solids may eventually release gases, what is especially important in the field of biomatter gaseification.
- Next we will couple a pair of reactors as conceived in the first example, which will be allowed to exchange energy through a pseudo-convective model; there is no major difficulty at this point as the coupling can be stabilized through a relaxation routine. We investigate how the ordering of reactor solution affects convergence rate.
- At the last level of complexity studied here, we add a shared common wall around the pair of reactors of the previous step; one might argue that such a system would require two *wall* temperatures to be described at steady state; here we will assume that this wall is *rotating* around the pair of reactors in such a way that it remains in contact with a reactor during part of a cycle, and a *statistically significant mean temperature* is solely used as a process measurable.

Because writting long classes is not didatic for the purposes of a tutorial, we will use a *semi-object-oriented* programming approach here. That means, there will be minimal classes just holding the basic elements representing the state of a problem, and functions that resemble class methods (receiving the class instance as first argument) to perform the required operations.

Below we start by importing all required tools at once:

Because some cells take some time to execute, during development you can set this to "true" so that they are not run. You can recognize these cells by the presence of magic `%%skipper`. Setting this to "false" will not skip the cells (this is the logic of the booleans here, we are saying true to skip, not true to execute!).

7.2 Shared functionalities

This section is not commented, variables and functions should speak by themselves.

7.3 Single plug-flow reactor

For the single reactor `ReactorModel` we write a simple wrapper around `PlugFlowChainCantera`; this model holds an instance of a `solution` object conceived with the same mechanism as provided to the PFR, and a simple source structure for the distributed source terms.

Both the `solution` and `source` are used in `add_source` to inject a mass flow rate `mdot` at temperature `T`, and composition `X` (given in mole fractions) at a set of cell indices where along the reactor. We will see later how discretization is performed and thus how to link where to an actual axial coordinate along the reactor.

The underlining PFR model has already its own solution method (which is actually called `update`) that *seeks* a steady state for the reactor given a certain source; sometimes it fails to converge and multiple calls may be required to find the steady state of a reactor. In `solve_reactor` below we wrap the aforementioned method and add some reporting.

7.3.1 Example - single PFR simulation

As this first case is already fully supported by the underlining library class, we are ready to solve a first example problem. Comments are provided in the function `sample_single` to guide you through the setup. A distributed power supply is added to the reactor as an analytical function as:

$$Q(z) = \begin{cases} A \left[1 - \exp\left(-\frac{z}{z_0}\right) \right] & z < \frac{L}{2} \\ 0 & \text{otherwise} \end{cases}$$

This is arbitrarily parametrized and is intended just to show that accessing `model.source.Q` allows to provide any heat source function to a reactor.

We said no convergence study was going to be performed; that was a lie. Below we inspect the evolution of final temperature in terms of reactor cell length in range $l \in [0.001; 0.1]$ m. Solution quickly approaches 340.6 K, and for practical purposes a discretization of 1 cm seems a good compromise between execution time and accuracy. Also notice that total supplied power may depend on discretization because of how it is provided.

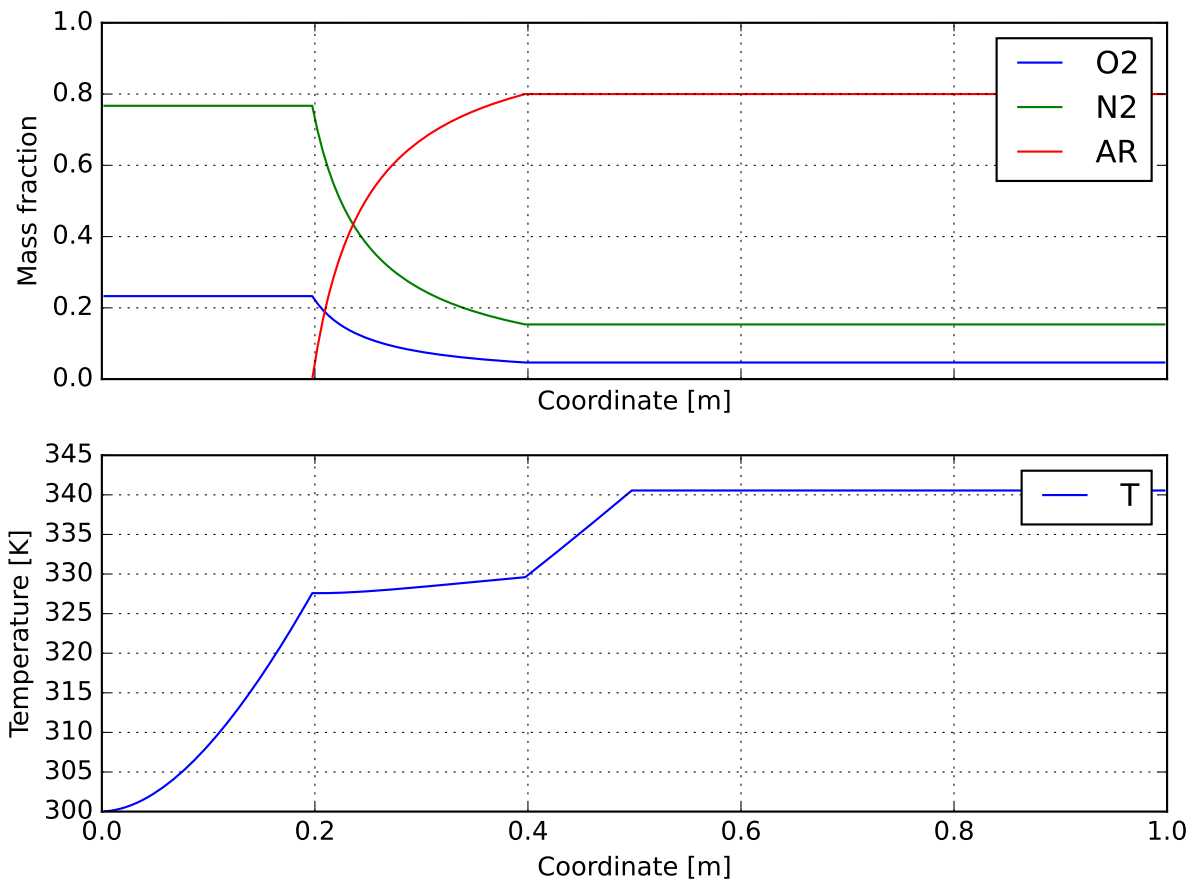
```
|-----|-----|
| Total mass flow rate | g/s | +2.500000e+02 |
| Total external power | W    | +6.035993e+03 |
| Final temperature    | K    | +3.404046e+02 |
```

```
|-----|-----|
| Total mass flow rate | g/s | +2.500000e+02 |
| Total external power | W    | +6.000357e+03 |
| Final temperature    | K    | +3.405258e+02 |
```

Below we show the report of first solution, which here is no better than an initial guess; sequential solution of reactor towards steady state is recommended in practice.

```
|-----|-----|-----|
| Total mass flow rate | g/s | +2.500000e+02 |
| Total external power | W   | +6.000085e+03 |
| Final temperature    | K   | +3.395247e+02 |
|-----|-----|-----|
| Total mass flow rate | g/s | +2.500000e+02 |
| Total external power | W   | +6.000085e+03 |
| Final temperature    | K   | +3.405501e+02 |
```

Finally we illustrate the composition and temperature profiles induced by the simultaneous mass and heat transfer:



You can also access the data in base reactor for post-processing as desired:

	0	1	2	3	4
z_cell	0.002500	0.007500	0.012500	0.017500	0.022500
V_cell	0.000039	0.000039	0.000039	0.000039	0.000039
Q_cell	1.177767	3.489592	5.744338	7.943414	10.088195
m_cell	0.000046	0.000046	0.000046	0.000046	0.000046
mdot_cell	0.050000	0.050000	0.050000	0.050000	0.050000
T	300.023321	300.092416	300.206156	300.363436	300.563179
density	1.171880	1.171610	1.171166	1.170553	1.169775

	0	1	2	3	4
Y_O2	0.232909	0.232909	0.232909	0.232909	0.232909
Y_N2	0.767091	0.767091	0.767091	0.767091	0.767091
Y_AR	0.000000	0.000000	0.000000	0.000000	0.000000

7.4 Counter-current flow reactors

The tooling required to compute a pair of reactors is already provided by `ReactorModel`; all we have to do is wrap a pair of reactors in `CounterCurrentReactors`. For convenience we store a copy of the axial coordinate (which is the same for both reactors). Notice here the introduction of `U`, a global heat transfer coefficient between phases, `Q`, the associated heat flux, and a relaxation manager.

One cannot compute heat fluxes between reactors before having an initial temperature profile established (see it as an initial guess). Function `initialize` is provided for that end and must be called after mass flows have been setup.

There are two ways one can solve the relaxation problem here, each with its strengths and weaknesses; one could simply calculate the flux, apply it to both reactors, and solve (`iter_direct`) or compose a series of flux and solution calculations (`iter_alternate`). When initial guess is not good enough, the alternating approach tends to deliver results in a more reliable fashion (because it is compatible with high relaxation factors), but it is slower. On the other hand, direct solution may converge faster as solution is already relaxed.

IMPORTANT: notice here that we get the reverse of `T2` to compute flux (as the reactors are in counter-current) and then `model.Q` is reversed again to set heat flux to reactor 2. This class could easily be generalized by adding a flag on whether temperature and flux should be treated as is (co-current) or reversed.

Based on the previous discussion, a mixed approach can be proposed: relax the problem with the alternating solution and move to direct one after a certain criteria or maximum number of iterations has been achieved. This is emulated in `solve_pair` provided below.

A simple standard plotting function is provided for the counter-current reactor pair:

7.4.1 Example - pair of coupled PFR

This second example replaces the user-defined heat flow function by a convective heat exchange through a fin separating the reactions; this is modeled through a heat transfer coefficient `h` which is made constant here for illustration purposes only. To test the robustness of the approach, a relatively high dilution flow is applied to a concentrated region in reactor 2 (hot stream).

After building the reactor pair, inlet and distributed conditions and the reactors are initialized, *i.e.* solved individually, before iterative solution is performed. The user may require a final solution step for printing reactor quantities in a table.

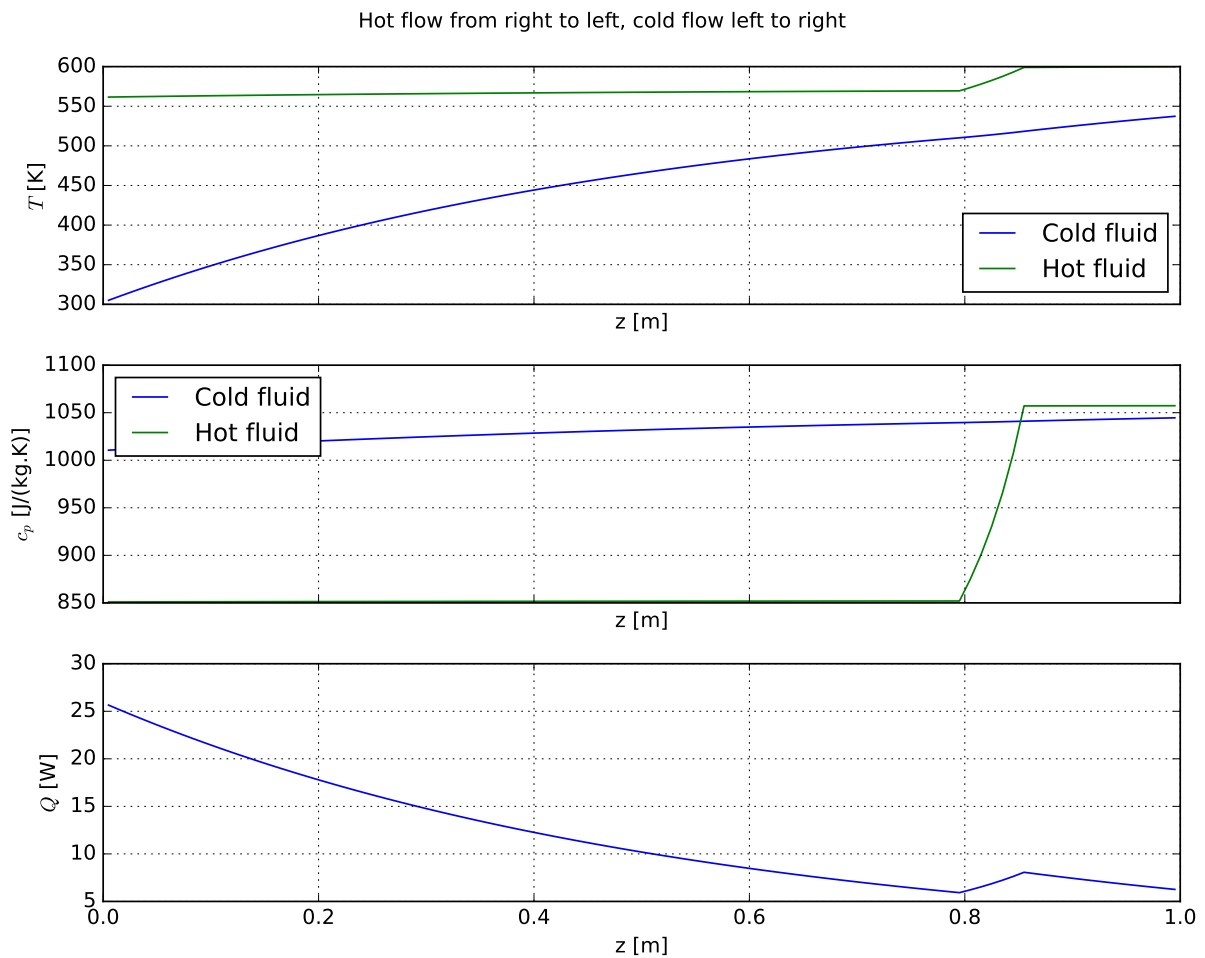
For low relaxation values ($\alpha < 0.5$), both methods may eventually converge under the same number of iterations.

```
|-----|-----|
| Total mass flow rate | g/s | +5.000000e+00 |
| Total external power | W    | +1.216871e+03 |
```

Final temperature	K	+5.372994e+02
----- ----- -----		
Total mass flow rate	g/s	+1.600000e+02
Total external power	W	-1.216871e+03
Final temperature	K	+5.615588e+02

Refining the grid for the present case has an impact of a few degrees over final solution (but takes much longer to solve), as illustrated below. Below 2 mm there is little change in the final solution and this value is a practical limit for grid refinement here.

Finally we display the results; for the hot stream we observe a sharp decrease in temperature (and associated specific heat capacity) due to argon dilution. Cold stream progressively heats up along the channel length as expected.



7.5 Adding losses to the environment

So far we have developed the ideas required to reach the final goal of conceiving a fully coupled system with a pair of reactors and walls. Again we proceed with a wrapper class which contains our problem variables, but in this case we fully develop it as a single block. Also, this time documentation is provided in the class itself as that seems clearer to the author in this case, and provides some closure to the study.

The main increment in problem solution here is the extra step of solving for the wall temperature, which can be found from the global flux definition:

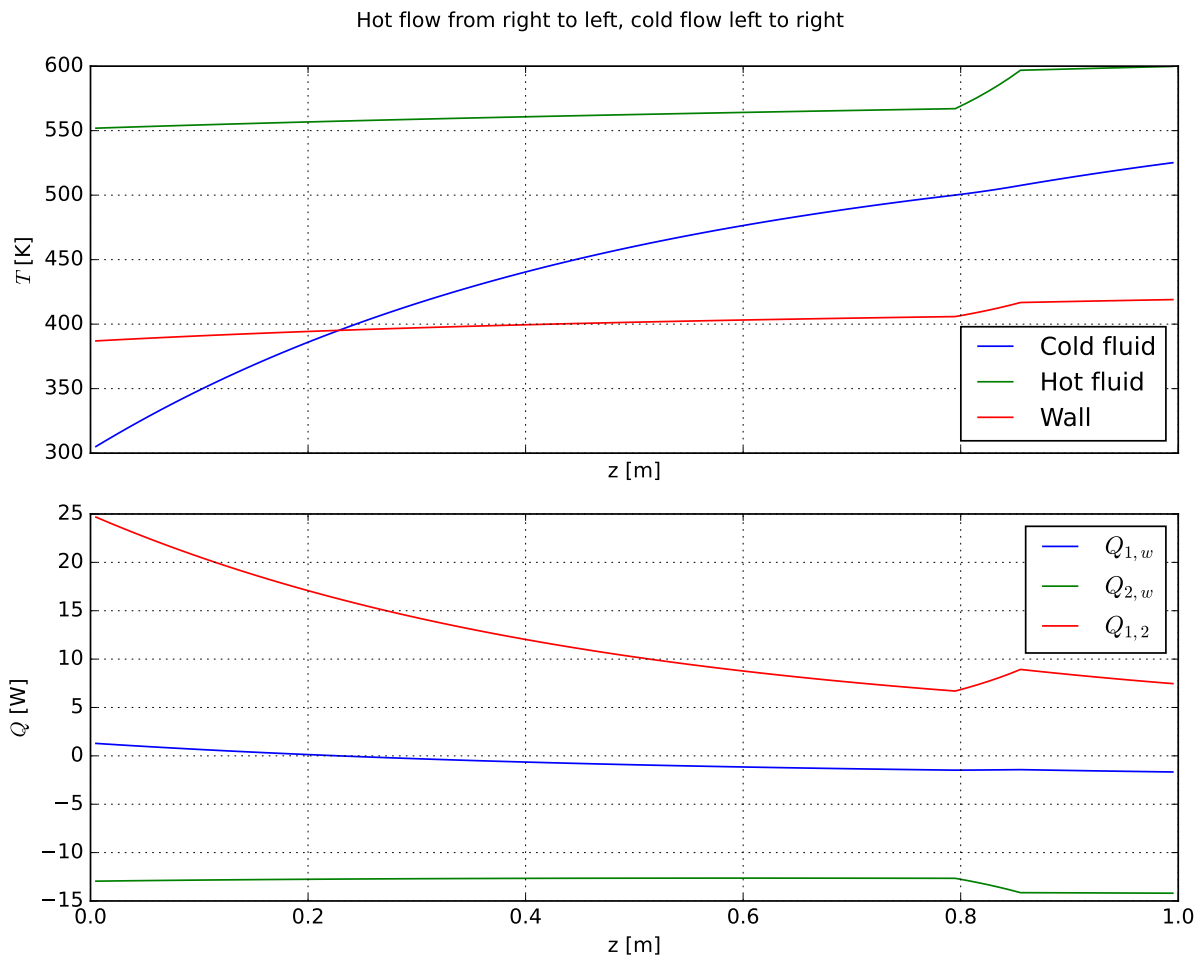
$$\dot{Q}_w = \frac{T_w - T_e}{R_{tot}} \quad \text{where} \quad R_{tot} = \frac{1}{hA} + \frac{1}{2\pi lk} \log \frac{R_{ext}}{R_{int}}$$

We make use of the main takes of the previous sections, which have shown that alternating reactor solution order is not always an advantage; here the problem has been programmed to use direct solution. Relaxation is applied to reactor fluxes, but not to wall flux, which is computed from a balance of the relaxed quantities for the sake of energy conservation.

7.5.1 Example - environment coupling

```
|-----|-----|
| Total mass flow rate | g/s | +5.000000e+00 |
| Total external power | W    | +1.003972e+03 |
| Final temperature    | K    | +4.963894e+02 |
|-----|-----|
| Total mass flow rate | g/s | +3.000000e+02 |
| Total external power | W    | -2.210504e+03 |
| Final temperature    | K    | +5.216989e+02 |
|-----|-----|

|-----|-----|
| Total mass flow rate | g/s | +5.000000e+00 |
| Total external power | W    | +1.153462e+03 |
| Final temperature    | K    | +5.251459e+02 |
|-----|-----|
| Total mass flow rate | g/s | +1.600000e+02 |
| Total external power | W    | -2.516963e+03 |
| Final temperature    | K    | +5.519201e+02 |
|-----|-----|
```

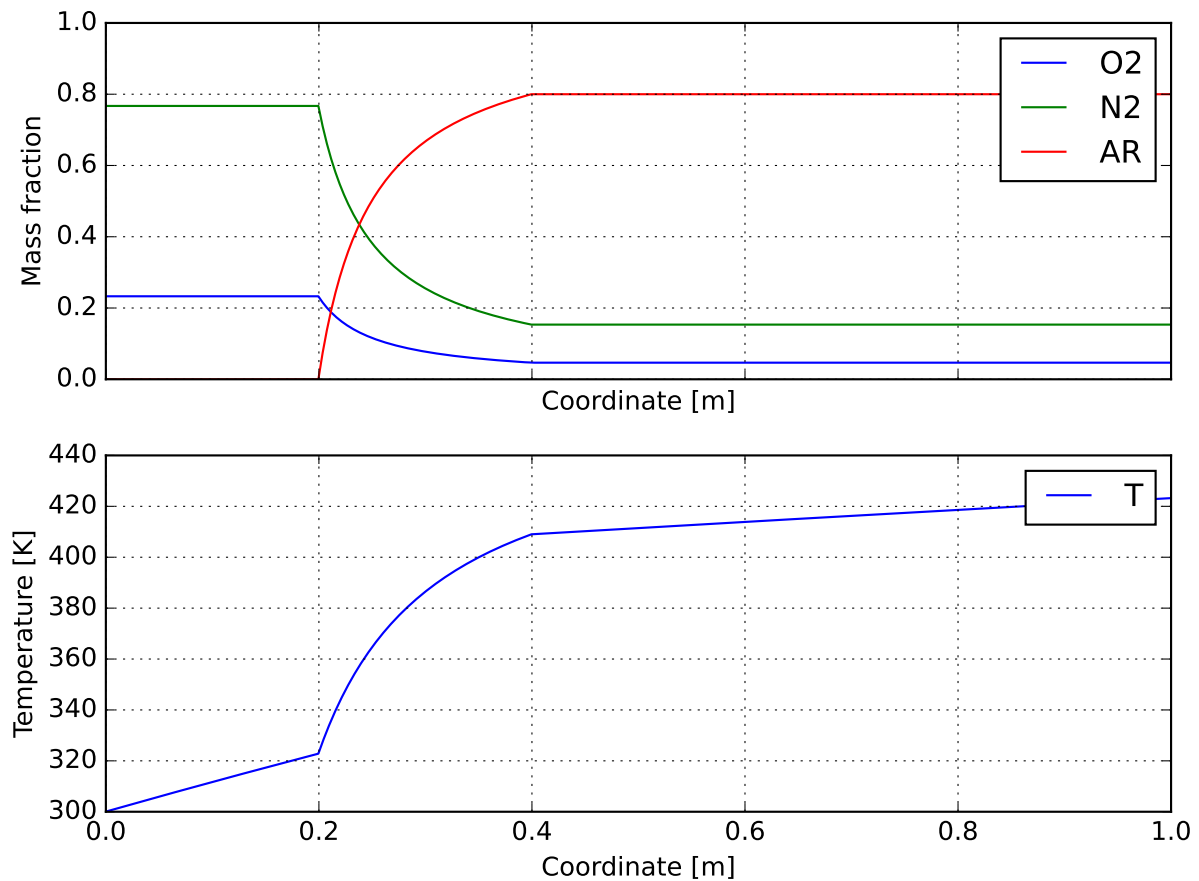


7.6 Registering flow for a single reactor

Below we develop further the idea into registering heat flows as an alternative to relaxation. The method may prove useful because heat flows are computed from current state of the reactors instead of an external guess.

7.6.1 Example - prototype with a single reactor

Total mass flow rate	g/s	+2.500000e+02
Total external power	W	+4.237303e+03
Final temperature	K	+4.232112e+02



7.7 Registering flow for a pair of reactors

As a last level of improvement, we generalize the approach for a pair of reactors. After some experimentation it was concluded that falling back to the alternating approach with intermediate wall solution lead to the best performance, probably because of decreased system stiffness. Relaxing the wall heat flow allows for faster convergence.

7.7.1 Example - extension to coupled system

```

|-----|-----|-----|
| Total mass flow rate | g/s | +5.000000e+00 |
| Total external power | W   | +1.153462e+03 |
| Final temperature    | K   | +5.251459e+02 |
|-----|-----|-----|
| Total mass flow rate | g/s | +1.600000e+02 |
| Total external power | W   | -2.516963e+03 |
| Final temperature    | K   | +5.519201e+02 |
CPU times: total: 15.5 s
Wall time: 15.6 s

```

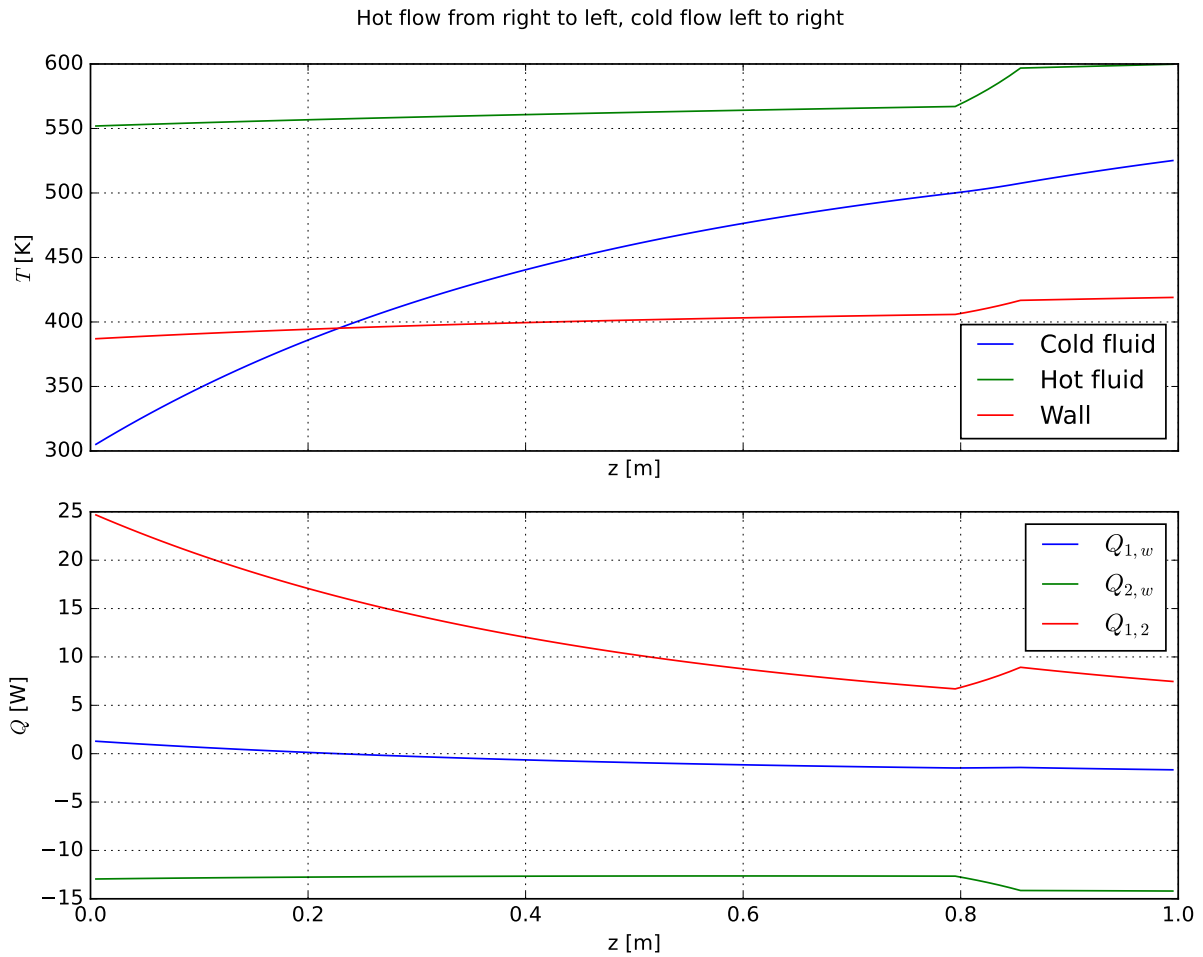
```

|-----|-----|-----|
| Total mass flow rate | g/s | +5.000000e+00 |
| Total external power | W   | +1.153462e+03 |

```

Final temperature	K	+5.251459e+02
----- ----- -----		
Total mass flow rate	g/s	+1.600000e+02
Total external power	W	-2.516963e+03
Final temperature	K	+5.519201e+02

CPU times: total: 17 s
Wall time: 17.1 s



----- ----- -----		
Total mass flow rate	g/s	+5.000000e+00
Total external power	W	+1.153462e+03
Final temperature	K	+5.251459e+02
----- ----- -----		
Total mass flow rate	g/s	+1.600000e+02
Total external power	W	-2.516963e+03
Final temperature	K	+5.519201e+02

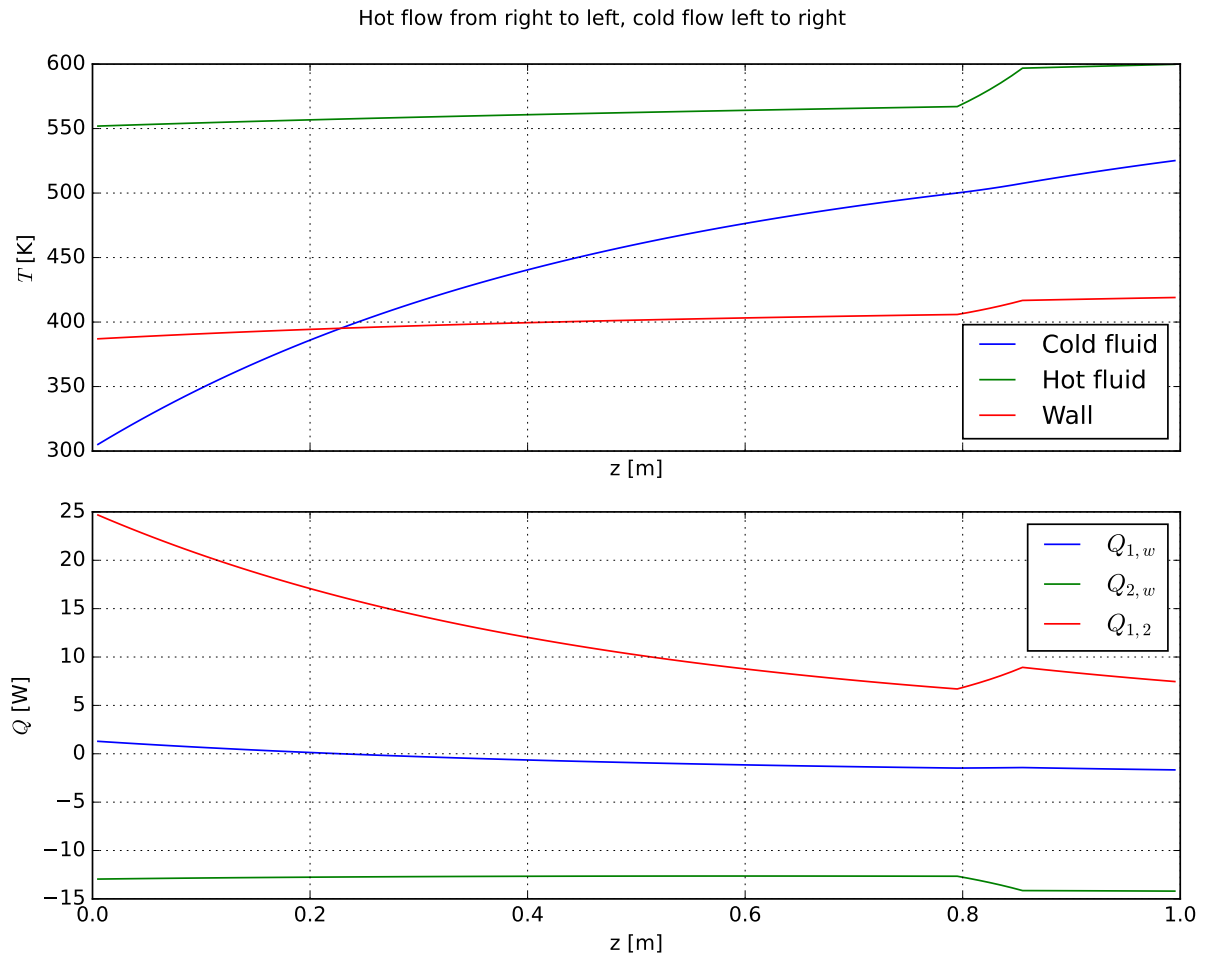
CPU times: total: 15 s
Wall time: 15 s

----- ----- -----		
Total mass flow rate	g/s	+5.000000e+00
Total external power	W	+1.153462e+03
Final temperature	K	+5.251459e+02


```

|-----|-----|-----|
| Total mass flow rate | g/s | +1.600000e+02 |
| Total external power | W   | -2.516963e+03 |
| Final temperature    | K   | +5.519201e+02 |
CPU times: total: 15.9 s
Wall time: 15.9 s

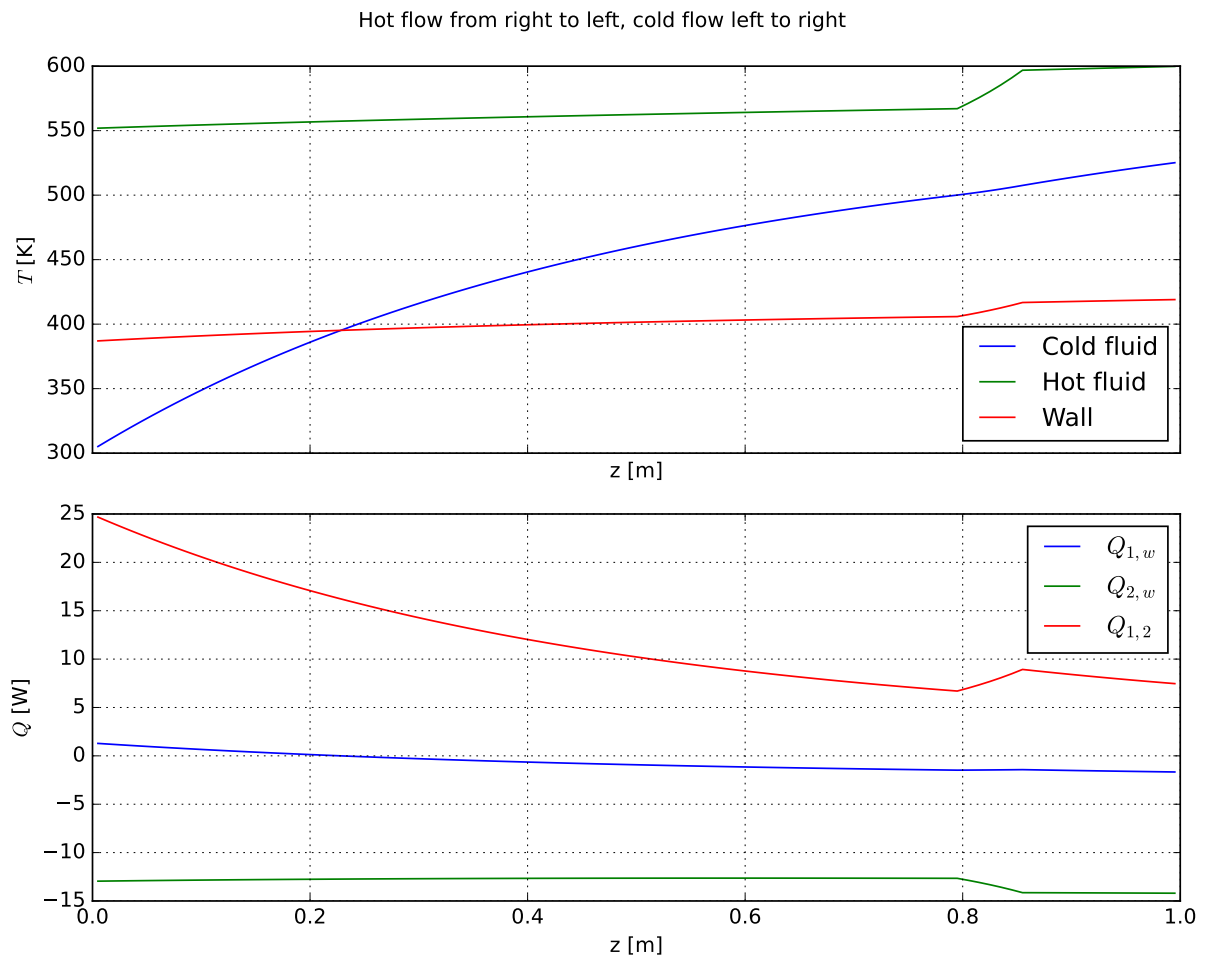
```



```

|-----|-----|-----|
| Total mass flow rate | g/s | +5.000000e+00 |
| Total external power | W   | +1.153462e+03 |
| Final temperature    | K   | +5.251459e+02 |
|-----|-----|-----|
| Total mass flow rate | g/s | +1.600000e+02 |
| Total external power | W   | -2.516963e+03 |
| Final temperature    | K   | +5.519201e+02 |
CPU times: total: 3.55 s
Wall time: 3.53 s

```



8 Fitting NASA9 data

Warning

Work in progress!

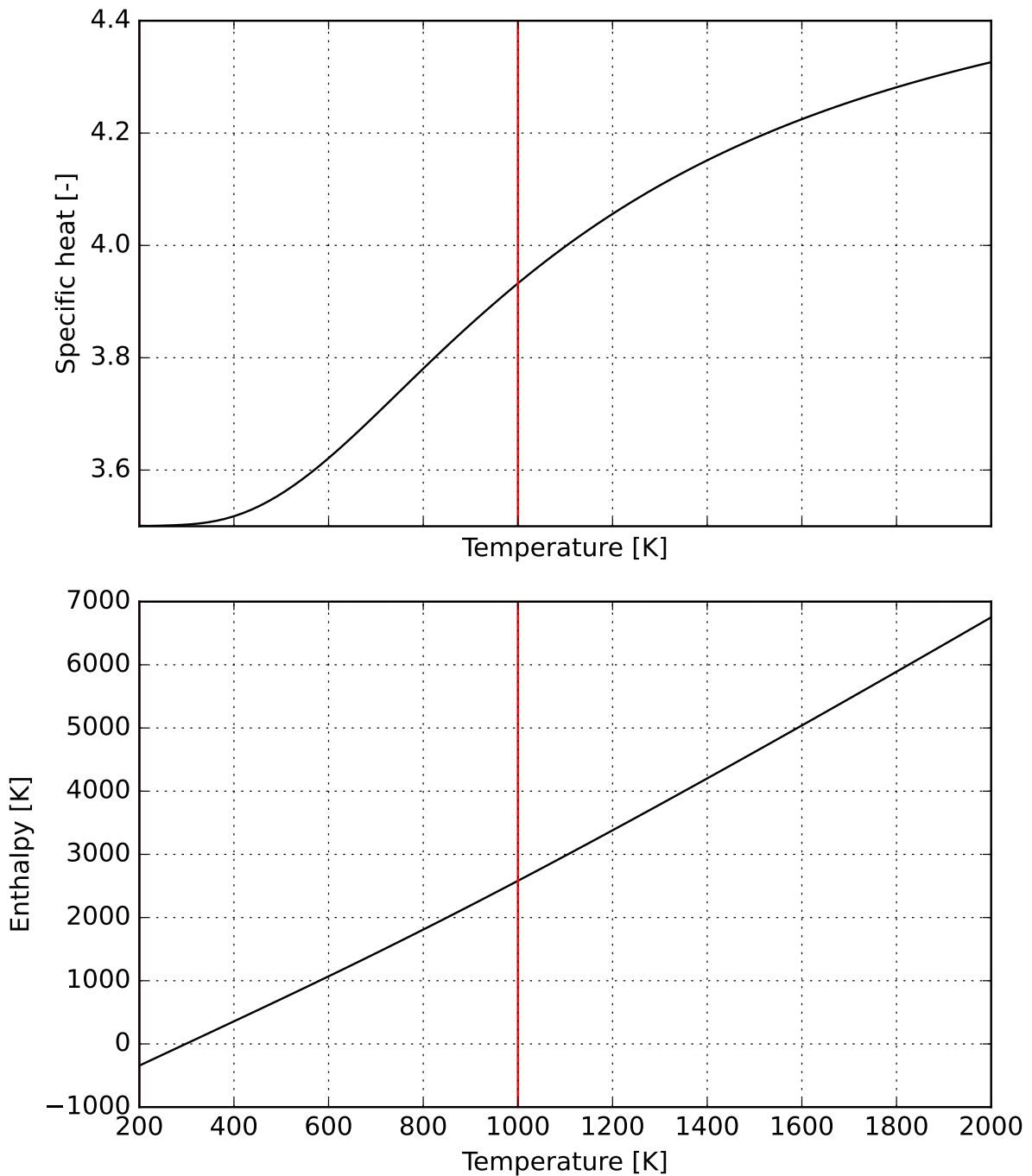
The goal of this document is to provide a better understanding on how to fit empirical data to [NASA9](#) parameterization of thermodynamic properties; more details are provided in the original [NASA report](#). A progressive level of complexity is used to provide better insights of how the species thermodynamics is encoded in this format.

8.1 Helper utilities

8.2 Nitrogen actual fit

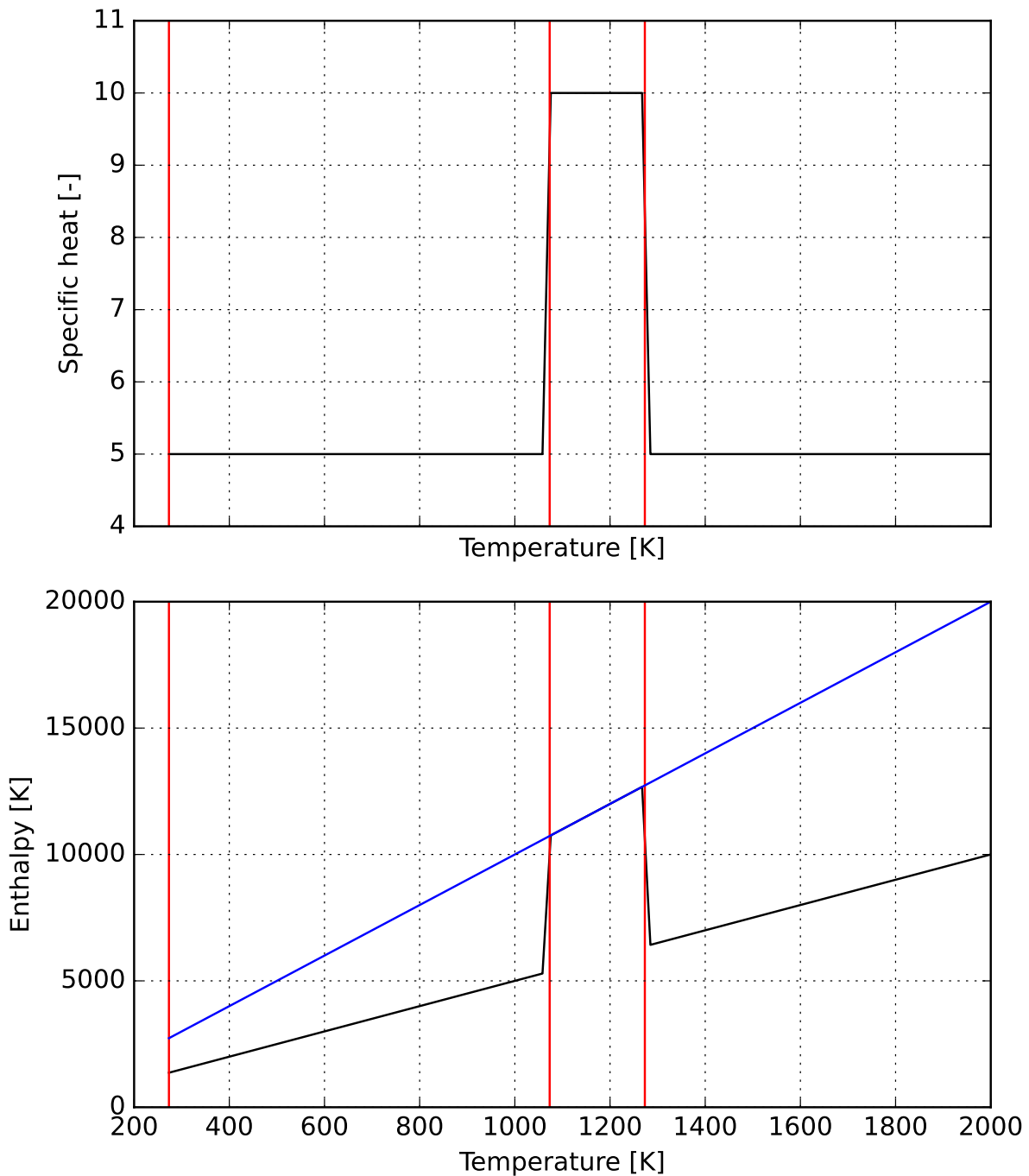
- Below we display in text format an actual NASA9 fit taken from `airNASA9.yaml` data file provided with Cantera; this is a fully parameterized species, making it a bit more difficult to visually understand the role of different coefficients
- The third coefficient a_2 is related to the specific heat intercept when $T = 0$ and the eighth coefficient a_7 represents the integration constant for providing enthalpy over a range
- Since NASA9 functions are provided in dimensionless form, it is worth dividing computed values by ideal gas constant, so that values are in the same *units* as the provided coefficients

The next cell illustrates the dimensionless specific heat and enthalpy over a typical industrial range.



8.3 Enforcing constant specific heat

- Next we create a dummy species (in Cantera composition needs to include *actual* atoms or you need to create a new database for elements) with all coefficients set to zero except those related to specific heat
- This way we establish a piecewise-constant specific heat function; notice that doing so will produce thermodynamic inconsistent steps in enthalpy



- First, notice that the temperature ranges are *closed-open* ranges, meaning the first point belongs to the range but the last not; we introduce T_{tol} to compute a *last* value of enthalpy over each range as close as possible to the next
- To ensure continuity of enthalpy function, we evaluate the gap $h_{hi}-h_{lo}$ at each range end and add that to the correction of the previous range (as enthalpy is integrated from the reference temperature)
- Using $dh0 = s.dimless_h(298.15)$ to compute the initial value is optional and simply enforces zero enthalpy at the reference state, but in principle it won't affect any computations as only *enthalpy changes* make sense physically

Note: as we are dealing with constant c_p here, we could have computed all the above enthalpy values straight from specific heat integration by hand, but this would be more complicated in general

cases and it was chosen to present the more general (and with least user interaction) approach. See below that the enthalpy at reference temperature fits nicely with the specific heat provided above.

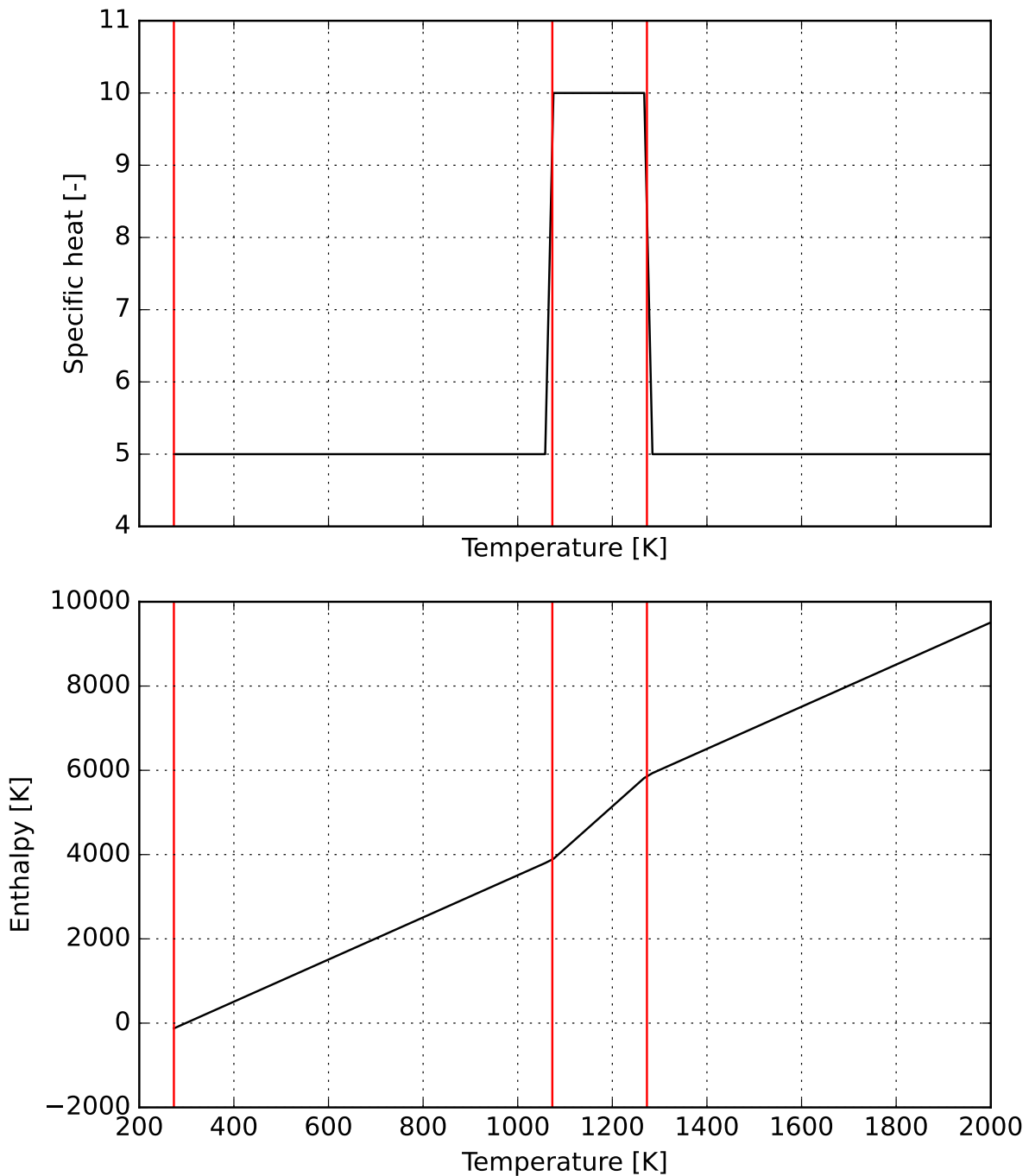
```
np.True_
```

Generalizing for all the steps, we could manually compute everything as follows:

```
((6856.5, np.float64(6856.5000000000005)),  
 (490.75, np.float64(490.75000000001455)))
```

8.4 Ensuring enthalpy continuity

- To wrap up, we use string interpolation to fix the integration constants required to ensure continuity of enthalpy function



8.5 Application to a real material

We start by providing actual physical values to be converted into NASA9 format.

Calculations involve three phases (1) convert into units used by NASA9 format, (2) smearing of reaction enthalpy into specific heat, and (3) corrections required for continuity of enthalpy function.

Alternative for using a constant specific heat over the *whole range*:

Next we generate the YAML input that will be used in Cantera database file.

```

name: meal
composition: {Si: 1, O: 2}
thermo:
  model: NASA9
  temperature-ranges: [273.15, 1073.15, 1273.15, 6000.0]
  data:
    - [ 0.00000000e+00, 0.00000000e+00, 7.51537686e+00,
        0.00000000e+00, 0.00000000e+00, 0.00000000e+00,
        0.00000000e+00, -2.24070961e+03, 0.00000000e+00]

    - [ 0.00000000e+00, 0.00000000e+00, 1.69746349e+01,
        0.00000000e+00, 0.00000000e+00, 0.00000000e+00,
        0.00000000e+00, -1.23919123e+04, 0.00000000e+00]

    - [ 0.00000000e+00, 0.00000000e+00, 9.82780051e+00,
        0.00000000e+00, 0.00000000e+00, 0.00000000e+00,
        0.00000000e+00, -3.29292018e+03, 0.00000000e+00]

```

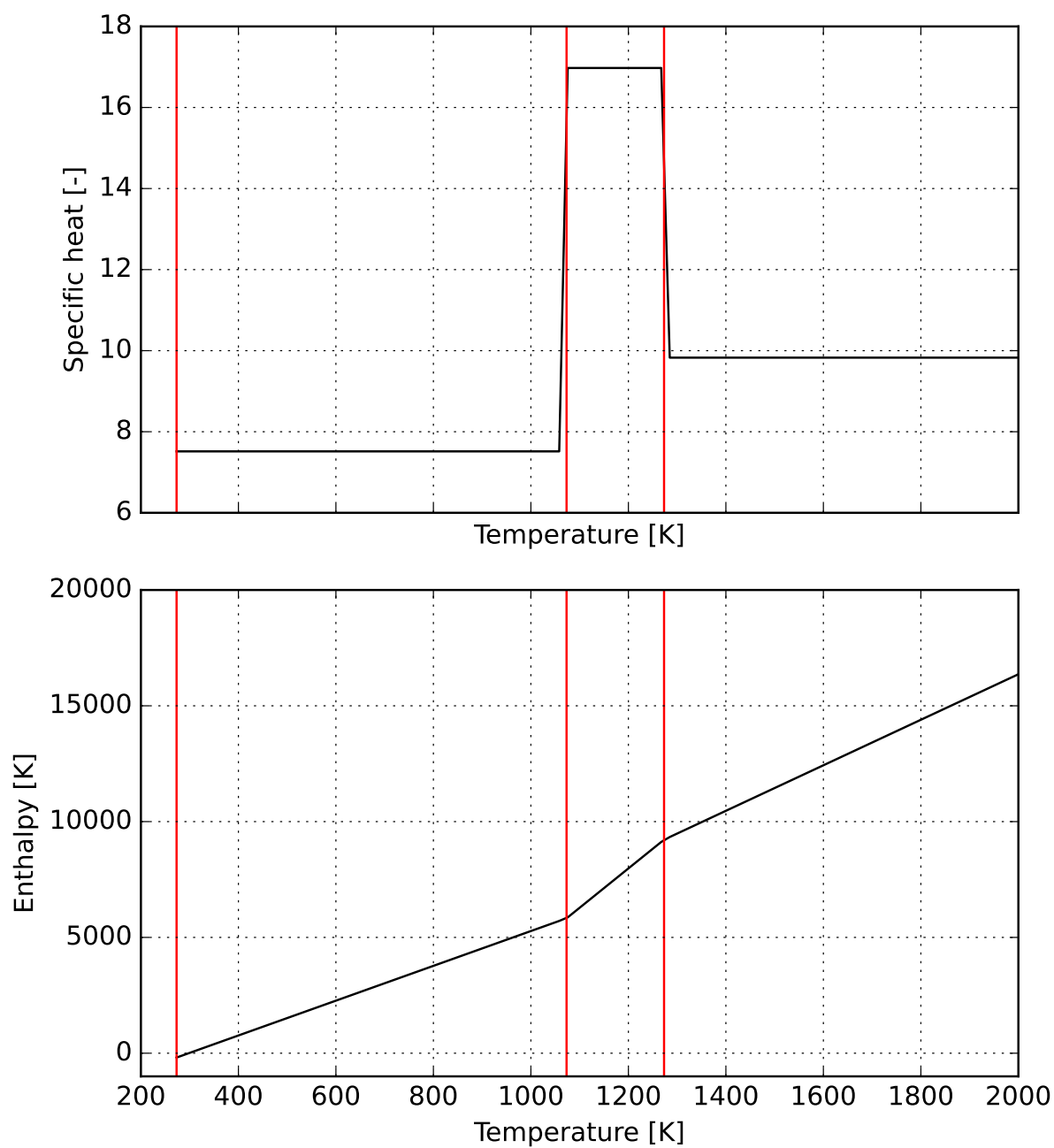
```

name: meal
composition: {Si: 1, O: 2}
thermo:
  model: constant-cp
  cp0: 95.5200 J/mol/K
  h0: 0.0 J/mol®

```

Finally we inspect results before proceeding to use of newly fit data.

Reaction enthalpy fitting - $\log_{10}(|error|) = -7.3$



Important: NASA9 is the only parameterization supporting an arbitrary number of temperature ranges in Cantera. Given the number of parameters, it is expected to slow down simulations. For more details, please check [the documentation](#).

8.6 Generalizing approach

Warning

Work in progress from here on... the code is right but the thermodynamics is pretty much sketchy!

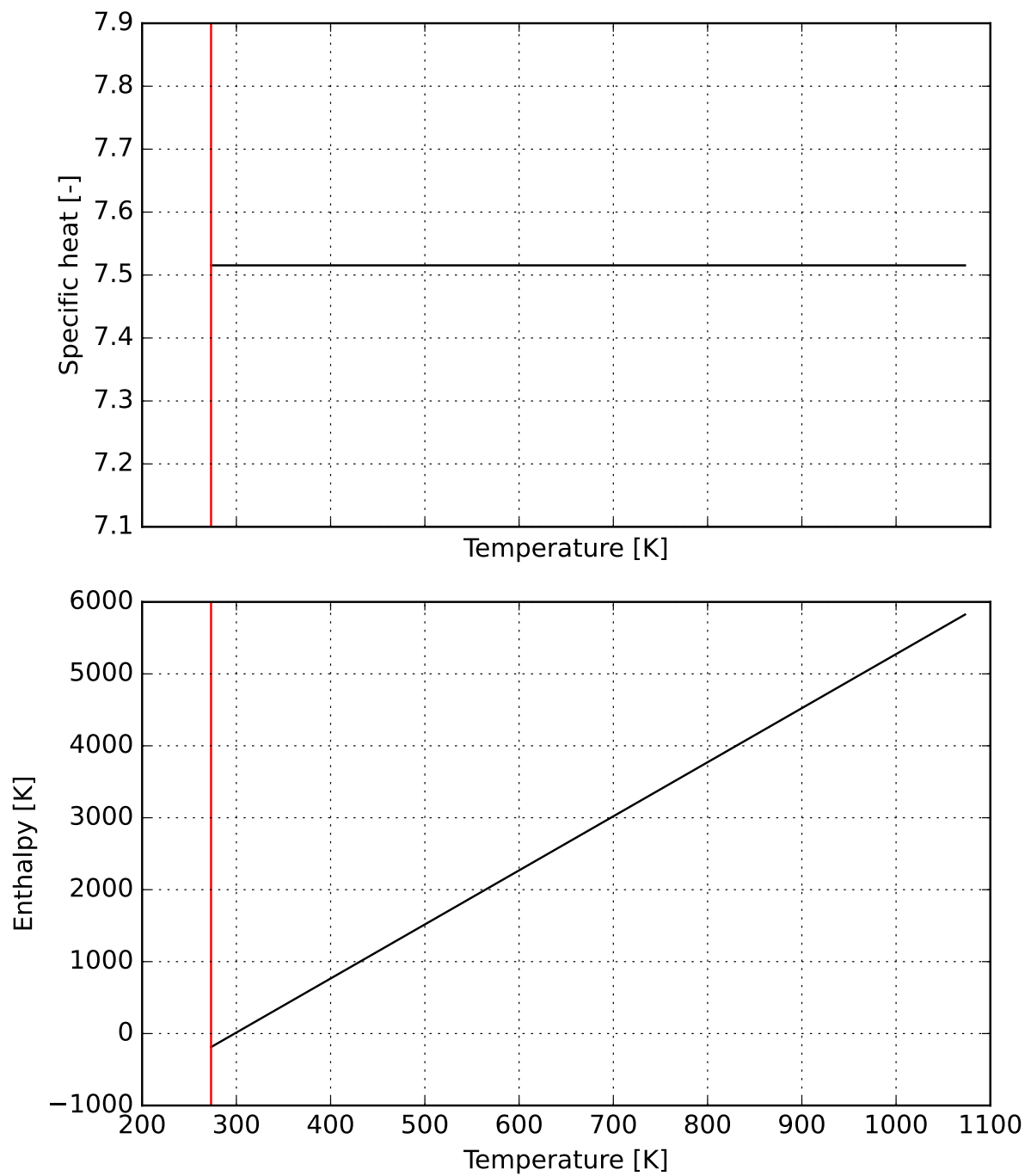
We start by declaring the symbols for temperature t and model coefficients a :

The next cell provides the representations of specific heat capacity and enthalpy as described [here](#). We also provide the derivatives of these functions computed algorithmically with help of `casadi.gradient`.

Below we illustrate the symbolic computational graph of operations for checking the functions:

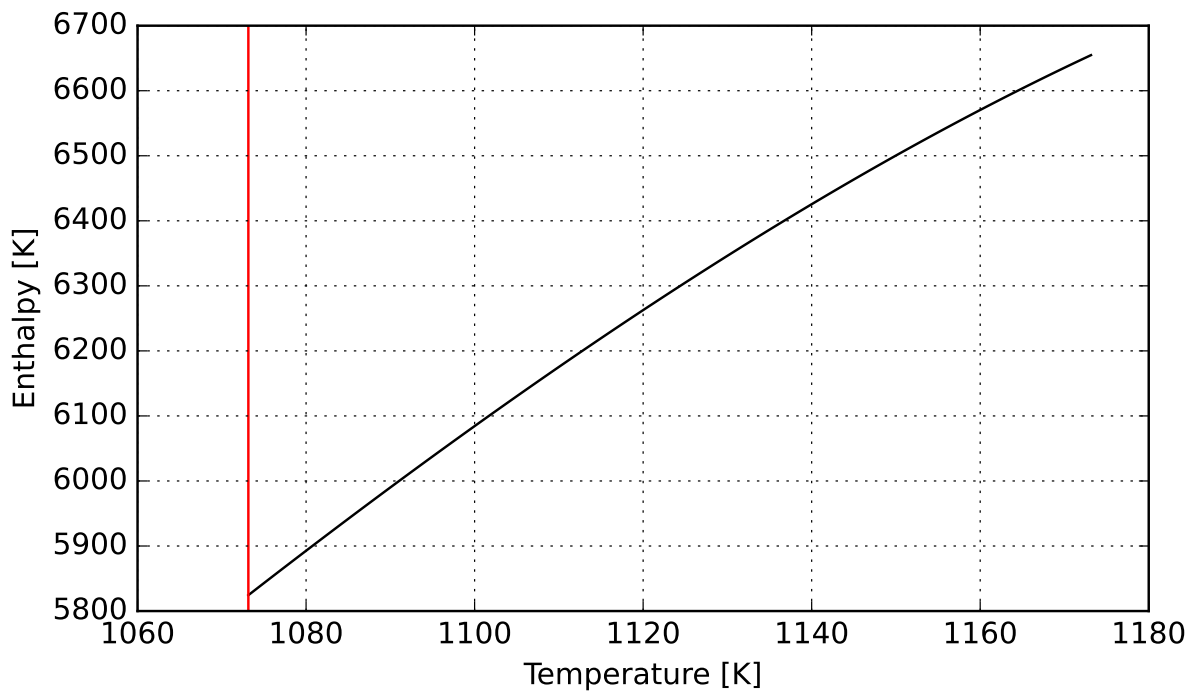
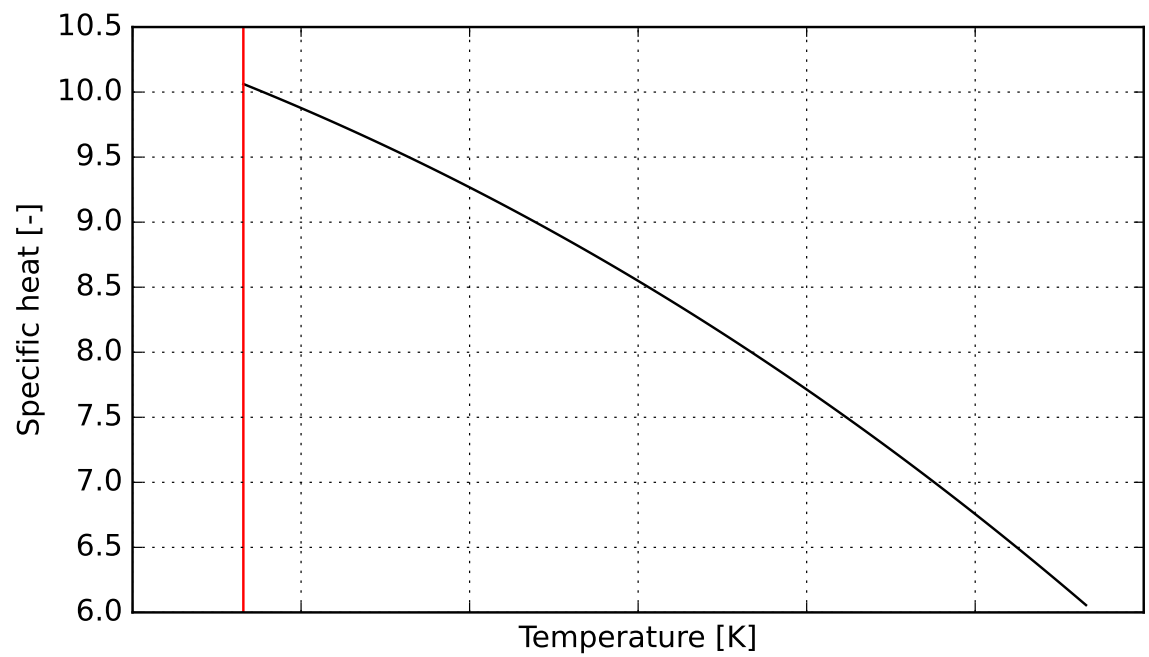
```
SX((((a_0+(a_1*t))+(a_2*sq(t)))+(a_3*(t*sq(t)))+(a_4*sq(sq(t)))))
```

Below the reaction start we enforce a constant specific heat with zero enthalpy at reference temperature:



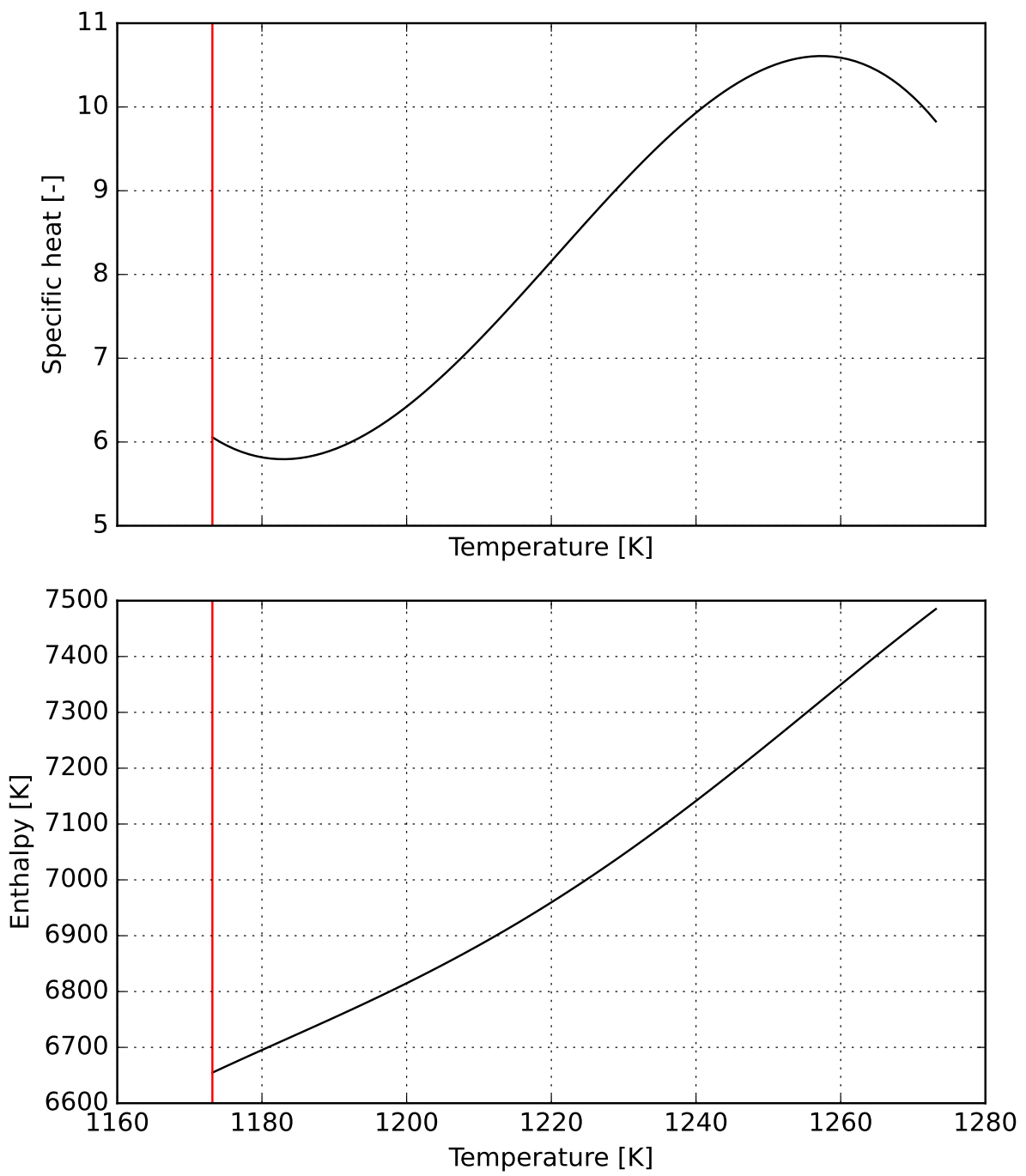
The reaction range is split in two parts, the first corresponding to a raising specific heat which reaches the maximum value.

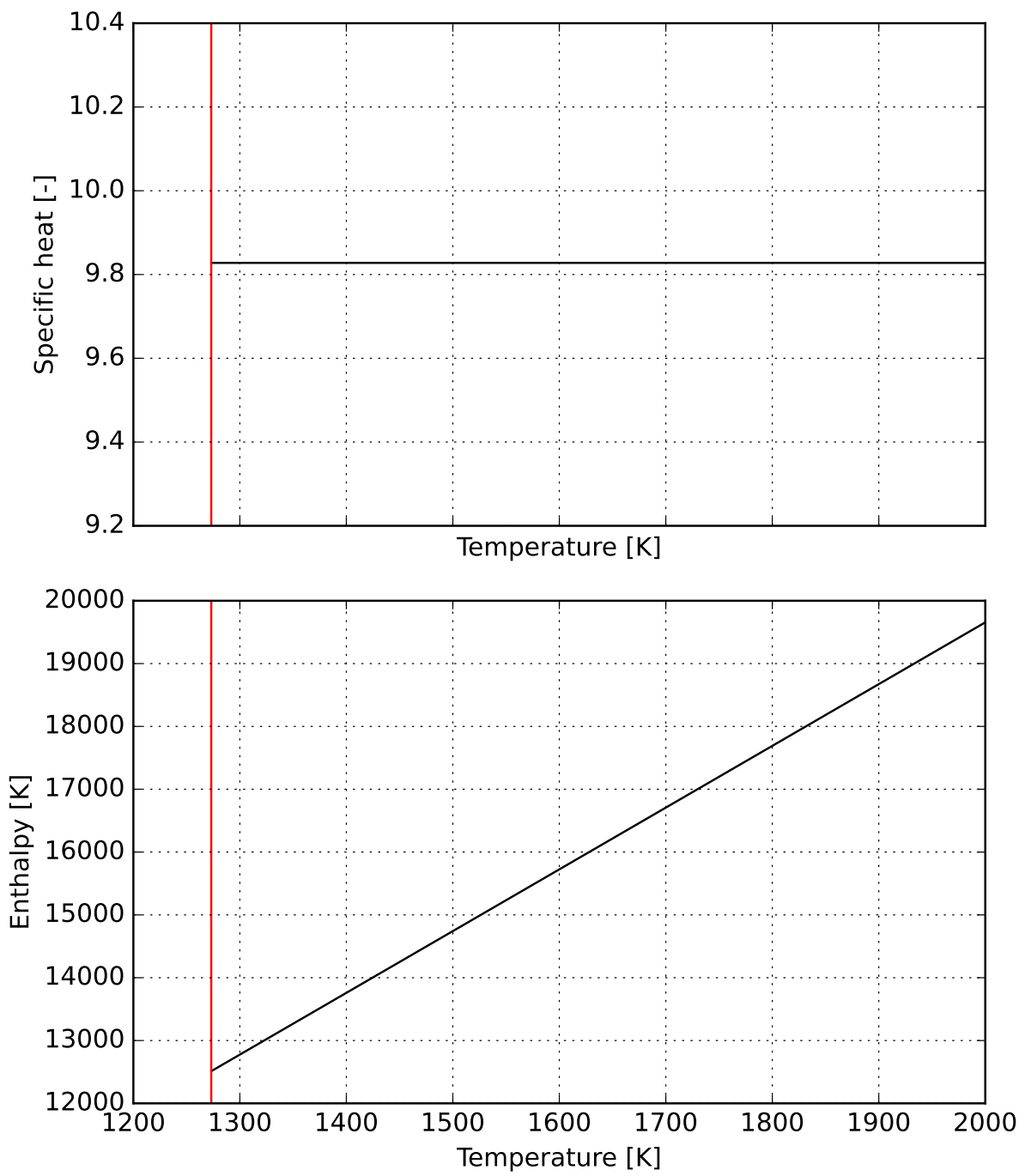
DM([1.81899e-12, -3.41061e-13])

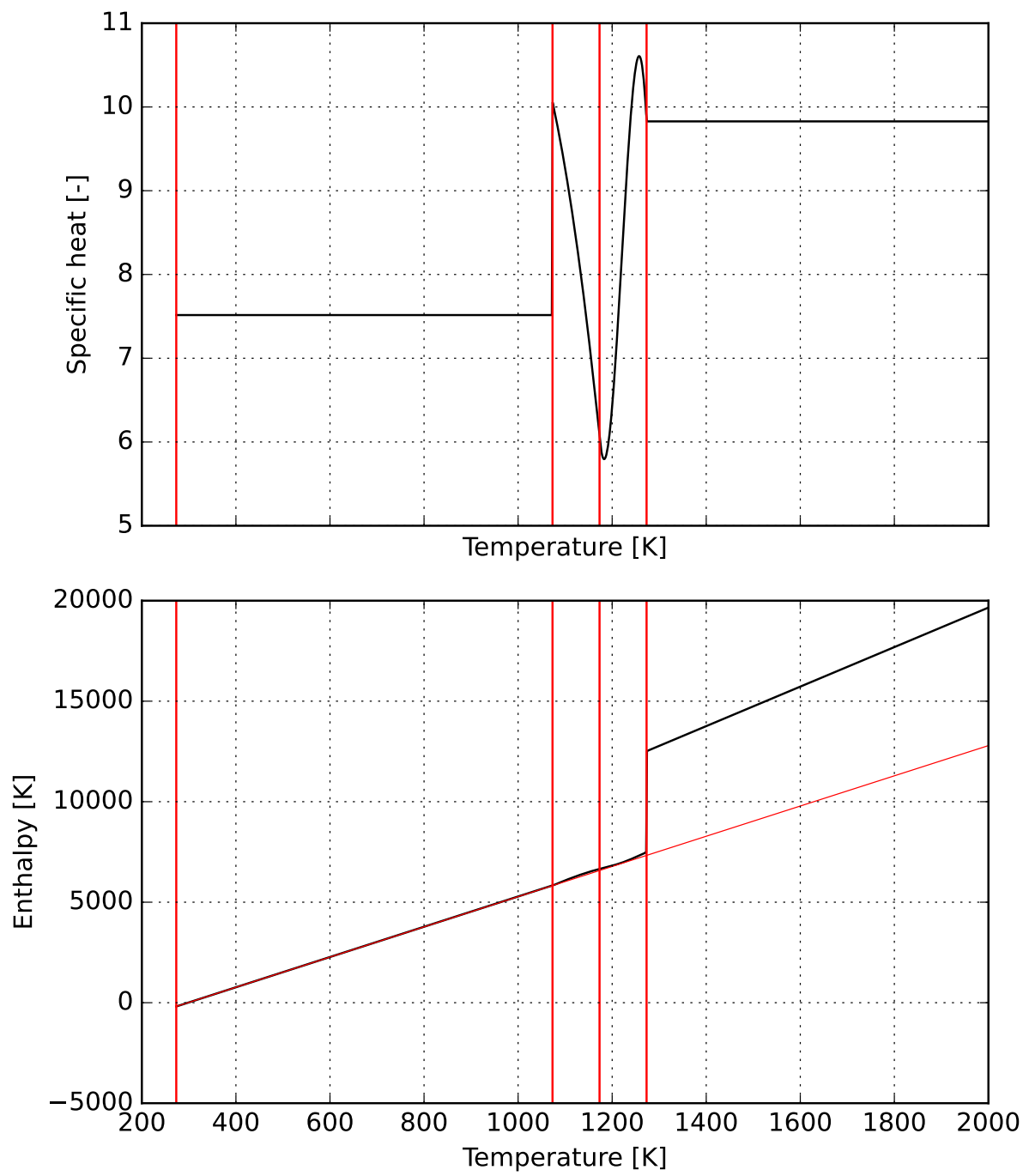


For the second part of the reaction we ensure continuity from the previous one

`DM([-6.74596e-09, -5.60689e-10, -2.45139e-10, 2.10093e-09, -1.1654e-09])`







Part III

Diffusion in Solids

9 Diffusion in solids

Assume a simple concentration-dependent diffusion in the absence of external driving forces; in this introduction we chose to *model* the diffusive flux through Fick's constitutive (first) law stated below, where D is the diffusion coefficient that may depend on position x and/or concentration c , *i.e.* $D \equiv D(x, c)$. It states that diffusive flux of an species has the composition gradient as a driver force; the negative sign indicates that diffusion occurs from high to low concentration regions.

$$J = -D \frac{\partial c}{\partial x}$$

We can anticipate here that this can be the case for simple diluted systems, but interactions between species must be considered in concentrated solutions, what is generalized by the *Irreversible Thermodynamics* by Onsager (1931a) and Onsager (1931b). Here we emphasize again the word *model*: empirical observation has led scientists to *represent* reality through such an approximation; different models can be used for the same physics with their applicability limited to certain scenarios. Properly selecting an applicable model is the role of the researcher/engineer, what is outside our scope here; in what follows we will focus on *how to solve* balance equations for a given model.

Simply put, diffusion equation solves for a local mass balance; and mathematically, a local balance leads to a divergence operation. Following this idea, it can be shown that the time derivative of concentration c at one location is given by the divergence - which collapses to the spatial derivative - of the negative of mass flux given by Fick's (first) law (or in general any other constitutive law/model describing the mass flux); sometimes this form is called Fick's second law in the literature. Given the possible non-linearity introduced by diffusion coefficient D , the derivative in the right-hand side is not expanded. In Cartesian coordinate system, the governing diffusion equation in one dimension writes:

$$\frac{\partial c}{\partial t} = \frac{\partial}{\partial x} \left(D \frac{\partial c}{\partial x} \right)$$

This is the form of the diffusion equation that is discussed in what follows.

9.1 System discretization

```
import pandas as pd
import sympy as sp
from abc import ABC
from sympy import expand, solve
from pandas import DataFrame
from majordome.utilities import sympy_symbols_factory

sp.init_printing(use_latex="mathjax")
```

Create all of the SymPy symbols used in the FVM derivations.

```

sympy_symbols_factory(
    # General FVM parameters and variables.
    "delta", "tau",

    # Boundary condition parameters and variables
    "h", ("c_inf", r"c_{\infty}"),

    # Cell-centered variables (compositions):
    "c_0", "c_E", "c_P", "c_W", "c_G", "c_B",

    # Cell-centered variables (diffusion coefficients):
    "D_E", "D_P", "D_W", "D_G", "D_B",

    # Face-centered variables (diffusion coefficients):
    "D_e", "D_w", "D_g", "D_b",

    # Face-centered variables (Fourier numbers):
    "Fo_e", "Fo_w", "Fo_g", "Fo_b",

    # Dump all of the above into the global namespace:
    scope = globals()
)

```

Create a helper class for representing the standard format of finite volume equations. This class extends `sympy.Eq` with some utilities for working with the equations in a more convenient way for our purposes. The main idea is to have a consistent way to represent the equations in the form $F(x) = 0$, where F is a function of the unknowns x . This allows us to easily extract coefficients and tabulate them in a consistent way across the notes.

```

class Eq(sp.Eq):
    """ Standard LHS format of finite volume equation. """
    def __new__(cls, lhs, rhs=0, **kwargs):
        return sp.Eq.__new__(cls, sp.expand(lhs), 0, **kwargs)

    def __truediv__(self, other):
        return Eq(self.lhs / other)

    def __rtruediv__(self, other):
        return Eq(other / self.lhs)

    def __mul__(self, other):
        return Eq(self.lhs * other)

    def __rmul__(self, other):
        return Eq(other * self.lhs)

    def subs(self, *args, **kwargs):
        """ Substitute into the equation LHS. """
        return Eq(self.lhs.subs(*args, **kwargs))

    def solve_for(self, var):
        """ Solve the equation for the specified variable. """

```

```

return sp.solve(self, var)

def coefficients(self, *, variables=None, op=lambda x: x):
    """ Extract coefficients for the specified variables. """
    if variables is None:
        variables = list(self.lhs.free_symbols)
        variables = [v for v in variables if str(v).startswith("c_")]

    variables = [v for v in variables if v in self.lhs.free_symbols]
    return {var: op(self.lhs.coeff(var)) for var in variables}

def tabulate(self, **kwargs):
    """ Helper to display a table of coefficients for the problem. """
    coefs = self.coefficients(**kwargs).items()
    return pd.DataFrame(coefs, columns=["Variable", "Coefficient"])

```

To solve the diffusion equation numerically, we employ the finite volume method (FVM). The first step to establish a FVM scheme consists of integrating the governing equation over a control volume P from w (west face) to e (east face) and over a time interval from 0 to τ . The left-hand side is first integrated over time as it represents a *storage* term, meaning that it measures how much the average concentration changes in the control volume over the time-step. The right-hand side is a *flux* term, and using similar arguments its *natural* integration is performed over space. Also, the order of integration can be exchanged due to the smoothness assumptions considered here. For more details, please check Patankar (1980).

$$\int_w^e \int_0^\tau \frac{\partial c}{\partial t} dt dx = \int_0^\tau \int_w^e \frac{\partial}{\partial x} \left(D \frac{\partial c}{\partial x} \right) dx dt$$

This double integration allows us to convert the partial differential equation into a form suitable for numerical discretization. Evaluating the innermost integrals leads to:

$$\int_w^e (c_P - c_P^0) dx = \int_0^\tau \left[\left(D \frac{\partial c}{\partial x} \right)_e - \left(D \frac{\partial c}{\partial x} \right)_w \right] dt$$

Here, c_P^0 and c_P represent the concentration at the center of the control volume P at times 0 and τ , respectively. A convention of not adding a superscript to time τ is adopted here, unless required by the context. The right-hand side represents the net flux into the control volume through its east and west faces. Assuming the concentration is uniform within each control volume and the fluxes are constant over the time interval, we obtain:

$$(c_P - c_P^0) \frac{\delta}{\tau} = \left(D \frac{\partial c}{\partial x} \right)_e - \left(D \frac{\partial c}{\partial x} \right)_w$$

where δ is the width of the control volume. This equation represents a discrete time-step of the diffusion equation for a single control volume, where one still needs to specify a discretization of the gradients, *i.e.* how the fluxes are evaluated at a cell face. A simple introductory approach is to use forward differences, where the composition gradients are approximated as first differences between neighbors.

To approximate the aforementioned gradients, the next cell introduces the expressions for the fluxes at east and west faces. These helper functions ensure the order of arguments is consistent with the finite difference definition given above.

```

def diffusion_flux(D, c_left, c_right, delta):
    """ Compute the forward difference diffusion flux between two points. """
    return -D * (c_right - c_left) / delta

def diffusion_flux_east(D=D_e, c_right=c_E, delta=delta):
    """ Compute the diffusion flux to the east cell. """
    return diffusion_flux(D, c_P, c_right, delta=delta)

def diffusion_flux_west(D=D_w, c_left=c_W, delta=delta):
    """ Compute the diffusion flux to the west cell. """
    return diffusion_flux(D, c_left, c_P, delta=delta)

```

For standardizing equations as organized as $F(x) = 0$ we override `sympy.Eq` with `Eq` class used in what follows. This allows us to easily extract coefficients and tabulate them in a consistent way across the notes. Below we compose the diffusion equation with help of [SymPy](#) and the above extensions. Notice that when compositing the right-hand side `rhs` we multiply by -1 because we are integrating the negative of the fluxes.

```

J_w = diffusion_flux_west()
J_e = diffusion_flux_east()

lhs = (c_P - c_0) * delta / tau
rhs = -1 * (J_e - J_w)

```

```

eq = Eq(lhs - rhs)
eq

```

$$-\frac{D_e c_E}{\delta} + \frac{D_e c_P}{\delta} + \frac{D_w c_P}{\delta} - \frac{D_w c_W}{\delta} - \frac{c_0 \delta}{\tau} + \frac{c_P \delta}{\tau} = 0$$

Here, c_E and c_W are the concentrations at the eastern and western neighboring control volumes, respectively, and D_e and D_w are the effective diffusion coefficients at the east and west faces of the control volume. Patankar (1980) shows that the evaluation of these diffusion coefficients is not arbitrary and for conservation (here the continuity of flux across the interface) to be respected one must take the harmonic mean of neighboring cells:

$$D_k = 2 \frac{D_i D_j}{D_i + D_j}$$

A final step to get a linear form of the problem is multiplying both sides by τ/δ and rearranging terms. This form clearly shows the contribution of each neighboring concentration to the rate of change at point P. The coefficient of c_P is negative, representing the outflow from the central node, while the coefficients of c_E and c_W are positive, representing inflow from the neighboring nodes.

$$-\frac{D_e c_E \tau}{\delta^2} + \frac{D_e c_P \tau}{\delta^2} + \frac{D_w c_P \tau}{\delta^2} - \frac{D_w c_W \tau}{\delta^2} - c_0 + c_P = 0$$

Introducing dimensionless coefficients For_k , the [Fourier number](#) for mass transfer (more generally referred to as $\mathcal{F}o_m$), we obtain the final discrete form. It represents the dimensionless diffusion number, which determines the stability and accuracy of the numerical scheme. The subscript k refers to either the east (e) or west (w) face.

$$Fo_k = \frac{\tau D_k}{\delta^2}$$

This definition is provided below for helping with the replacements that follow.

```
def fourier_number(D, tau, delta):
    """ Compute the Fourier number for the given parameters. """
    return D * tau / delta**2
```

```
SUBS_FOURIER = [
    (fourier_number(D_e, tau, delta), Fo_e),
    (fourier_number(D_w, tau, delta), Fo_w),
    (fourier_number(D_g, tau, delta), Fo_g),
    (fourier_number(D_b, tau, delta), Fo_b),
]
```

Performing the replacements with help of SymPy:

```
eq = eq.subs(SUBS_FOURIER)
eq
```

$$-Fo_e c_E + Fo_e c_P + Fo_w c_P - Fo_w c_W - c_0 + c_P = 0$$

Notice that so far nothing has been said with respect to the instant at which the right-hand side is evaluated. The obvious solution would be to take $c_0 \equiv c_P^0$ as it is currently known, but for numerical stability reasons we will continue here with an implicit scheme so that all values other than c_0 are defined to be evaluated at time τ , *i.e.* $c_K \equiv c_K^\tau$, in phase with the notation convention introduced above.

Because it is more convenient for symbolic manipulations with SymPy, the problem is currently written as $F(c) = 0$, *i.e.* all elements are on the left-hand side. Care must be taken to properly interpret the signs of the coefficients below.

```
coefs = eq.coefficients(variables=[c_0, c_E, c_P, c_W, c_G, c_B])
coefs
```

$$\{c_0 : -1, c_E : -Fo_e, c_P : Fo_e + Fo_w + 1, c_W : -Fo_w\}$$

Because it is more readable, a helper utility is provided to tabulate them:

```
eq.tabulate(variables=[c_0, c_E, c_P, c_W, c_G, c_B])
```

	Variable	Coefficient
0	c_0	-1
1	c_E	-Fo_e
2	c_P	Fo_e + Fo_w + 1
3	c_W	-Fo_w

9.2 Boundary conditions

So far all we have derived are the internal balances of the system, we still lack proper boundary conditions. It should be evident by now that we have formulated a tridiagonal matrix for representing the 1-D system: there are 3 unknowns at step τ (c_W , c_P , and c_E) and a currently known *field* c_0 .

In what follows we will work out the three families of boundary conditions: Dirichlet (type 1), Neumann (type 2), and the more general Robin (type 3). Following this sequence will create the required intuition for deriving the progressively more complex conditions. In practice, for diffusion in solids often only the implementation of Robin conditions is required, as it will be discussed later. Our goal at presenting types 1 and 2 is simply to introduce the tooling required for working out type 3 boundary condition; they will not be treated in full detail, leaving the full discussion for later.

The implementation of boundary conditions requires special treatment of the control volumes at the domain boundaries. The general form for a boundary control volume is:

$$(c_P - c_P^0) \frac{\delta}{\tau} = \text{flux}_{\text{interior}} - \text{flux}_{\text{boundary}}$$

It is worth here repeating the partial development derived in the previous section for an analogy:

$$(c_P^\tau - c_P^0) \frac{\delta}{\tau} = D_e \frac{c_E - c_P}{\delta} - D_w \frac{c_P - c_W}{\delta}$$

Please, notice that from now on we make use of class `StandardProblem` for the end of representing a canonical discrete equation. It encodes the results obtained in the previous section and is also repeated below:

```
class StandardProblem:
    """ Standard format of finite volume diffusion problem in 1-D. """
    @staticmethod
    def get_lhs(**kw):
        """ Standard format of inner LHS of problem. """
        return (c_P - c_0) * delta / tau

    @staticmethod
    def get_rhs(**kw):
        """ Standard format of inner RHS of problem. """
        J_e = kw.get("J_e", diffusion_flux(D_e, c_P, c_E, delta))
        J_w = kw.get("J_w", diffusion_flux(D_w, c_W, c_P, delta))
        return -1 * (J_e - J_w)

    @staticmethod
    def full_expression(subs_lhs=[], subs_rhs=[], subs_fourier=False, **kw):
        """ Canonical form of problem set as `f(c)=0`. """
        lhs = StandardProblem.get_lhs(**kw).subs(subs_lhs)
        rhs = StandardProblem.get_rhs(**kw).subs(subs_rhs)
        eq = Eq(lhs - rhs) * tau / delta
        return eq.subs(SUBS_FOURIER) if subs_fourier else eq
```

`StandardProblem.full_expression()`

$$-\frac{D_e c_E \tau}{\delta^2} + \frac{D_e c_P \tau}{\delta^2} + \frac{D_w c_P \tau}{\delta^2} - \frac{D_w c_W \tau}{\delta^2} - c_0 + c_P = 0$$

9.2.1 Type 1 BC - Dirichlet

Let's start by the Dirichlet, or fixed concentration, boundary condition (type 1), for which the concentration at the boundary face is prescribed, i.e. $c_B = c_{\text{prescribed}}$. Notice that before using this we need to make a choice regarding the computational grid (mesh); you could have half-volumes on boundaries, fictitious volumes (sometimes called ghost cells), or immersed nodes over the boundary.

Partial control volumes on boundaries are not desirable as they introduce implementation complexities in code and lack conservation (especially in higher dimensions), see a discussion by Maliska (2004) (section 3.7) for details. In what follows we discuss immersed nodes and extended domains with *ghost* cells. This last solution implies considerably more unknowns in 3-D problems, for which different variants of FVM will be discussed at a more advanced point.

Immersed boundary: Assume the last cell on *west* boundary is indexed as P ; over its *west* boundary we find a node where prescribed composition c_B is enforced - and the corresponding diffusion coefficient D_b . Notice that this node does not belong to the cell, nor represents a half-cell, but lies over its boundary. To keep it simple, let's assume c_B is constant (time-dependency would slightly complicate the development and it is worth postponing it to the algorithm development discussion), so keeping our fully implicit scheme leads to We start by computing the mass flux from the boundary node to the first cell center; this flux is computed using half-cell length $\delta/2$ for the derivative approximation.

$$J_w = \text{diffusion_flux_west}(D_b, c_B, \delta/2)$$

$$J_w = -\frac{2D_b(-c_B + c_P)}{\delta}$$

This expression of flux is used as replacement for the *west* boundary in the full equation:

```
eq = StandardProblem.full_expression(J_w=J_w, subs_fourier=True)
eq
```

$$-2Fo_b c_B + 2Fo_b c_P - Fo_e c_E + Fo_e c_P - c_0 + c_P = 0$$

The coefficients for the first row of the matrix problem are summarized below:

```
eq.tabulate()
```

	Variable	Coefficient
0	c_P	$2*Fo_b + Fo_e + 1$
1	c_E	$-Fo_e$
2	c_0	-1
3	c_B	$-2*Fo_b$

There are two main features in this equation: the coefficient of cell P now depends on the prescribed composition through α_b and a modification is applied to the right-hand side at boundaries where solution is prescribed. That's a good reminder that the array of coefficients must correspond to the number of interfaces, not cells - more on this later.

The immersed boundary approach presented above is simple. Nonetheless, it requires a different calculation of boundary nodes with respect to the volume, but that comes with the advantage that no additional equation has been added to the system. Let's now explore the extended domain alternative.

Ghost cell: As above, assume the last cell on *west* boundary is indexed as P . Consider another *ghost* cell to its left being denoted G (fictitious cell in place of W). The node c_B where composition is enforced and the same details presented above still apply here. Here we repeat the immersed node flux computation and do the same with respect to *ghost* cell G .

Our goal is to identify an equation for c_G so that it is solved along with the interior points and ensures the same flux as the immersed node. To identify the compatible *ghost* composition we equate the fluxes by both definitions and solve for c_G . The following steps repeat what we have seen so far and should be self-explanatory.

```
J_w1 = diffusion_flux_west(D_b, c_B, delta/2)
J_w2 = diffusion_flux_west(D_g, c_G, delta)
```

```
Eq(J_w1 - J_w2).tabulate()
```

	Variable	Coefficient
0	c_P	$-2D_b/\delta + D_g/\delta$
1	c_B	$2D_b/\delta$
2	c_G	$-D_g/\delta$

Care must be taken for interpreting the above result; here c_G takes the diagonal position and c_P is the first *east* cell in the first row of the problem's matrix. There is no previous state (no coefficient on c_0), but a source term based on c_B that must be placed on the right-hand side for solution. This is the equation we are looking for to extend the system.

Since this equation has D_g as a coefficient, an initial guess for c_G is required. The boundary B being the same interface for evaluation of D_g , for both fluxes to equate one needs to make sure the same diffusivity is evaluated there. Thus we make $D_b \equiv D_g$, what seems physically reasonable but so far not supported by any mathematical argument:

```
c_G_sol = Eq(J_w1 - J_w2).solve_for(c_G)[0]
c_G_sol = c_G_sol.subs(D_b, D_g)
c_G_sol
```

$$\frac{2D_g c_B - D_g c_P}{D_g}$$

This simply says that the composition is linearly extrapolated, with the distances from cell centers to the boundary being used as weighting factors. It ensures the problem is initialized the same as the immersed boundary as we made $D_g = D_b$.

Common mistake

One could think about solving for c_G and using the result in the standard balance equation, but doing so falls back to the immersed boundary approach. Replacing the solution in J_w2 simply gets back to the initial point and does not produce one extra equation!

```
J_w = J_w2.subs(c_G, c_G_sol)

eq = StandardProblem.full_expression(J_w=J_w, subs_fourier=True)
eq.tabulate()
```


	Variable	Coefficient
0	c_P	Fo_e + 2*Fo_g + 1
1	c_E	-Fo_e
2	c_0	-1
3	c_B	-2*Fo_g

9.2.2 Type 2 BC - Neumann

Next, let's handle Neumann boundary condition (type 2). For a fixed flux boundary condition, the flux at the boundary is prescribed at q_B . A special case arises when one imposes zero flux (impermeable boundary for mass transfer, adiabatic boundary for heat transfer) where $q_B = 0$. It is worthless formulating the problem for this case as the equation is trivial, so we will assume a general q_B in what follows.

$$-D \frac{\partial c}{\partial x} \Big|_B = q_B$$

When dealing with Neumann boundary conditions one must take care of the sign of fluxes, especially in 1-D. In higher dimensions, when applying Gauss theorem to convert the volume integral into a surface integral one normally works with vector calculus formulations; here we have dropped the vector notation when integrating over dx , but the concept of surface orientation must be recalled at this point. At the *west* boundary the flux into domain is positive and conversely, at the *east* boundary the flux out of domain is positive.

$$\text{(west)} \quad (c_P^\tau - c_P^0) \frac{\delta}{\tau} = D_e \frac{c_E - c_P}{\delta} + q_B$$

$$\text{(east)} \quad (c_P^\tau - c_P^0) \frac{\delta}{\tau} = -q_B - D_w \frac{c_P - c_W}{\delta}$$

We will not elaborate more as the most relevant condition represented by this class, the impermeable condition, can be represented with a type 3 boundary condition by setting the mass transfer coefficient h to zero.

9.2.3 Type 3 BC - Robin

Finally, we generalize the above discussion for a Robin boundary condition (type 3). Also known as a convective or mixed boundary condition. In this case the flux is proportional to the concentration difference between the boundary and a reference external composition c_∞ . This often models mass transfer between a reacting fluid and a solid, but the discussion of boundary layer formation is out of the present scope and we summarize it simply by the mass transfer coefficient h .

$$-D \frac{\partial c}{\partial x} \Big|_B = h(c_B - c_\infty)$$

i Fallback to other kinds

In fact, this is not the canonical form of such a boundary condition, but the reasonable engineering statement. Below we transform the equation to a format more commonly found in the mathematical literature, where all elements should be interpreted as being evaluated at boundary B .

$$\alpha c + \beta \frac{\partial c}{\partial x} = g$$

where

$$\alpha = 1 \quad \text{and} \quad \beta = \frac{D}{h} \quad \text{and} \quad g = c_\infty$$

Under this form we can easily jump to conclusions with respect to its relationship to the other types of boundary conditions. First, simply by setting $\beta = 0$ we find ourselves with a type 1 condition where $c = \alpha^{-1}g$ - or in engineering notation $c_B = c_\infty$; this happens as mass transfer efficiency is ideal, as $\beta \rightarrow 0$ when $h \rightarrow \infty$. In computational practice it is enough to enforce that $h \gg D$ to approach a Dirichlet boundary condition.

A type 2 condition would be established with $\alpha = 0$, but that is not possible straight from the way we have posed the equation. By multiplying everything by h and setting its value to zero we get to an impermeable Neumann condition, the only one we consider here for the mass transfer context (because in general it is not simple to enforce a mass uptake by a material).

That said, type 3 condition is *all we need* for a first solid state diffusion solver. As for the type 2 condition, we need to take care of signs here.

The (scalar) convention we used so far implies a convective flux that is positive *outwards* the domain, $J = h(c_b - c_\infty) > 0$ if $c_b > c_\infty$, *i.e.*, the chemical potential in the solid is higher than the environment. The west boundary being oriented towards the $-y$ axis, we need to account for its sign; in later chapters we will see that the boundary condition reasoning is actually simpler to follow in higher dimensions as we implicitly treat the face normal vectors through the inner product.

```
def convection_flux(cb, h, c_inf):  
    """ Compute the convection flux between environment and surface. """  
    return h * (c_inf - cb)
```

```
q_bw = -convection_flux(c_B, h, c_inf)  
q_be = +convection_flux(c_B, h, c_inf)
```

In what follows we derive the immersed boundary equation for type 3 condition. The extension to *ghost* cells is trivial and can be worked out with the ideas presented in section Section 9.2.1. We solve for c_B at both sides of the domain by equating the convective flux to the one established internally with the immersed boundary node:

```
J_w = diffusion_flux_west(D_b, c_B, delta/2)  
J_e = diffusion_flux_east(D_b, c_B, delta/2)  
  
c_Bw = Eq(J_w - q_bw).solve_for(c_B)[0]  
c_Be = Eq(J_e - q_be).solve_for(c_B)[0]
```

This leads to

```
c_Bw
```

$$\frac{2D_b c_P - c_\infty \delta h}{2D_b - \delta h}$$

and

c_{Be}

$$\frac{2D_b c_P - c_\infty \delta h}{2D_b - \delta h}$$

If we take, *e.g.*, the *west* boundary equation with the previous definition of flux evaluated from the immersed boundary node, we retrieve the corresponding equation for the system. Notice that c_B is a known value at the current iteration, so in solution it can be moved to the right-hand side and act as a flux term. This gives a grasp of the solution algorithm that makes the subject for the next section.

```
eq = StandardProblem.full_expression(J_w=J_w, subs_fourier=True)
eq
```

$$-2Fo_b c_B + 2Fo_b c_P - Fo_e c_E + Fo_e c_P - c_0 + c_P = 0$$

	Variable	Coefficient
0	c_P	$2*Fo_b + Fo_e + 1$
1	c_E	$-Fo_e$
2	c_0	-1
3	c_B	$-2*Fo_b$

9.3 Arbitrary grid spacing

For establishing the equations for an arbitrary grid spacing in 1-D, we move forward to a more general derivation. Instead of expressing the problem over a single coordinate, we introduce the control volume V that will be revisited again when solving for 3-D geometries. For convenience the minus sign is moved outside the integral in what follows.

$$\int_V \int_0^\tau \frac{\partial c}{\partial t} dt dV = - \int_0^\tau \int_V \nabla \cdot \vec{J} dV dt$$

The inner integration on the right-hand side is treated first. Using the divergence theorem, we convert the volume integral of the divergence of flux into a surface integral over the control volume's faces. Since in FVM we are dealing with a finite number of faces, we can express the surface integral as a sum over all faces f of the control volume, where J_f is the flux through face f and A_f is the area of that face.

$$\int_V \nabla \cdot \vec{J} dV = \int_S \vec{J} \cdot \vec{n} dS = \sum_f J_f A_f$$

Substituting this result back into the original equation and integrating the left-hand side over time gives:

$$\int_V (c_P - c_P^0) dV = - \int_0^\tau \sum_f J_f A_f dt$$

And finally integrating the last level:

$$(c_P - c_P^0) \frac{V}{\tau} = -\vec{J}_f \cdot \vec{A}_f$$

In 1-D, there are only two faces: the west face (w) and the east face (e). Thus, the summation over faces simplifies to just these two contributions:

$$(c_P - c_P^0) \frac{V}{\tau} = -J_w(-\hat{i})A_w(-\hat{i}) - J_e(+\hat{i})A_e(+\hat{i})$$

Notice that as we are dealing with the actual surfaces, the signs of the terms will be naturally handled by the flux definitions. The final step consists of substituting the expressions for the diffusive fluxes at each face, which can be approximated using finite differences. Below we account already for the face orientation (sign). For the west face:

$$-J_w(-\hat{i})A_w(-\hat{i}) = D_w \frac{c_W - c_P}{\delta_w} A_w$$

And for the east face:

$$-J_e(+\hat{i})A_e(+\hat{i}) = D_e \frac{c_E - c_P}{\delta_e} A_e$$

Substituting these flux expressions back into the integrated equation gives:

$$(c_P - c_P^0) \frac{V}{\tau} = D_w \frac{c_W - c_P}{\delta_w} A_w + D_e \frac{c_E - c_P}{\delta_e} A_e$$

Rearranging terms leads to the final discrete equation for an arbitrary grid spacing in 1-D implicit formulation:

$$-\frac{D_w A_w}{\delta_w} c_W + \left(\frac{V}{\tau} + \frac{D_w A_w}{\delta_w} + \frac{D_e A_e}{\delta_e} \right) c_P - \frac{D_e A_e}{\delta_e} c_E = \frac{V}{\tau} c_P^0$$

or assuming a unit cross-sectional area A so that $V = A\delta_P$, we have:

$$-\frac{D_w \tau}{\delta_w \delta_P} c_W + \left(1 + \frac{D_w \tau}{\delta_w \delta_P} + \frac{D_e \tau}{\delta_e \delta_P} \right) c_P - \frac{D_e \tau}{\delta_e \delta_P} c_E = c_P^0$$

The interpolation of the diffusivities is no longer a simply harmonic mean as in the case of uniform grid spacing, but it can be computed using the diffusivity values at the neighboring nodes and the distances to the faces. For interface ij we have:

$$D_{ij} = \frac{\Delta_{ij}}{D_i \delta_i^{-1}/2 + D_j \delta_j^{-1}/2} = \frac{(\delta_i + \delta_j)/2}{D_i \delta_i^{-1}/2 + D_j \delta_j^{-1}/2} = \frac{(\delta_i + \delta_j)}{D_i/\delta_i + D_j/\delta_j}$$

The above expression can be easily derived by considering the flux through the face as a series of two resistances (one for each neighboring node) and applying the concept of equivalent resistance in series. It falls back to the harmonic mean in the case of uniform grid spacing.

For the boundary conditions we apply the same approach as before:

$$-J_b(-\hat{i})A_b(-\hat{i}) = D_b \frac{c_B - c_P}{\frac{1}{2}\delta_P} A_b = hA_b(c_\infty - c_B)$$

Applying this expression to the full problem, we have that the length squared to be used in the Fourier number for the boundary is $\delta_P^2/2$. Before simplifying this expression to find out c_B it is worth introducing the **Sherwood number** Sh defined as follows, where L represents a characteristic length of the problem and D the diffusion coefficient; here $L = \delta_P/2$ as half-cell length was used in flux computations and

$$Sh = \frac{hL}{D} \quad \text{and} \quad Sh_b = \frac{h\delta_P}{2D_b}$$

leading to the final boundary node concentration

$$c_B = \frac{c_P + Sh_b c_\infty}{1 + Sh_b}$$

Using a segregated solution of non-linear boundaries or immersed nodes:

$$\begin{bmatrix} A_{(0)} & A_E & 0 & & \dots & 0 & 0 \\ A_W & A_P & A_E & & \dots & 0 & 0 \\ 0 & A_W & A_P & A_E & 0 & \dots & 0 & 0 \\ 0 & 0 & \ddots & \ddots & \ddots & \ddots & \vdots & \vdots \\ \vdots & \vdots & \ddots & \ddots & \ddots & \ddots & 0 & 0 \\ 0 & 0 & \dots & 0 & A_W & A_P & A_E & 0 \\ 0 & 0 & \dots & 0 & 0 & A_W & A_P & A_E \\ 0 & 0 & \dots & 0 & 0 & 0 & A_W & A_{(n)} \end{bmatrix} \cdot \begin{bmatrix} c_{(0)}^\tau \\ c_{(1)}^\tau \\ c_{(2)}^\tau \\ c_{(3)}^\tau \\ \vdots \\ c_{(N-3)}^\tau \\ c_{(N-2)}^\tau \\ c_{(N-1)}^\tau \end{bmatrix} = \begin{bmatrix} c_{(0)}^0 \\ c_{(1)}^0 \\ c_{(2)}^0 \\ c_{(3)}^0 \\ \vdots \\ c_{(N-3)}^0 \\ c_{(N-2)}^0 \\ c_{(N-1)}^0 \end{bmatrix} + \begin{bmatrix} B_{(0)} \\ 0 \\ 0 \\ 0 \\ \vdots \\ 0 \\ 0 \\ B_{(N-1)} \end{bmatrix}$$

10 Slycke's Models

Educational code for simulating diffusion processes in solids. Example application based on [Slycke's](#) models of carbonitriding processes in steel.

10.1 Preliminary calculations

The problem as formulated by Slycke and Ericsson (1981) is expressed in terms of concentrations, while the diffusion coefficients (which are based on geometric exclusion principles) require molar fractions. There may be some overhead in simulations if handling of unit conversion is not done carefully, so in this section we will derive the necessary relationships to convert between the two.

The problem may seem nonlinear, as density might be considered as a function of composition (in general for carbonitriding it is reasonable to assume that the density of the steel is constant, but we will not make this assumption here). From the definition of molar fraction, one can derive the mean number of atoms of a kind per unit cell in terms of the total number of atoms per unit cell and the molar fractions. For instance, for carbon and nitrogen we have:

$$x_c = \frac{N_c}{N_c + N_f + N_n}, \quad x_n = \frac{N_n}{N_c + N_f + N_n}$$

With a little manipulation, we can express the number of atoms per unit cell of carbon and nitrogen as a function of the molar fractions and the total number of atoms of iron N_f (which depends only on the crystal structure, here for FCC $N_f = 4$) per unit cell:

$$N_c = -\frac{N_f x_c}{x_c + x_n - 1}, \quad N_n = -\frac{N_f x_n}{x_c + x_n - 1}$$

In the absence of interstitial atoms, the density of plain iron expressed in terms of the unit cell parameters (see any introductory text on Materials Science, e.g. Callister (2007)) can be expressed as:

$$\rho_0 = \frac{M_f N_f}{A_v V_c}$$

Including the additional interstitial atoms - and neglecting their effect on lattice parameter, which is not a bad hypothesis at high temperature - the density of the system can be expressed as:

$$\rho = \frac{N_f (-M_c x_c + M_f (x_c + x_n - 1) - M_n x_n)}{A_v V_c (x_c + x_n - 1)}$$

Notice the common factors between the two expressions, which allows us to express the density of the system as a function of the density of pure iron and the molar fractions:

$$\rho = \frac{\rho_0 (-M_c x_c + M_f (x_c + x_n - 1) - M_n x_n)}{M_f (x_c + x_n - 1)}$$

The concentration is the ratio between the solution density and its mean molar mass. Writing the molar mass of the system as a function of the molar fractions and molar masses of the individual components is straightforward:

$$\bar{M} = M_c x_c + M_f (-x_c - x_n + 1) + M_n x_n$$

With this expression we find the trivial result:

$$C = -\frac{\rho_0}{M_f(x_c+x_n-1)}$$

and for the individual components:

$$C_c = -\frac{\rho_0 x_c}{M_f(x_c+x_n-1)}, \quad C_n = -\frac{\rho_0 x_n}{M_f(x_c+x_n-1)}$$

With these expressions, we can express the molar fractions as a function of the concentrations:

$$x_c = \frac{C_c M_f}{C_c M_f + C_n M_f + \rho_0}, \quad x_n = \frac{C_n M_f}{C_c M_f + C_n M_f + \rho_0}$$

The preliminary conclusions from the above calculations are:

1. Problem is to be initialized in terms of concentrations, which are computed from the molar fractions and the density of plain iron.
2. Solution is carried iteratively, so we do not need to handle the non-linearity arising from the dependence of composition with both species.
3. The model is responsible by converting from concentrations to mole fractions internally, so there is no impact on the solver.

10.2 Convergence analysis

10.3 Discretization methods

10.4 Comparison to validation data

Part IV

Development

11 Core library

The development focuses on portability and ease of setup, so by default any dependency relying on a C/C++ compiler is forbidden. This means that the development environment can be set up on any platform without needing to install additional compilers or tools.

Majordome code is documented using Numpydoc style docstrings, and the documentation is generated by module `majordome.documents` using Quarto markdown. This design choice reflects the academic aim of the package, with reporting and notebook support as first-class citizens.

The environment for developing Majordome requires at least a reasonably modern version of Python and Rust; documentation depends on Quarto and `uv` is used for managing the environment. The exact versions may change over time, but as of the last update, the following (minimum) versions are recommended:

- [Python 3.12+](#)
- [Rust toolchain](#)
- [Quarto CLI](#)
- [uv](#)

Tip

Ubuntu users can bootstrap the environment with the following commands; make sure to check the latest versions of the tools and update the commands accordingly. Podman is required for building the manylinux wheel inside a container for compatibility and release in PyPI.

```
sudo apt-get install -y podman
sudo apt-get install -y python3-dev

curl --proto '=https' --tlsv1.2 -sSf https://sh.rustup.rs | sh -s -- -y
source "$HOME/.cargo/env"

curl -LsSf https://astral.sh/uv/install.sh | sh
source "$HOME/.local/bin/env"

quarto_repo="https://github.com/quarto-dev/quarto-cli"
quarto_deb="quarto-1.9.37-linux-amd64.deb"

wget $quarto_repo/releases/download/v1.9.37/$quarto_deb
sudo apt-get install -y ./.$quarto_deb
```

11.1 Release process

In addition to what is described in the manual builds section, developers can also need to publish the docs, update the [version](#), and publish the new version to GitHub and PyPI. This is fully automated by `release.ps1` to ensure consistency in actions.

- Make sure the package builds in both Windows and Linux, and that the documentation builds (many tests are included in the documentation, so this is a good sanity check for the code as well). The following commands provide a good starting point for the verifications in Windows:

```
# Check the sanity of the package:
```

```
cargo check
```

```
# Run Rust tests:
```

```
cargo test
```

```
# Run Python tests:
```

```
uv sync --all-extras
```

```
pytest -v
```

```
# Manually generate a wheel and check it:
```

```
uv build --wheel
```

```
python -m zipfile -l dist/*.whl
```

- From WSL or a Linux machine, repeat the above but also try generating the Linux wheel inside the compliant manylinux container:

```
./release.sh
```

- Once verifications are done, run one of the following:

```
# - You fixed something small:
```

```
./release.ps1 patch
```

```
# - You added a new (non-breaking) feature:
```

```
./release.ps1 minor
```

```
# - You made a breaking change:
```

```
./release.ps1 major
```

```
# - Avoid this, unless you know exactly what you are doing:
```

```
./release.ps1 1.0.0
```

11.2 Manual release steps

This script is useful but can be fragile; it is better to have a fall-back plan in case something goes wrong.

- Select the new version to release:

```
$version = "major"
```

```
$version = "minor"
```

```
$version = "patch"
```

- Update the version in both `Cargo.toml` and `pyproject.toml`:

```
# First, test with --dry-run to check the changes that will be made:
uv version --bump $version --dry-run
cargo set-version --bump $version --dry-run
```

```
# After testing with the above, remove --dry-run:
uv version --bump $version
cargo set-version --bump $version
```

- You can manually check the changes in the version files; alternatively, the following snippet can be used to extract the version from Cargo.toml and use it for the commit message and GitHub release tag:

```
$inSection = $false
$version = $null

foreach ($line in Get-Content "Cargo.toml") {
    if ($line -match '^\[([.+]*)\]\s*$') {
        $inSection = $Matches[1] -eq "package"
        continue
    }

    if ($inSection -and $line -match '^version\s*=\s*"([~"]+)"') {
        $version = $Matches[1]
        break
    }
}
```

- Generate the command for WSL build and inspect it before running:

```
$winPath = (Get-Location).Path.Replace("\", "/")
$wslPath = (wsl wslpath $winPath).Trim()
$nixCmd = "cd $wslPath && source ~/.bashrc && ./release.sh"
```

- Build the wheels in both Windows and Linux (inside the compliant manylinux container):

```
Remove-Item -Force -Recurse "dist/" -ErrorAction Ignore

uv build --wheel
wsl -d Ubuntu -- bash -c $nixCmd
```

- Inspect the wheels (especially if new data was included):

```
python -m zipfile -l dist/*.whl
```

- If everything is fine, commit the changes, push to the repository:

```
git add "Cargo.toml" "pyproject.toml" "Cargo.lock" "uv.lock"
git commit -m "Release version $version"
git tag "v$version"
git push origin main --tags
```

- Create and activate a virtual environment:

```
uv venv --python python3.12 .venv
```

Activate on Windows:

```
. .venv/Scripts/activate
```

Activate on Unix:

```
. .venv/bin/activate
```

- Install the package before generating the documentation:

```
uv pip install .[docs]
```

- Point Quarto to the virtual environment:

Windows:

```
$env:QUARTO_PYTHON = ".venv/Scripts/python.exe"
```

Unix:

```
export QUARTO_PYTHON=".venv/bin/python"
```

- Generate the documentation and publish it to GitHub Pages:

```
quarto render docs --to html
quarto publish gh-pages --no-prompt --no-browser docs
```

- Create the release on GitHub:

```
gh release create "v$version" dist/*.whl --generate-notes
```

- Publish the new version to PyPI (owners only) is done manually with uv:

```
uv publish
```

12 .NET Extras

12.1 Design and Implementation Notes

- All files other than the main `Library.fs` use fully-qualified modules on top level instead of namespace declarations. This allows for avoiding one level of nesting and makes it easier to find the relevant code for each module.

12.2 Requirements

- [.NET 10](#) or later
- (Optional) [Gnuplot](#) for plotting results
- (Optional) [Quarto](#) for building documentation
- (Optional) [uv](#) for managing Python dependencies
- (Optional) [Python >3.12](#) for building documentation and running samples

12.3 Building and running the code

From the repository root, build the project using the .NET CLI:

```
dotnet build
```

In development mode you can run `dotnet restore` before building to ensure all dependencies are up to date and then build without restoring to save time:

```
dotnet build --no-restore
```

Tests can be run on a project basis using one of the following:

```
dotnet test src/Diffusion.Core/Diffusion.Core.Tests.fsproj
dotnet test src/Diffusion.Slycke/Diffusion.Slycke.Tests.fsproj
```

Sample programs are located in the `samples/` directory. They are supplied as [Verso Notebooks](#) and can be run directly from VS Code with the [Verso extension](#) installed. You can also run the samples from the command line using `dotnet fsi`. This allows you to interact with the code and modify it on the fly by copying snippets and executing them in the F# interactive session. Alternatively, you can export the samples as `.fsx` files and run them directly or with VS Code support.

12.4 Building the documentation

The API documentation is generated using `fsdocs` and is available in the `dotnet/` directory. Before building the API documentation, you need to install the `fsdocs-tool` global tool:

```
dotnet tool install -g fsdocs-tool
```

Then the documentation do be deployed in a server can be built using the following command:

```
fsdocs build --input docs/dotnet --output output
```

For navigating locally you need to use `watch`, as the generated documentation relies on client-side routing which doesn't work properly when opened directly from the file system:

```
fsdocs watch --input docs/dotnet
```

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